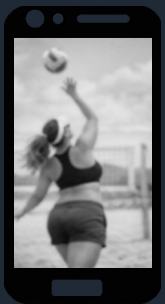
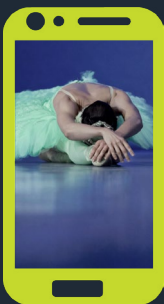
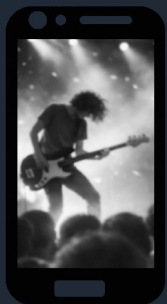
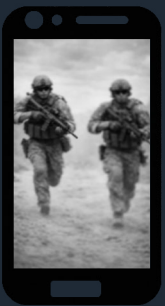
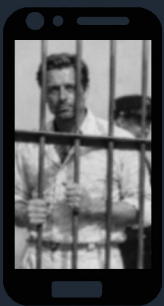
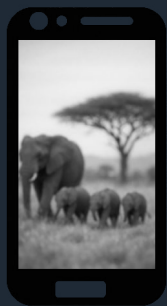
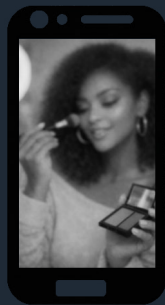
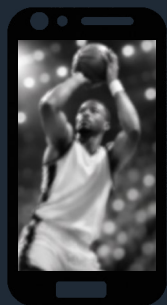


MOBILE PERSONALIZATION

The history and theory of
mobile streaming media



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JUSTIN GRANDINETTI
CHARLES ECENBARGER

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*The History and Theory of
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Justin Grandinetti &
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Mobile Personalization: The History and Theory of Mobile Streaming Media

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INTRODUCTION

Mobile Matters

On April 6, 2020, an app launched in the Apple iOS and Android marketplaces in the United States, Canada, and select European countries with the goal of revolutionizing the streaming media experience by capturing mobile “on the go” audiences. Quibi (a fitting portmanteau of “quick bites”) was backed by a significant \$1.75 billion dollars in investment capital from Disney, 21st Century Fox, Sony, Viacom, Warner Media, Lionsgate, MGM, NBC, Alibaba, Wall Street banks, and the billionaire Walton family, and featured Hollywood-produced content with recognizable stars. The app appeared, at least on paper, to have what it takes to battle it out in the “streaming wars” against other established and upstart platforms. Setting Quibi apart was a mobile-only focus on which viewers could watch shows and movie-length content in ten to twenty-minute segments. Now just a few years later, Quibi is a defunct platform—its library of seventy-five programs sold for a comparatively meager \$100 million and rebranded as “Roku Originals.” Instead of taking a seat at the increasingly crowded table of mainstay streaming platforms, Quibi has the ignominy of being a mere footnote in the history of streaming (or perhaps a great trivia night question).

In the broadest sense, the expeditious rise and fall of Quibi is a reminder of the impermanence of technologies, apps, and platforms in a digital age. Few send faxes, for instance, in the era of email. Despite newfound hipster status, cassette tapes have become a rarity due to compact disks, which were subsequently displaced by MP3s, which were then later displaced by streaming. The erosion of Facebook’s (now Meta) social media dominance felt impossible until TikTok came around. In short, the technologies, apps, and platforms we rely on today can become obsolete or replaced tomorrow.

Returning to Quibi, many have attempted a postmortem on the platform—the incredibly poor timing of releasing an on-the-go streaming app coinciding with the immobility and lockdowns of a global pandemic, a lack of clear market, and limited mobile-only functionality are all plausible reasons for the platform’s failure. Executives at Quibi got a lot wrong when it came to how much their product would revolutionize the streaming world; however, the people behind Quibi *did* correctly identify the growing prevalence and importance of mobile streaming around the world.

This book isn’t about Quibi, but instead what Quibi was about—the ever-expanding practices of streaming media via mobile devices. Our aim is to untangle the complexity of the contemporary landscape of mobile streaming media. To do so, we consider both the past and present. Emphasis on current mobile streaming means considering the changing mobile devices, new generations of wireless infrastructures, and innovative platforms and applications that make these sociotechnical practices possible. We also want to assess how this moment of mobile streaming came about in the first place, which requires attention to both antecedent technologies and the trajectory of a cultural desire for a mobile and portable personalization and monetization of ever-growing times and spaces. We aim to define the characteristics of contemporary mobile streaming. By extension, we take seriously the challenge of telling a more complete story about mobile streaming by bringing together scholarship on streaming, mobility and mobile media, infrastructure, platforms, screens, and spatial relations. Put perhaps most simply, this book is about how and why we have come to engage in mobile streaming in a complex and ever-shifting sociotechnical landscape.

Just how ubiquitous are the wide variety of mobile streaming practices? While it’s difficult to quantify exactly how much streaming occurs through mobile devices around the world, surveys and studies have shown growth in mobile engagement year after year. Ownership of mobile devices has been overtaking fixed alternatives for some time. As of 2019, just over 51% of the 3.9 billion worldwide mobile users accessed the web solely through smartphones.¹ These numbers have continued to grow steadily each year, as Cisco estimates that over 70% of the global population (roughly 5.7 billion people) have mobile web connectivity as of 2023.² Due to this reliance on mobile media, some estimates, like that from a 2018 report by Intel,

¹ Lucy Handley, “Nearly Three Quarters of the World Will Use Just Their Smartphones to Access the Internet by 2025,” *Yahoo! Finance*, January 24, 2019.

² “Cisco Annual Internet Report (2018–2023),” Cisco, March 9, 2020.

predict that video will account for 90% of all 5G wireless traffic by 2028.³ It should come as little surprise, then, that official data from popular platforms conforms to these predictions. Roughly 90% of Twitter video views and 70% of YouTube content are viewed through mobile devices.⁴ The most downloaded app of 2020 was the short-form mobile streaming social media app TikTok, with over two billion global downloads and 732 million monthly active users, who engage with the app an average of nineteen times and eighty-nine minutes per day.⁵ Taken as a totality, it is clear that mobile matters now more than ever.

The fact that individuals are streaming more and more content from mobile devices is not surprising, considering the host of options available for use on smartphones and tablets. Platforms like Spotify, Tidal, and Apple Music allow users to stream algorithmically curated playlists and suggested podcasts to their smartphones. Audio-visual content providers like Netflix, Hulu, and Disney+ have increasingly optimized their movie and televisual content for mobile functionality. Video conferencing and video calls through apps such as FaceTime, Zoom, and Google Meetings, which saw a marked increase in use during the Covid-19 pandemic, have become integrated into the regular fabric of life. The previously mentioned TikTok has quickly threatened the grip of established social media giants like Facebook and Instagram—something that seemed near impossible only a few years ago. Feeling the pressure, Instagram has launched a clone of TikTok via Instagram Reels, whereas other platforms like Twitter and YouTube increasingly emphasize mobile engagement. So inescapable is mobile streaming that even the popular dating app Tinder has experimented with its own interactive, live streamed TV-like series. What these highly varied examples make clear is that it's hard to draw clear boundaries around what mobile streaming can be and that an emphasis on streaming through mobile devices has taken a prominent role in the contemporary media landscape.

³ "Intel Study Finds 5G Will Drive \$1.3 Trillion in New Revenues in Media and Entertainment Industry by 2028," *Intel Newsroom*, October 11, 2018.

⁴ "New Research: Twitter Users Love to Watch, Discover, and Engage with Video," *X Blog*, Twitter, March 13, 2015; "Marketing Strategies," Think with Google, Google, April–June 2018.

⁵ Rebecca Bellan, "The Top Social Media Apps of 2020, According to Apptopia," *Forbes*, December 3, 2020; Tim Ingham, "TikTok Users Watch the App for an Average of 89 Minutes per Day—and There's 732M of Them Worldwide," *Music Business Worldwide*, April 14, 2021; Tracy Qu, "TikTok and China Version Douyin Surpass Two Billion Download Milestone, Underlining Continued Appeal," *South China Morning Post*, April 30, 2020.

Of course, the growing adoption of mobile practices is well-documented by mobile media scholars attentive to the Global South. Mobile web connectivity is the normative means of access for most people around the world, largely because wireless infrastructures are easier and cheaper to create, while mobile devices are generally more affordable and useful than their fixed and immobile counterparts.⁶ Streaming platforms typically monetize through one of three strategies: the ad revenue-based advertising video-on-demand (AVOD), the monthly access membership of subscription video-on-demand (SVOD), or the individual purchase-based transactional video-on-demand (TVOD). Despite these differences in how platforms make money, these services are locked in an unending battle for viewers, their money, and their data. Countries in the Global South are no stranger to being used as a site of resource extraction for Western powers, and streaming media unfortunately has some parallels with colonial projects of the past, as platform providers have looked for new sources of revenue in these countries.⁷ For example, Netflix has trialed a mobile device-only subscription plan in select Asian countries, while the Walmart-owned and India-based app Flipkart launched in-app streaming in 2019 in response to tech giant Amazon's encroachment into Indian markets with discounted Prime subscriptions and streaming. Like the aforementioned experiments in mobile streaming, these examples are merely some of the many new ways that streaming has on a global scale become inexorably intertwined with mobile devices, technologies, infrastructures, apps, platforms, and new spatial and mobile practices.

This complexity also means that holistic scholarly accounts of mobile streaming are challenging. Instead, various media studies approaches provide a prism that refracts different elements of streaming. For instance, some scholars focus on streaming media industry shifts, including how streaming has displaced traditional cable television subscriptions, altered our orientation to owning music, and upended movie theater release cycles that have been in place for decades.⁸ Others consider the changing features of mobile devices that have enabled new bodily practices and relations to space.⁹ Mobile streaming also has overlap with studies of the different generations

⁶ Jonathan Donner, *After Access: Inclusion, Development, and a More Mobile Internet* (MIT Press, 2015), 25.

⁷ Paola Ricaurte, "Data Epistemologies, the Coloniality of Power, and Resistance," *Television & New Media* 20, no. 4 (2019): 351.

⁸ Amanda D. Lotz, *Media Disrupted: Surviving Pirates, Cannibals, and Streaming Wars* (MIT Press, 2021), 12.

⁹ Jason Farman, *Mobile Interface Theory: Embodied Space and Locative Media* (Taylor and Francis, 2021), 4.

of wireless infrastructure that facilitate all kinds of everyday streaming practices.¹⁰ Additionally, mobile streaming connects to academic conversations on platforms, including critical accounts of corporate consolidation, control, platformization, and datafication in a world where there's always "an app for that."¹¹ Complicating matters even further is how streaming and, by extension, mobile streaming fit within historical narratives. For example, scholars have considered the technological antecedents of television, film, and radio in relation to contemporary streaming.¹² Others still add to this archeological mix precursors that include the telephonoscope, long distance radio DXing, and even canals, aqueducts, and pipelines as models of infrastructures of flow.¹³ And, as if mobile streaming needed yet *another* dimension, these practices are facilitated by screen technologies, which have long been the subject of scholarship on the role of how screens structure spaces of everyday life.¹⁴ Taken together, mobile streaming brings together quite the big tent of interrelated scholarship.

Although we draw from this wealth of scholarship throughout the book, our lighthouse for this project is Raymond Williams's canonical work on the televisual.¹⁵ Williams is an enduring theorist of cultural studies, and his account of the emergence of television is often referenced for good reason. Perhaps what is most influential about Williams is that he rejects technological determinism—the notion that technology determines cultural change—to instead trace the emergence of television as a cultural technology. In doing so, Williams shows the formation of the televisual as intertwined with cultural needs and shaped by systems of power. Key here is the concept of mobile privatization, in which Williams theorizes how both the television and automobile were part of a deep contradiction of centralized transmission and privatized reception in the 1950s. During this time, families became

¹⁰ Heather A. Horst, "The Infrastructures of Mobile Media: Towards a Future Research Agenda," *Mobile Media & Communication* 1, no. 1 (2013): 147.

¹¹ Jean-Christophe Plantin et al., "Infrastructure Studies Meet Platform Studies in the Age of Google and Facebook," *New Media & Society* 20, no. 1 (2018): 298.

¹² Doron Galili, *Seeing by Electricity: Cinema, Moving Image Transmission, and the Emergence of Television, 1878–1939* (Duke University Press, 2020): 108.

¹³ Benjamin Burroughs, "A Cultural Lineage of Streaming," *Internet Histories* 3, no. 2 (2019): 2; Justin Grandinetti and Chris Ingraham, "Infrastructures of Flow: Streaming Media as Elemental Media," *Critical Studies in Media Communication* 39, no. 2 (2022): 2.

¹⁴ Jeremy Packer and Kathleen F. Oswald, "From Windscreen to Widescreen: Screening Technologies and Mobile Communication," *The Communication Review* 13, no. 4 (2010): 312; Lev Manovich, "A Screen's Genealogy," in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017), 125.

¹⁵ Raymond Williams, *Television: Technology and Cultural Form* (Fontana, 1974).

both more mobile and more private and isolated; the television emerged as a sociotechnical response to the need for contact outside the privatized space of the home.

We want to make clear that while streaming and mobile streaming are influenced by legacies of the televisual, streaming is far more than merely “the next stage of TV.” We do, nevertheless, take inspiration from Williams in a few crucial ways. First, Williams’s research on television eschews both determinism and “eureka” moments. Applied to our own project, mobile streaming did not just appear as the result of one singular thought, invention, or decision. This brings us to the second key inspiration from Williams’s work on the televisual: the emergence of technologies as intertwined with cultural and historical trajectories. Like Williams’s project, ours is a mix of historical and contemporary examples that help unpack how and why mobile streaming engagement has become so quotidian, embedded, and important.

Inspired by Williams’s mobile privatization, we call this book “mobile personalization.” With this term we refer to the historical trajectory of personalization via mobile and portable media to better understand how these technologies are critical components in transforming times and spaces in providing entertainment and information for users while simultaneously operating toward the goals of monetization and control. This is perhaps where we believe Williams’s influence to be the most important—such a focus permits us to consider historical antecedent technologies and discourses, as well as the interplay of technological, industry, and infrastructural changes that facilitate our contemporary mobile moment. In turn, we aim to decenter the primacy of one single technology (often the smartphone or tablet) to instead consider the emergence and development of our contemporary practices as shaped by a variety of discourses, imaginings, power structures, economic models, technological changes, and infrastructural development.

Such an approach helps consider mobile streaming not as a discrete present-day technology or event, but instead as a shifting sociotechnical constellation of humans, nonhumans, and technologies. In turn, this emphasis allows us to reflect on emergent relationships of power (contemporarily manifesting via the growing reach of consolidated tech companies and platform infrastructure providers), as well as the economic implications of mobile streaming. Stated differently, it took many different interdependent social and technical shifts to bring us the current mobile streaming landscape, and we aim to unpack how this moment came to be.

The following chapters employ a material-discursive framework for understanding the social-cultural-technical emergence of contemporary mobile streaming. We consider the discursive: the imaginings, discussions, and discourses on mobile and portable technologies so powerful in shaping what is seeable, sayable, and knowable about these multivariant practices over time. Furthermore, we are attentive to material changes that facilitate both historical antecedents and contemporary mobile streaming through the connected interplay of platforms, apps, infrastructures, devices, APIs, big data and artificial intelligence, codes, and users. Consequently, we are interested in mobile streaming not only in relation to mobile phones, tablets, and screens, but also the complex mix of historical trajectories, antecedent technologies, cultural imaginaries, and infrastructural technologies that enable these practices.

In chapter 1, we begin with the historical roots of mobile personalization. We first detail our approach to studying mobile streaming before providing an overview of technologies commonly considered precursors to streaming media, such as film and television. As our focus is on mobile streaming, we complicate these historical narratives with attention to the drive-in cinema and the portable television—technologies often overlooked or on the margins of commonly accepted historical narratives. We outline a brief history on the emergence of these technologies, with particular attention to the monetization of spaces and times through novel portable and mobile technologies.

Chapter 2 examines architectures of mobility. Here, we go beyond the smartphone to also consider the infrastructures and platforms that enable mobile streaming media. This includes the role of platform models and wireless infrastructures, and how platforms and infrastructures of mobile streaming have come to merge. Moreover, we consider some of the key technical underpinnings of mobile streaming like big data, algorithmic models, and APIs. Smartphones are certainly the point of access when it comes to streaming content, but infrastructures of data flows also play a pivotal role in creating, sending, and receiving streaming media.

Chapter 3 moves from the historical considerations of chapter 1 and architectural focus of chapter 2 to assess the varied practices of contemporary mobile streaming. From TikTok to Twitch, Instagram Reels to Apple Music, YouTube to videocalls, Switch to Stadia, we bring together the wide array of how mobile streaming has become embedded in everyday life around the world. We specifically focus on the different strategies of monetization emergent through mobile streaming. While once billed as a disruptor that

would end cable television, streaming has come to look (and cost) quite a bit like its predecessor. Additionally, the number of emergent streaming platforms belies the industry consolidation of big tech, both in the United States and around the world. We highlight numerous examples, including how US streaming platforms have attempted to grab international audiences as well as new geopolitical proxy battles between companies like Meta and TikTok.

Chapter 4 brings together our focus on the material-discursive historical trajectories of mobile streaming, as well as the host of architectures and monetization strategies to consider how mobile streaming necessitates a reimagining of streaming literature. Scholarship about streaming as spatially fixed has provided excellent conceptual anchors, yet it is critical to reassess and reconceptualize these attributes in the era of mobile streaming. Specifically, we revisit and refine some of the characteristics commonly associated with streaming media: mobile streaming flow, liveness and sharedness, screens and space, mobility and hybridity, and algorithms and big data.

We conclude our book with final thoughts on the past, present, and future of mobile streaming. Reflecting upon the interconnected strands of history, infrastructures, monetization, and platforms, we consider what's next for mobile streaming and streaming generally. Futurists failed to anticipate a landscape where we're glued to our phones more than ever before. We hope that by creating a more holistic look at the trajectories that brought us to this moment, we can, at the very least, better understand the present.

CHAPTER ONE

The History of Mobile Personalization

On January 6, 2016, Netflix's co-founder and then-chief executive Reed Hastings delivered a keynote announcement to mark the platform's expanded global presence to more than 130 countries. Speaking to the Las Vegas Computer Electronics Show crowd, Hastings boasted:

Today you are witnessing the birth of a new global Internet TV network. With this launch, consumers around the world—from Singapore to St. Petersburg, from San Francisco to Sao Paulo—will be able to enjoy TV shows and movies simultaneously—no more waiting. With the help of the Internet, we are putting power in consumers' hands to watch whenever, wherever and on whatever device.¹

Within the year, Hastings continued the “Netflix-as-media-disrupter” narrative, this time targeting cinema. Commenting on a question about the future of cinema, Hastings juxtaposed the stagnant innovation of the movie industry with Netflix's desire to “unleash film” and grow the movie business by upending the traditional release model of films.²

Just a few years later, founder Jeffrey Katzenberg and CEO Meg Whitman of the aforementioned and much-maligned short-form streaming app Quibi similarly positioned the platform as a milestone in televisual evolution. Katzenberg made the following comments in a 2018 interview in which he attempted to generate buzz for Quibi:

¹ “Netflix Is Now Available Around the World,” *Netflix Newsroom*, Netflix, January 6, 2016.

² Janko Roettgers, “Netflix CEO Reed Hastings: Movie Theaters Haven't Innovated Beyond Popcorn,” *Variety*, March 17, 2017.

You take a TV in your pocket every day and ten minutes it happens all the time, waiting at the doctor's office and on your commute and it's taking longer and waiting for a friend for dinner, and we want to take that ten minutes and want to make those moments extraordinary by the service we're offering . . .³

In what ended up being an impressive publicity cycle relative to actual platform impact, the founder/CEO duo further emphasized that Quibi would also feature "cinematic content" broken down into quick, bite-sized portions for mobile streamers.⁴ Quibi, however, never took off, and the platform's content was eventually sold piecemeal to other services.

Aside from being streaming platforms, Netflix and Quibi aren't much alike. Netflix started as a snail mail delivery DVD rental service, while Quibi was born digital. Netflix operates globally and its name remains synonymous with streaming, whereas Quibi fizzled in under a year. And unlike Quibi, Netflix is not mobile specific (even though estimates show that the percent of mobile device access of the platform is growing each year). Nonetheless, what both of these platforms *do* have in common is a discursive connection with the televisual and cinematic. Put another way, there is clearly a link between our popular understanding of streaming today and audio-visual antecedent technologies—and major platform representatives have gone out of their way to cultivate this mindset.

As noted in the introduction, contemporary mobile streaming takes many forms, even when just limiting focus on the audio-visual. Not all of this content is televisual or cinematic in nature; for example, social media posts and mobile game streaming also complicate any attempt at a neat delineation. This conceptual morass leaves scholars of mobile streaming in a sticky situation. What do we make of statements by major platform representatives and our colloquial proclivities to situate mobile streaming as the latest step of a linear evolution of television? Or is streaming more akin to the movies—especially as prestige programming situates itself as cinematic? What about the social aspects of many mobile streaming platforms? And that's not to ignore mobile audio-visual game streaming . . .

It certainly makes logical sense that platform providers would invoke the popular and established media of television and cinema to facilitate the embrace of streaming. Television and cinema are familiar audio-visual de-

³ Jacob Wolinsky, "Katzenberg and Whitman Discuss Their New Streaming TV Venture, NewTV," *ValueWalk*, January 31, 2020.

⁴ Nicole Sperling, "What is Jeffrey Katzenberg's Quibi All About, and Why Should You Care?" *Vanity Fair*, June 14, 2019.

livery mechanisms that have stood the test of time and delivered countless hours of entertainment for mass audiences. Who wouldn't want to engage with the "future" of these familiar entertainment media?

And it also follows that there are some material similarities in terms of content and, increasingly, a convergence of delivery mechanisms. There are, of course, strands of media studies research that account for the seemingly inescapable televisual and cinematic lineage of streaming. As noted in the introduction, scholars have examined parallels and distinctions between television and new media, in turn rethinking the ontology and epistemology of the televisual itself.⁵ Put differently, to know the future of television one needs to understand what television *is* and *was* in the first place. There are others who focus on how streaming movies has heavily disrupted established cinematic release models—a phenomenon precipitated by the uneven recovery of cinema following Covid-19.⁶ Others still decenter the content of the televisual by instead examining the function of the screen itself as part of everyday activities and the structuring of space.⁷ Some even go so far as to claim that digital screens are best understood outside the paradigm of specific media; instead, film, televisions, computers, and telephones should no longer be treated as discrete.⁸ Contrary to the corporate bluster from streaming platform representatives, scholars of televisual, film, streaming, and screens do not merely accept that streaming is the direct and inevitable next form of television and cinema; nevertheless, these conceptual frameworks attempt to demystify the link between established mass media formats and contemporary streaming.

This book, however, is about *mobile* streaming. The history of the televisual is most often linked to the fixed site of the living room since the 1950s. Major film premiers have taken place in brick-and-mortar cinemas beginning in

⁵ Lynn Spigel, "Introduction," in *Television After TV: Essays on a Medium in Transition*, ed. Jan Olsson and Lynn Spigel (Duke University Press, 2004); Lisa Parks, "My Media Studies," *Television & New Media*, 10, no. 1 (2019); William Boddy, "Is It TV Yet? The Dislocated Screens of Television in a Mobile Digital Culture," in *Television as Digital Media*, ed. James Bennett and Niki Strange (Duke University Press, 2011); James Bennett, "Introduction: Television as Digital Media," in *Television as Digital Media*, ed. James Bennett and Niki Strange (Duke University Press, 2011).

⁶ Michael D. Smith and Rahul Telang, *Streaming, Sharing, Stealing: Big Data and the Future of Entertainment* (MIT Press, 2017), 34.

⁷ Anna McCarthy, *Ambient Television* (Duke University Press, 2001); Anna McCarthy, "From Screen to Site: Television's Material Culture, and Its Place," in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017), 389.

⁸ Kathleen Oswald and Jeremy Packer, "Flow and Mobile Media: Broadcast Fixity to Digital Fluidity," in *Communication Matters: Materialist Approaches to Media, Mobility, and Networks*, ed. Jeremy Packer and Stephen B. Crofts (Wiley (Routledge, 2012), 277.

the early 1900s. Yet, alongside the stationary imaginings of the audio-visual is an oft-ignored mobile and portable lineage. Our goal is to consider how mobile streaming has become so quotidian, embedded, and important. Doing so means taking seriously historical antecedent technologies to disrupt neat linear narratives. We believe that we must account for the persistent connection between streaming and television and film. But that doesn't mean that mobile streaming is merely the next evolution of these technological compositions. In essence, it's critical that we be cautious of industry hype cycles about the future of televisual and cinema via streaming as well as colloquial linear narratives of technological advancement from cinema to television to streaming to mobile streaming; at the same time, we also can't completely ignore, write-off, or disregard these persistent discursive connections between streaming and historical audio-visual antecedent technologies.

To make sense of this situation, we begin with the notion that mobile streaming is not merely a cause nor effect, but a complex and shifting mix of material technologies and discursive imaginings of audio-visual content on-the-go function toward mobile personalization. That is, how mobile and portable technologies are critical components in the ongoing transformation of times and spaces, by providing entertainment and information for users while simultaneously operating toward the goals of monetization and control.

In this chapter, we trace some mobile and portable antecedents to contemporary audio-visual mobile streaming: the portable television and the drive-in cinema. These technologies are often overlooked or on the margins of a more fixed and static history; nonetheless, they are critical to grasping how we arrived at the current moment of mobile and portable personalization. In what follows, we first detail this book's approach before examining each of these two technological antecedents, with particular attention to the monetization spaces and times through novel portable and mobile technologies.

Tracing Mobile Personalization

Our entry point for considering the historical lineage of mobile streaming comes from Raymond Williams, whose canonical work on the emergence and integration of the televisual into everyday life situates television as a cultural technology. In other words, Williams examines how culture and technology are intimately intertwined.

Media theory before Williams was often characterized by notions of technological causality and determinism. For instance, the Payne Fund Studies of the late 1920s through early 1930s, thought to be some of the earliest scholarship in the emerging field of media studies, used qualitative and quantitative social science methodologies to decipher the connection between juvenile delinquency and violent cinema. These studies set the tone for scholarly accounts designed to assess the consequences that the burgeoning mass media landscape had on individuals.⁹ Media effects research, born out of the new twentieth-century paradigm of mass media, often focuses on the relationship between “media” as content and how audiences make sense of what they see. Similar scholarship focusing on the relationship between content and societal effects has continued for decades.¹⁰ Relatedly, some strands of critical theory from the mid-twentieth century, though varied in how these theorists study the tête-à-tête between culture and technology, is similarly concerned with the impact of a powerful media on easily influenced populations. The Frankfurt School, for example, wrote in response to what they saw as an erosion of individual thought when the individual in a hyper-consumerist landscape internalizes mechanisms of social control.¹¹ Taken together, there is a near one-hundred-year-old proclivity to focus on media as content, to research how this content affects society (or vice versa), and to treat this relationship as relatively unidirectional.

By contrast, Marshall McLuhan is an enduring theorist whose writings eschewed attention to content to instead provide an expansive definition of media as mediator. His radical redefinition of “media” included clothes, money, and roads.¹² Instead of questioning how individuals interacted with mass media content, McLuhan focused on the scale, pace, and pattern new technologies introduced into human affairs. While McLuhan’s proto materialist accounts of technology provide an interesting and important direction for media studies, his work is tinged with a different kind of determinism—that resulting from his belief in technology’s capacity to increase individual freedom.¹³ As explained by Tiziana Terranova, McLuhan’s vein of techno optimism also characterized many discussions of new media in the

⁹ Herbert Blumer and Philip M. Hauser, *Movies, Delinquency, and Crime* (Macmillan, 1933).

¹⁰ Elizabeth M. Perse, *Media Effects and Society* (Laurence Erlbaum Associates, 2001), 1.

¹¹ Herbert Marcuse, *One-Dimensional Man: Studies in the Ideology of Advanced Industrial Society* (Beacon Press, 1964), 9.

¹² Marshall McLuhan, *Understanding Media: The Extensions of Man* (MIT Press, 1964).

¹³ James Bennett, “Introduction,” in *Television as Digital Media*, ed. James Bennett and Niki Strange (Duke University Press, 2011), 9.

early 2000s.¹⁴ Although McLuhan helped popularize an important avenue of ingress for media studies by going beyond content, his work is mired in determinist thinking.

Countering McLuhan, Raymond Williams rejects the notion that technology itself causes cultural change. Instead, Williams examines how the televisual emerged as a provisional outcome of complex social processes—in large part the increased mobility and privatization of American family life in the 1950s. By tracing the emergence of a particular technology as entangled with cultural structures, Williams provides a model for understanding how the emergence and acceptance of a technology cannot be divorced from its historical time period.

Of course, Williams's scholarship is not without some well-earned criticism of its own. Like many scholars of the time, Williams prioritizes a human-centric notion of agency. Media studies work has, over time, come to understand agency as relational, highlighting the role of technical machines as contingent components in larger societal assemblages.¹⁵ Whether it's assemblage theory, actor-network theory, object-oriented ontology, or vital materialism, media studies scholarship after the so-called materialist turn often highlights the importance of non-humans in expressing, maintaining, and transforming relationships of power.¹⁶ In other words, more materialist accounts strike a balance between the passive inevitability of McLuhan's technological determinism and Williams's overly human-centered and culturally driven focus.

One approach to assessing the intricacies of the relationship between culture and technology over time is what's known as media archaeology. Most generally, media archaeological techniques disrupt canonized narratives of media history to trace how the specific affordances, constraints, and discourses of technical media are constitutive of power relations in a given cultural

¹⁴ Tiziana Terranova, "Cyberculture and New Media," in *The Sage Handbook of Cultural Analysis*, ed. Tony Bennett and John Frow (Sage, 2008), 597.

¹⁵ Jennifer Daryl Slack and J. MacGregor Wise, *Culture and Technology: A Primer*, 2nd ed. (Peter Lang Publishing, 2015), 138.

¹⁶ Bruno Latour, *Science in Action: How to Follow Scientists and Engineers Through Society* (Harvard University Press, 1987); Donna Haraway, *A Manifesto for Cyborgs: Science, Technology, and Socialist Feminism in the 1980s* (Center for Social Research and Education, 1985); Gilles Deleuze and Felix Guattari, *A Thousand Plateaus: Capitalism and Schizophrenia* (University of Minnesota Press, 1980); Jane Bennett, *Vibrant Matter: A Political Ecology of Things* (Duke University Press, 2010); Gregory J. Seigworth and Melissa Gregg, "An Inventory of Shimmers," in *The Affect Theory Reader*, ed. Melissa Gregg and Gregory J. Seigworth (Duke University Press, 2010).

field.¹⁷ In other words, media archaeology is historically attuned in looking not for a linear evolution of specific technologies, but instead for moments of suppressed, neglected, and forgotten discursive and material manifestations of cultural technologies.¹⁸ As such, media archeological accounts are driven by a question of power: how clashes of competing technologies and discourses are posed as solutions to a given problem, eventually emerging as a temporarily stabilized sociotechnical apparatus.¹⁹

Media archaeology is often characterized by a focus on either discursive accounts indebted to Michel Foucault or more techno hardware materialist attention inspired by Fredrich Kittler. Foucault's archaeology is attentive to what he calls discursive formations. This approach looks beyond the discourse itself to understand how discourse functions as a locus of power. Take for instance this chapter's opening, which features statements by powerful industry executives about streaming as the future of television and cinema. These statements aren't the be-all and end-all final word on streaming; simultaneously, these discourses are powerful in shaping the colloquial understanding of how streaming fits into media history. By contrast, a materialist approach considers how technology changes over time, with specific focus on the dead ends, the forgotten, and the understudied. Televisual history, for example, is shaped by a competition between transmission mechanisms—the spinning mechanical disk versus the cathode ray tube.²⁰ Mechanical televisions were ultimately less successful and failed to reach market status. This oft-forgotten material struggle over format, function, and design is critical in understanding the development of the televisual.

Attention to the material *and* the discursive vis-à-vis media archaeology can go hand-in-hand. Technologies are more than materiality; they are a method of addressing problems systematically. That is, technical media arise out of distributed discursive networks (bodies of knowledge and techniques), but then intermix and recombine over time.²¹ Viewed from a media archaeological perspective, the history of mobile streaming can be

¹⁷ Erkki Huhtamo and Jussi Parikka, "Introduction: An Archaeology of Media Archaeology," in *Media Archaeology: Approaches, Applications, and Implications*, ed. Erkki Huhtamo and Jussi Parikka (University of California Press, 2011), 15.

¹⁸ Huhtamo and Parikka, "Introduction," 39.

¹⁹ Alexander Monea and Jeremy Packer, "Media Genealogy and the Politics of Archaeology," *International Journal of Communication* 10 (2016): 3150.

²⁰ Doron Galili, *Seeing by Electricity: Cinema, Moving Image Transmission, and the Emergence of Television, 1878–1939* (Duke University Press, 2020), 23.

²¹ Alexander Monea and Jeremy Packer, "Media Genealogy and the Politics of Archaeology," *International Journal of Communication* 10 (2016): 3147.

understood as part of continuities and discontinuities with both the fixed and mobile imaginings and material attempts at a mobile and portable televisual and cinema.

As a moment of brief recap and respite, our evaluation of mobile streaming is influenced by accounts of media that go beyond attention to content to examine how culture and technology are intertwined. Doing so requires threading a needle while walking a narrow tightrope. On one hand, we don't want to write off the persistent connection between the televisual, cinematic, and streaming. On the other hand, we want to challenge and shake up simple linear narratives of technological evolution that all-too-often fall into the trap of cultural and/or technological determinism.

Fortunately, we aren't blazing this academic trail on our own. We follow in the footsteps of recent scholarship that traces the cultural lineage of streaming technology, noting the long line of antecedent technologies as well as their accompanying cultural meanings and practices.²² Other accounts go beyond an electrical technology lineage to examine how streaming has roots in infrastructural and environmentally indebted technologies that attempted to control flow.²³ Our historical and mobile focus is inspired by Erkki Huhtamo's call for more attention to the mobile, portable, wearable, and nomadic when it comes to media archaeology.²⁴ Greater attention to the historical examples of mobile antecedent technologies has gained traction with mobile media scholars.²⁵ And in particular, our own examination of the history and development of mobile streaming intertwines with mobile historical scholarship that emphasizes the emergence of customizable media environments, portable personalization, and continuous consumption habits.²⁶

In the remainder of this chapter, we begin our historical inflection point with the context of the electrical audio-visual technological experiments of the turn of the twentieth century industrial modernity. Specifically, we

²² Burroughs, "A Cultural Lineage," 13.

²³ Grandinetti and Ingraham, "Infrastructures of Flow," 2.

²⁴ Erkki Huhtamo, "Pockets of Plenty: An Archaeology of Mobile Media," in *The Mobile Audience: Media Art and Mobile Technologies*, ed. Martin Rieser (Rodopi, 2011), 23.

²⁵ Meryl Alper, "Portables, Luggables, and Transportables: Historicizing the Imagined Affordances of Mobile Computing," *Mobile Media & Communication* 7, no. 3 (2018); Hugh Davies, "Fanology: Hand-fans in the Prehistory of Mobile Devices," *Mobile Media & Communication* 7, no. 3 (2019); Jordan Frith and Didem Özkul, "Mobile Media Beyond Mobile Phones," *Mobile Media & Communication* 7, no. 3 (2019); Mary Morley Cohen, "'Always on and Always on Them': Portable Radios and the Creation of the Mobile Media Experience," *Mobile Media & Communication* 4, no. 3 (2016).

²⁶ Morley Cohen, "'Always on and Always on Them,'" 371.

trace the understudied history of portable television and the mobility-linked cinema of the drive-in.

The Birth of Electrified Entertainment

Recent decrees of “the death of cinema” can make it easy to forget just how important and impactful the movie distribution model has been to mass communication for well over one hundred years. Movie theaters, much like the imagined site of the televisual, evoke a fixed and stable site. That is to say that cinemas are defined as much by the film and later digital projection technologies as they are by the structure of ritual practice these sites provide. People come to experience a copresence of what’s known as “one-to-many” audio-visual consumption.

It’s impossible to predict the future, but a betting person would likely take the position that the pinnacle of cinema is over. Cinematic spending peaked during the 1940s. Adjusted for inflation, box office rates have fallen year after year in the twenty-first century, with major hits to the traditional cinematic distribution model coming during and after Covid-19 lockdowns.²⁷ The popularity of cinema is, of course, dependent on how this technology is linked with societal events, other technologies, and changing distribution models. Even at a time when audiences seem more content than ever to wait for movies to come to streaming video so that they can enjoy them from the comfort of their own home, it’s important not to downplay cinema’s importance.

Recent innovations like “direct to streaming,” shorter theater release cycles, the demise of purchasing DVDs (and before that, VHS tapes) have fundamentally disrupted existing distribution models, and these changes are relevant to our contemporary understanding of what movies entail. In the era of mobile streaming, cinematic content can be accessed on-the-go via a variety of locations and mobile devices just as easily as it can be accessed from the couch at home. Cinema has, for much of its history, been viewed as a fixed audio-visual practice; but an examination of cinematic history reveals that portable and even mobile connotations have long existed alongside this more site-locked interpretation of cinema.

One such pre-cinema technology that complicates the notion of film as fixed is the magic lantern. The magic lantern is a relatively simple projec-

²⁷ “Domestic Movie Theatrical Market Summary 1995 to 2026,” *The Numbers*, <https://www.the-numbers.com/market/>.

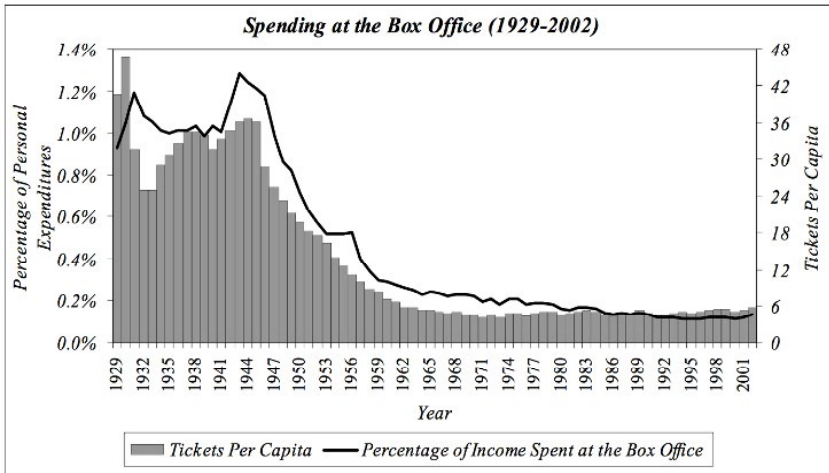


Figure 1.1. Spending at the box office (1929–2002). Orbach & Einay (2007, fig. 4).

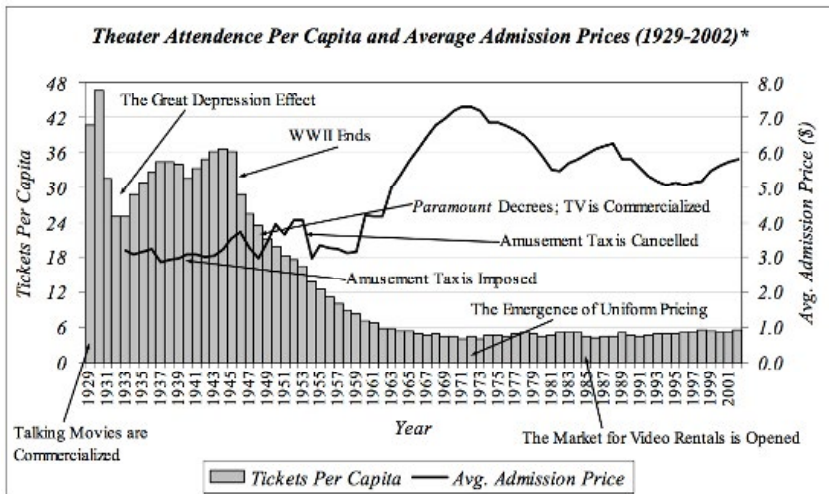


Figure 1.2. Theater attendance per capita and average admission prices (1929–2002). Orbach & Einay (2007, fig. 3).

tion technology, consisting of a box or chamber in which a light source is brought to focus to project light through an image painted onto glass.²⁸ The magic lantern was popular in eighteenth-century Europe, with the earliest known lantern show in America taking place in 1743.²⁹ Magic lantern shows

²⁸ Thomas Lamarre, "Magic Lantern, Dark Precursor of Animation," *Animation* 6, no. 2 (2011): 129.

²⁹ "A Brief History of the Magic Lantern," Magic Lantern Society, <https://magiclantern.org.uk/history/>.



Figure 1.3. The nineteenth-century magic lantern. Engraving courtesy of the Museum of London.

transformed a variety of spaces into sites of entertainment and education; by 1895, there were between 30,000 and 60,000 lantern showmen in the US, with an estimated 75,000 to 150,000 performances a year.

The magic lantern is just one of many burgeoning technologies of the nineteenth century that integrated a greater awareness of perception, namely the notion that stimuli can be divorced from their external referent, and that perception itself exists in time.³⁰ Stated differently, early visual technologies like the magic lantern and optical toys like the thaumatrope, stereoscope, and zoetrope marked the discovery that images can be linked together and the human mind will create a continuity of motion. Relevant to the history of cinema, these optical devices and techniques that were often initially devised for the purpose of scientific discovery were soon adapted into forms of popular entertainment. Furthermore, although devices like the magic lantern were part of a transformation of space into visual (with accompanying live audio) centers of entertainment and education, the technologies themselves are not fixed nor mobile but *portable*.

These precursors of visual entertainment set the stage for public fascination with the audio-visual. In the United States, a variety of inventors in the late 1800s patented the kinoscope (motion picture camera) and kintograph

³⁰ Jonathan Crary, *Techniques of the Observer: On Vision and Modernity in the Nineteenth Century* (MIT Press, 2001), 13.

(peephole viewing machine) in 1891.³¹ Early kinetoscope exhibition was in many ways neither completely mobile nor fixed, as viewers watched short clips through the peephole of a large device. While inventor Thomas Edison patented the kinetoscope, he did not initially ascribe substantial commercial value to the device.³² However, the kinetoscope and early short films gained popularity, and the first kinetoscope parlor opened on Broadway in New York City in April 1894, and the first European kinetoscope parlor followed in October of the same year.³³ The kinetoscope is often considered a direct precursor to cinema, but unlike the cinema structures that would soon take hold, these exhibitions did not always occupy a permanent space, as phonograph parlors, arcades, hotels, restaurants, etc. also attempted to capitalize on the novelty of the kinetoscope by hosting the technology in a transient and temporary sense.

Chances are most people living today have never seen a kinetoscope. That's largely because initial public fascination with the technology was quickly exhausted. That said, the kinetoscope and other cinematic precursors are notable to our account of streaming media for two key reasons. First, interest in the technology established to inventors that there was a market for projection technologies. Second, these optical technologies make clear that cinematic precursors were not necessarily stationary and site-fixed media forms. While the cinema itself soon became housed in more permanent brick-and-mortar structures, the notion of cinema and its lineage as fixed in place is complicated by these optical experiments. It takes time to build infrastructure—in this case, the cinema structure itself. Simultaneously, the move from more portable to fixed is more than just preference; what the shift shows is that the site-specific evolution of early cinema was part of strategies of legitimation gained by tying the electric theater to burgeoning urban modernity. This evolution is critical to the connection between streaming and cinema, in that it becomes possible to view some of the historical antecedents to the mobile and portable audio-visual entertainment achieved by contemporary mobile streaming.

Rivalries and competing technologies were part of the continued construction of the Western cinema space in the coming decades. As public interest

³¹ Charles Musser, *Before the Nickelodeon: Edwin S. Porter and the Edison Manufacturing Company* (University of California Press, 1991), 30.

³² Norman O. Keim and David Marc, *Our Movie Houses: A History of Film & Cinematic Innovation in Central New York* (Syracuse University Press, 2008).

³³ Musser, *Before the Nickelodeon*, 44; Charles Musser, *The Emergence of Cinema: The American Screen to 1907* (University of California Press, 1994).

in early projection technologies took hold, peephole devices were replaced by storefront movie theaters dubbed nickelodeons.³⁴ These small spaces were often converted from the same arcades that exhibited the kinetoscope. Early displays often had a transient and mobile nature. For example, traveling projection exhibitors would often find an empty story space, set up in parks, or use the side of a downtown building.³⁵

Even though there is no inherent need for permanent structures for cinema exhibition, the cinema space soon became a fixture of urban modernity. Profitability and public interest in cinema were part of the construction of movie theaters, and New York theater-building reached its zenith by 1914.³⁶ Even so, popularity and profitability alone are not the only forces that led to the creation of movie theaters. An early discursive positioning of movies as “photoplays” gave the medium more legitimacy, akin to a night out at the theater.³⁷ Along with discourses that legitimated cinema, the materiality of audio-visual capture is important; movie projection can be and was done in a more temporary and transient form, but grandiose cinema spaces gave film an air of acceptability akin to that of a play. The fixed structure of the cinema precipitously took hold in the United States beginning at the turn of the twentieth century. Known early as “the electric theater,” the cinema became emblematic of the evolving urban spaces of modernity and industrialization in the Western world. In essence, there was (and perhaps still is) a power to building temples to cinema that one can trace back hundreds of years to pre-electrical theaters.

As mentioned at the start of this section, movie theaters still matter. According to the National Association of Theatre Owners, there were 39,798 indoor and 595 drive-in movie screens in the United States in 2017.³⁸ Simply stated, cinema spending per capita may have peaked in the 1940s, but the cinema structure endures as not only a Western but increasingly global fixture of urban spaces. While the first few decades of the twentieth century were dominated by the cinema, interest in the broadcast television, followed later by cable television, and eventually the internet created audio-visual competition to cinema. Yet, even in a more diverse landscape of media

³⁴ Keim and Marc, *Our Movie Houses*, 23.

³⁵ David Welling, *Cinema Houston: From Nickelodeon to Megaplex* (University of Texas Press, 2011), 10.

³⁶ Keim and Marc, *Our Movie Houses*, 27.

³⁷ Keim and Marc, *Our Movie Houses*, 25.

³⁸ NAC Editorial Team, National Association of Theatre Owners Report, 2024.

content, the ritual function of “going to the movies” persists alongside other fixed and mobile forms of entertainment.

At the Drive-in

Combining both cinema and automobility, the drive-in theater represents an alternate imagining of cinema. Unlike streaming from a smartphone, the drive-in cinema is not portable; the drive-in is, however, a conjunction of cinema, space, and mobility. Scholars of cinema consider the relationship between the materiality of the cinema and urban space largely because electrical projection technologies quickly became a fixture of urban space, dominating leisure time as part of expected movement through the city.³⁹ Throughout the past century, the cinema has, in turn, been part of urban modernity and industrialization, post-industrialization, and contemporary global relations. Much like the fixed spaces of urban cinema structures, the drive-in organizes movement to and from the cinema site itself. But the drive-in marks a departure from more traditional cinema spaces through its imbrication of mobile technology and how the drive-in offers a reconstruction not of urban, but instead *rural* and *suburban* spaces. As such, the drive-in represents an alternative construction of audio-visual entertainment that emphasizes mobility and a new engagement with space—a cinematic lineage relevant to mobile steaming.

Although commonly associated with the 1950s and 60s, the first drive-in theater opened on June 6, 1933, in Camden, New Jersey.⁴⁰ The idea for the drive-in is credited to Richard Hollingshead, who began his initial experimentation in 1928 using a Kodak projector, a sheet stretched between two trees, and a radio for sound.⁴¹ Hollingshead took inspiration from his mother’s discomfort sitting in traditional movie theaters, and was granted a patent for the drive-in theater in 1933.⁴² Though not as popular as the

³⁹ Bill Marshall, “Montreal Between Strangeness, Home, And Flow”; Adrian Fielder, “Poaching on Public Space: Urban Autonomous Zones”; J. Paul Narkunas, “Streetwalking in the Cinema of the City: Capital Flows Through Saigon”; Ronaldo B. Tolentino, “Cityscape: The Capital Infrastructuring and Technologization”; and Geoffrey Nowell-Smith, “Cities: Real and Imagined,” in *Cinema and the City: Film and Urban Societies in a Global Context*, ed. Mark Shiel and Tony Fitzmaurice (Blackwell, 2001).

⁴⁰ “First Drive-in Movie Theater Opens,” History.com, November 13, 2009.

⁴¹ Ben Ikenson, *Patents: Ingenious Inventions* (Black Dog & Leventhal, 2004), 241.

⁴² “First Drive-in Movie Theater.”

fixed site of the urban cinema, the number of drive-in theaters peaked in the United States at 4,063 in 1958.⁴³

In recreating the cinema experience, the drive-in captured the growing American interest in automobility. In describing this elision between the cinema and automobility, Hollingshead called the drive-in “a novel construction to outdoor theaters whereby the transportation facilities to and from the theater are made to constitute an element of seating facilities of the theater.”⁴⁴ To this end, the drive-in represents a relationship to space that was often quite different from its urban theater counterparts.

Because of the need to accommodate the space of automobiles, drive-in theaters required multiple acres of land. Most of the pre-1960s drive-in theaters were built on plots bigger than twelve acres, the largest of which was located on twenty-eight acres in Copiague, New York, with space for 2,500 cars and a full-service restaurant.⁴⁵ Since the drive-in was an open-air venue, these theaters had to be located far outside urban spaces and away from major highways. Consequently, the drive-in theater was part of an alternate idea of the cinema, instead structuring movement to and from rural spaces. Although perhaps not directly “mobile” during the motion picture projection itself, the drive-in theater does explain how cinematic technologies, along with a connection to automobility, could recreate not only cities, but other spaces as well.

The height of the drive-in’s popularity coincided with the rise of broadcast television and high rates of US auto ownership. Here again, Williams’s concept of mobile privatization is critical: Families became more mobile, increasingly moved to the emerging suburbs, and became more isolated. These social shifts necessitated new forms of contact outside the home, and these same needs were served via the drive-in theater. Drive-ins, at least initially, offered family-friendly entertainment and interconnection. In this regard, the drive-in theater represented a non-urban imagining of the theater space that directly implicated the automobility required to reach the theater into the construction of the theater space itself. This connection between the theater and the automobile maintained cultural underpinnings of mobile privatization—the family isolated within privatized space via their car as part of the theater.

⁴³ Steve Robrahn, “Drive-in Theaters Slowly Fade Out of Business,” *LA Times*, July 10, 1988.

⁴⁴ Ikenson, *Patents*, 241.

⁴⁵ Robrahn, “Drive-in Theaters.”

This family-centric construction of the drive-in space rapidly changed following the 1960s. As an outdoor venue, the drive-in is beholden to natural cycles of weather, temperature, and the requirement of the night sky to view movies in the first place. These limitations in scheduling were not lost on film distributors, who realized that drive-ins could only show so many movies per day. Soon, movie studios sent first-run releases to indoor theaters only. Drive-ins were left to show B-movies, and eventually, X-rated films.⁴⁶ This revision of the drive-in concept did allow some to survive, while simultaneously inscribing the mobile cinematic construction with youthful and “deviant” associations.

The decline in drive-in theater popularity was also precipitated by other elements. First, the relentless expansion of suburbanization rendered drive-in theaters unsustainable. As cities expanded outward, formerly cheap land became far too valuable to devote to a seasonal business model.⁴⁷ In addition, as the novelty of the family drive and automobility wore off, so did interest in the drive-in. One final contributor to the decline in drive-in theaters came in the construction of shared time. Daylight savings time went into effect in the United States in 1918, was repealed after World War I, and was subsequently reinstated during World War II. From 1973 to 1974, the impact of a fuel crisis on the United States led to a temporary national adoption of a year-round daylight savings time as a fuel-saving strategy.⁴⁸ One consequence of this additional hour of daylight was that drive-in showtimes were pushed later into the evening, and this further separated the drive-in space from the family-centric focus of previous decades. This confluence of global, local, economic, political, and environmental forces is accordingly part of the history of the drive-in cinema.

The drive-in theater offers a unique articulation between the cinema space, non-urban geography, and automobility. While scholarship on the early years of the cinema often focuses on the theater as part of an urban reconstruction of space, the drive-in offered a form of audio-visual entertainment adapted for areas considered on the periphery of modernization, industrialization, and electrification. Cinematic precursors certainly demonstrated that visual (with accompanying audio) entertainment could be set up almost anywhere. But it was the novel interest in automobility that

⁴⁶ Robin T. Reid, “The History of the Drive-in Movie Theater,” *Smithsonian Magazine*, May 27, 2008.

⁴⁷ Reid, “History of the Drive-in”; Robrahn, “Drive-in Theaters.”

⁴⁸ Paul B. Murray, “Did Daylight Savings Time Make the Drive-in Theater Go Dark?,” *Southern Outdoor Cinema*, March 31, 2012.

allowed drive-in cinema to gain traction as a mobile reconstruction of cinema space. This same desire to watch “anywhere” is directly relevant to the potentiality of mobile streaming, even if the technologies, infrastructures, and the content itself have changed. At the same time, factors like daylight savings time, changing distribution models, and urban/suburban expansion show that it is impossible to divorce technologies from their sociotechnical milieu. The fact that drive-in theaters have dwindled over time while the “traditional” cinema has thrived also shows the power of a fixed urban site.

Despite its decline in popularity, the drive-in offers a critical example of desire for an alternate, mobile orientation to cinema. The fixed site of early movie theaters was part of a new structuring of time and spaces in modern cities. The decades of rapid social and technological change that accompanied industrial modernity forever altered the patterns and pace of everyday life. Such changes were not limited to work hours; technologies of entertainment like the cinema changed the temporality and spatiality of leisure as well. And as shown by the example of the drive-in cinema, experimentations imbricated with the mobile extend and transform these new sociotechnical patterns to spaces outside urban life.

The Spaces of the Televisual

Just as Quibi founder Jeffrey Katzenberg positioned smartphones as a “TV in your pocket,” a 2017 AT&T Wireless commercial shows potential customers the possibilities of mobile streaming through a smartphone with the dialogue:

With AT&T you can get your entertainment right here, right now. When you get the incredible iPhone 7 from AT&T you can get unlimited data. And live TV. The channels you love. Your favorite shows and movies. Making your iPhone more of a . . . (iPhone ring tone and call screen interrupts the commercial flow) . . . oh, my TV is ringing.

The quip “My TV is ringing” is a joke about the multiplicitous uses and media convergence of mobile devices.⁴⁹ Just as importantly, this advertisement, while humorous in nature, reveals how the televisual and smartphone technologies have become increasingly intertwined. Mentioned in the introduction of this chapter, Netflix made the grandiose proclamation that its service was

⁴⁹ “Ringling,” AT&T Wireless TV advertisement, July 2017, https://www.ispot.tv/ad/w1_N/at-and-t-wireless-ringling.

the world’s first global television network. Due to discursive positionings such as these, it should be unsurprising that streaming distributors have been construed as part of a linear evolution of the televisual.

Television history itself is commonly divided into periods of TVI, TVII, and TVIII. The chart below shows the common temporal boundaries and general characterization of each. If one was to subscribe to such a rigid history, then it would naturally follow that streaming is a form of “TVIV.” Media theorists have nonetheless challenged such a position, noting that Netflix and other streaming providers might be more accurately considered a “matrix media” in which viewing patterns, branding strategies, industry structures, and media content interact and intermix in ways that shift away from the traditional television set.⁵⁰ Here, the materiality of media and use practices complicate established definitions and disrupt linear histories of the televisual. Or, put differently, it is important to recognize that film and cable, satellite, and broadcast television still co-exist with fixed and mobile streaming. Mobile streaming does not replace nor render these alternatives vestigial. Mobile streaming is more popular than ever, and although mobile streaming practices are not always strictly “televisual,” the connection between the two media forms cannot be ignored. Specifically, thorough examinations of televisual history emphasize the medium’s early domestic and fixed history. However, there are material and discursive precedents of mobile streaming that require greater inquiry.

TV Periods	Years	Characterization
TVI	mid 1950s to early 1980s	Channel scarcity, mass audience, three-network hegemony
TVII	early 1980s–late 1990s	Channel/network expansion, quality television, network branding strategies
TVIII	late 1990s–present	Proliferating digital platforms, further audience fragmentation, shift in commodity relations

Figure 1.4. Televisual periodizations. Adapted from Jenner (2014).

Locating the moment of origin for television is difficult, as, much like with the cinema, the televisual arose out of a convergence of invention

⁵⁰ Mareike Jenner, “Is This TVIV? On Netflix, TVIII and Binge-Watching,” *New Media & Society* 18, no. 2 (2014): 260.

and innovation across multiple decades. One potential starting point is the 1897 creation of the cathode ray tube, a specialized vacuum tube capable of transferring refracted electron beams into images.⁵¹ This technology is still utilized in televisions, video cameras, and other image technologies. In 1927, the television was patented by Philo Farnsworth, who later sold the rights to RCA for one million dollars.⁵² Alongside these material milestones, the emergence of the televisual is as much a story of how the technology entered popular imagination. As noted by theorists, the televisual follows a trajectory of technologies like the camera obscura, telescope, telephone, and telegraph that sought to connect two distant points in real time.⁵³ Key to the televisual, then, are notions of simultaneity and storage—especially in relation to optics. In essence, the televisual and the cinema became linked to one another. Television offered “live” moving picture entertainment, whereas the cinema became associated with stored content.

Over the coming decades, the concept of the televisual changed more than the technology itself did, with various political and economic forces driving different ideas about what television should be, all while maintaining a focus on live television’s ability to engage simultaneous presence. The television model was eventually dominated by one-to-one transmission—a center to periphery model of communication—focusing on either simultaneity or storage.⁵⁴ In addition to these histories, the televisual and broadcasting are intertwined with specific social needs. As previously explained, Raymond Williams’s mobile privatization is a period-specific theorization of the deep contradictions between the televisual and automobility, wherein centralized, public transmissions was tied together with privatized reception. As families became more mobile, they also became more private, and television was part of a sociotechnical response to the need for contact outside the privatized space of the home.

This deep connection between the televisual and the domestic spaces remained both the material reality and dominant discursive positioning of television practices for decades, despite network proliferation, expanding times of televisual interaction, and infrastructural advances. For example, Lynn Spigel similarly examines the televisual in conjunction with the con-

⁵¹ Ikenson, *Patents*, 62.

⁵² Ikenson, *Patents*, 271.

⁵³ William Uricchio, “Television’s First Seventy-Five Years: The Interpretive Flexibility of a Medium in Transition,” in *The Oxford Handbook of Film and Media Studies*, ed. Robert Koller (Oxford Academic, 2008), published online September 18, 2012.

⁵⁴ Uricchio, “Television’s First Seventy-Five Years.”

struction of public and private, noting the way that televisual technology provided new connections while collapsing prior distinctions between public and private spaces. With television, the home is no longer divorced from the external world: recreation, consumerism, the outside filtered in.⁵⁵ At the same time, there were attempts to create a mobile and portable television as early as the 1950s, indicating that the goal of a ubiquitous audio-visual relation to mass communication media is also fundamentally intertwined with televisual lineage.

This same desire is apparent in the mobile and portable cinematic examples, making it evident that contemporary mobile streaming was not as a sudden shift that occurred with the creation of smartphones but rather a longstanding goal of audio-visual media within capitalist landscapes. The example of the portable television—specifically, the discursive and commercial positionings of this technology—highlights this understudied aspect of television as an antecedent to contemporary mobile streaming, affirming the connection between portability, youth, dynamism, futurism, and a reconstruction of the world into spaces of the televisual.

Mobile, Portable, and Televisual

Discursive imaginings and material experiments in mobile and portable television have existed for decades. The portable television, like many other technologies, has military roots. The electronic camera was the first video technology to become portable, with the camcorder and videorecorder following soon after.⁵⁶ Strapped to soldiers and TV cameraman, these portable technologies created a mobile orientation to television—the notion that TV could go anywhere.

In the 1950s, the US Army, Navy, and the National Defense Research Committee contracted RCA to improve upon wartime small and lightweight airborne television systems.⁵⁷ The idea of these portable devices was to create a television system that could pick up and transmit enemy movements from the front line to command centers. Of course, the production technologies

⁵⁵ Lynn Spigel, *Make Room for TV: Television and the Family Ideal in Postwar America* (University of Chicago Press, 2004), 11.

⁵⁶ Ricardo Cedeño Montaña, *Portable Moving Images: A Media History of Storage Formats* (De Gruyter, 2017), 80.

⁵⁷ Cedeño Montaña, *Portable Moving Images*, 137.

of television as mobile differs from the commercial technologies of home and portable television sets.

From a commercial perspective, the black and white picture portable era for television began in 1952, with the release of the first battery-operated transistorized television for sale in 1960.⁵⁸ While television manufacturers such as Sentinel produced what could be considered “portable” televisions due to their size (for example, the 1948 Sentinel 400TV 7”), the reliance on electric outlets limited the actual mobile uses of such models. Preceding Sony’s TV-8-301 model, television console sizes began to shrink. Following Sony’s success and public interest, myriad portable models were produced throughout the 1970s and 1980s.

Advertisements for various models of portable television provide insight into some of the popular discursive positionings and imaginings of the televisual untethered from the space of the living room. For example, Sony’s 1976 advertisement for a 13” portable model depicts a nuclear family watching television while lying together in bed (fig. 1.5). Although still a domestic image of the televisual, the portable nature of the television allows a reconstruction of home media engagement beyond the space (and normalized temporal schedule) of the living room.

The 1968 AWA advertisement depicted in figure 1.8 similarly highlights shifting family relations, imploring consumers to “be a two TV family” for “twice the fun,” as “today’s living demands” portability. Here, the living room televisual scene is juxtaposed with a bikini-clad young woman in a more exotic locale. The connection between youth and portability is again emphasized in General Electric’s advertisement, in which a son tells his father that their fixed television, and by extension living room space, is “all yours” (fig. 1.6).

Collectively, these advertisements position portability as associated with youth, dynamism, and the possibility of the televisual beyond the living room. Other advertisements, such as Sony’s 1960s promotions, engage more directly with the reconfiguration of space via the portable television. In this specific example, Sony tells customers that they can take the television with them to the beach, yacht, or cottage, or use the portability to watch television in the kitchen, bedroom, park, or garden (fig. 1.7). These selections are just some of many commercial discursive positionings of portable television; however, they display explicit engagement by corporate interests to sell the idea of a new dynamic and mobile televisual.

⁵⁸ “Television History: A Timeline 1878–2005,” Tarlton Law Library, University of Texas at Austin, <https://tarlton.law.utexas.edu/mason-and-associates/timeline>.

**NOW
A TUMMY TELEVISION
THE WHOLE FAMILY CAN
GET A BELLYFUL OF**



My how we've grown!
Time was when we made a Tummy Television
barely big enough for one person to watch.
Now we have a black-and-white portable that's all of 13"
(screen measured diagonally). And since it's 100% solid state,
has a glare-free screen for indoor/outdoor viewing
and an energy-saving system that shuts the power off completely
when the set is not in use, we expect the biggest problem you'll ever have
is agreeing on which program to watch.

**INTRODUCING OUR 13" PORTABLE
"IT'S A SONY!"**

©1976 Sony Corp. of America. SONY is a trademark of Sony Corp. Model TV-131 Black and White Portable. TV picture simulated.

Figure 1.5. Advertisement for Sony Model TV-131 Portable TV Tummy Television, 1977.

Much like the father-son ad example of figure 1.6, other advertisements connect portable television to futurism. Sony's dramatic call for individuals to "Hold the future in your hand" via portable television claims that the portable model is the "personal set predicted for the family of the seventies" (fig. 1.9). In addition to these bold claims, Sony notes some of the technical specifications and advances that make the model possible. Of course, the

brief text ends with a return to the forthcoming technological future, telling readers that someday all televisions will look like Sony's micro model, so why wait? One final example, yet again from Sony, characterizes their portable model as "personal" television (fig. 1.10). This discursive move is one alluded to in some of the previous examples—that portable television allows greater individuality, choice, and freedom. At the same time, the explicit engagement with personalization as synonymous with portability is a powerful imagining that has been revisited via contemporary mobile streaming practices.

It's all yours, Dad... I'll take the General Electric Portable

No wonder this is TV's most popular portable... it weighs only 26 pounds, performs like a console!

Just now a General Electric Portable can add to fun at your home—

Now you can enjoy your favorite programs while the kids see theirs. And you can sit and watch anywhere you choose to be. For this set—aptly named the "Companion"—goes where you go, from room to room, outdoors in summer, even away on trips and vacations. You see, it weighs only 26 pounds. Yet, wherever you plug it in, it gives a

clear, sharp picture. That's because it's equipped with an advanced picture tube, a dark safety window which helps to deepen contrast, and a built-in antenna. Just like high-priced models.

So start thinking about this new idea in television, and see your G-E dealer soon.

Ask about other sizes, too—the "Big-Screen" which weighs only 32 pounds, and G-E's handy new "Personal"—weighs less

than 12½ pounds, costs as little as \$89.95.

There are 3 handsome color combinations to choose from—take your pick. Be HURRY—the Christmas rush'll soon be on General Electric Company, Television Receiver Dept., Syracuse 6, N. Y.

Progress Is Our Most Important Product

GENERAL ELECTRIC

Figure 1.6. Advertisement for General Electric's portable televisions, 1956.



9" TV BBC 1, BBC 2 and ITV. 43" telescopic aerial. Mains or battery operated (battery optional extra), 85 Gns.

THE TV THAT'S REALLY PORTABLE!

In just twenty years the Sony Corporation has become the world's leading name in electronics. Sony pioneered miniature transistorised radios and televisions. And the result of many years of research is shown above. It's the TV that's really portable. Take it with you: on the beach, the yacht, to the cottage. Use it in the kitchen, bedroom, park or garden; wherever you are take a Sony.

Sony make a complete range of transistorised radios and tape recorders. Full details from your Sony dealer or write for a colour brochure to: SONY U.K. SALES DIVISION, Dept. T1, Mercia Road, Gloucester. Tel: Gloucester 28841. London Showroom, 70-71 Welbeck Street, London, W.1. Tel: HUNter 2143



SONY®
RESEARCH MAKES THE DIFFERENCE

AP1268-B. Printed in U.K.

Figure 1.7. Advertisement for Sony portable TV, 1960.

Though these advertisements provide only a small selection of how corporations positioned the portable television, the relation of portability to dynamism, freedom, personalization, and new spaces of televisual engagement are commonplace throughout the 1950s, 1960s, and 1970s. Taken as a totality, these advertisements shed light on the connection between the televisual and the reconstruction of space, mobility, and personalization.

Consequently, it becomes possible to view mobile streaming media not as a wholly novel technological interplay, but one rooted in historical continuities and disruptions.

Be a two TV family - IT'S TWICE THE FUN!

Solve TV programme arguments in the family and have twice the fun from your TV entertainment... be a two TV family with an AWA "Marconi" Portable! Only AWA gives you the superb quality of Deep Image picture, rich natural sound, and a choice of screen sizes... 11", 19", 21" and now the completely new 17" Teletar Deep Image PA!

The "Teletar" glimmers with space-age technology. A sound system that features the new integrated circuit as used in space vehicles, plus a sound deflection panel for bigger full-front sound. Fingertip-smooth channel changing. Superb 17" Deep Image picture. Twin telescopic aerials. Carrying handle. AC operated. In Woodgrain Blue, Shell White.

AWA 19" Stellarite Portable P2 - full front sound, up-front controls, twin telescopic aerials, private earpiece AC operated. Blue-Gal, Shell Blue, Maple or Teak Woodgrain.

AWA 21" "Master" P3 - full-front sound, top-front tuning, twin telescopic aerials. Remote grille in for easy carrying. Woodgrain top with simulated Rosewood Walnut Woodgrain.

AWA 11" Lightweight Portable "The Marconi" P1 - a "space-ship" portable with biggest performance and full-front sound. Australia's top-selling portable! Features twin telescopic aerials, private earpiece for personal listening. AC operated. In teak-look Blue, teak-look Red or Shell White.

Today's living demands AWA DEEP IMAGE PORTABLE TV

Manufactured and guaranteed by AMALGAMATED WIRELESS (AUSTRALASIA) LIMITED, Australia's national Wireless Organisation
 Price \$2

THE AUSTRALIAN WOMEN'S WEEKLY - November 27, 1968

Figure 1.8. "Be a two TV family." Advertisement for AWA televisions, 1968.

The decline in popularity of portable television sets occurred long before the advent of smartphones and mobile streaming. The television was able to achieve rapid household penetration in large part due to its limited infrastructural requirements; existing electrical power coupled with analog



hold the future in your hand
with **SONY**
RESEARCH MAKES THE DIFFERENCE

THIS IS TELEVISION OF THE FUTURE. This is the personal set predicted for the decade of the Seventies. So light and compact you carry it with you like a book, wherever you go. Put it beside your bed, on your desk at the office, outdoors for picnicking on the patio, in the back of the car or on the boat. It plays anywhere on its own rechargeable battery pack, auto battery or AC, with a picture so bright and sharp ordinary sets pale by comparison. Weighing only 8 lbs., it is hardly larger than a telephone, yet it out-

performs standard receivers in sensitivity and durability. Available only in limited quantities, SONY brings it to you today through its advanced research in the epitaxial transistor, so powerful and sensitive it is used only in computers and other advanced electronic equipment—and the new Micro-TV. It would be no exaggeration to say that someday all TV will look like SONY Micro-TV. But why wait for someday? See it today at selected dealers. SONY Micro-TV list \$229.95. Optional battery pack.

See and hear the world famous SONY all-transistor, battery operated radios at selected dealers.
SONY CORPORATION OF AMERICA 514 Broadway, New York 12, N.Y.
In Canada: General Distributors Ltd. 701 Notre Dame Ave., Winnipeg

Figure 1.9. "Hold the future in your hand." Advertisement for Sony televisions, 1962.

broadcast meant that most American households owned a television by the end of the 1950s.⁵⁹ However, early analog broadcast television, otherwise

⁵⁹ Woojin Kim, "Television and American Consumerism," *Journal of Public Economics* 208 (2022): 3.

known as TVI, is marked by channel scarcity and three-network hegemony. This limited channel structure continued until the early 1980s, when cable television infrastructure allowed growth in both the number of channels and picture quality itself. The infrastructural limitations for portable television further situated the portable version of the television as inferior, a sentiment still often echoed today.

The tepid response to the portable television reveals the complex nature of technological dissemination. For example, the videophone, another audio-visual experiment that can be considered part of both the televisual and telephonic lineages, also failed to gain traction. Despite its conceptualization as early as the 1920s, as well as pre- and post-war experiments, including by Bell Laboratories, the expensive nature of service and infrastructural requirements halted the videophone.⁶⁰ The videophone is not a mobile technology; nevertheless, like the portable television, the videophone shows how technological development is itself a complex composition of often competing forces. By contrast, the portable radio saw commercial success beginning in 1939, with manufacturing beginning decades earlier, the imaginings of which began as an idea in science fiction narratives in 1890.⁶¹ These examples show how the success (or lack thereof) of technologies is deeply connected to their historical context. Returning, then, to the instance of the portable television, infrastructural and material limitations meant that the “personalization” of space through mobile televisual engagement was not yet possible. By extension, television continued for decades as a predominately fixed technology.

Television, historically connected to the space of the living room, has in the decades since its inception expanded beyond the domestic sphere. Scholarship on the televisual has positioned the television as connected to liveness, storage, and simultaneity, while screen studies often investigate the place of the material screen as part of a structuring of space.⁶² As it was clear from the advertisements analyzed above, mobile experiments

⁶⁰ Matt Novak, “A Brief History of the Videophone That Almost Was,” *Tech News* (blog), Gizmodo, August 28, 2013.

⁶¹ Michael Brian Schiffer, *The Portable Radio in American Life* (University of Arizona Press, 1991), 5.

⁶² Packer and Oswald, “Flow and Mobile Media,”; Paul Frosh, “The Face of Television,” in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017); Ethan Tussey, “Connected Viewing on the Second Screen: The Limitations of the Living Room,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014); Anna McCarthy, *Ambient Television* (Duke University Press, 2001).



SONY Now Offers a Choice In Personal Television

Regardless of the model you select, you choose the finest when you buy SONY personal, truly portable television. Whether it's the remarkable 8-lb. Micro-TV with 5" screen or the new 12-lb. Model 9-304W with 9" screen, it operates on its own rechargeable battery pack, auto/best power and AC. Designed by SONY, developer of the transistorized pocketable radio and leader in the field of transistor electronics, the 27-transistor Model 9-304W introduces innovations which make it

unique. Included are a new rectangular picture tube with sensitivity and resolution superior to ordinary sets (a special accessory filter provides unsurpassed resolution even in bright daylight); new Mesa transistors for increased sensitivity; automatic gain control for better picture stability; 1/10th the power consumption of ordinary sets and practically no heat. A full series of accessories, including battery pack and UHF adapter, is available for both models.

SONY
RESEARCH MAKES THE DIFFERENCE

SONY CORPORATION OF AMERICA • 580 FIFTH AVE., N.Y. 26, N.Y. • REGIONAL OFFICES: LOS ANGELES • CHICAGO

Figure 1.10. "A choice in personal television." Advertisement for Sony televisions, 1964.

of the television were untethered from the infrastructural constraints of domestic life. Portable television highlights the mobile-fixed dialectic also present at various points in film history. While the living room at the time of television's rise may have been a space best understood through Williams's mobile privatization, the portable television reflected a desire for the transformation of space. This portability and mobility were discursively linked to notions of futurism—specifically the transformation of any space into one of personalized convenience. With regard to portable television, what matters is that there have long existed mobile experiments in the televisual as a deterritorialization of space and mobile capture of desire.

As previously mentioned, Williams's mobile privatization provides a conceptualization of how the televisual encouraged a way of life that was both mobile and home-centered; however, his notion is specific to a TVI paradigm. In engaging with the work of Williams, Lynn Spigel offers "privatized mobility" to understand how miniature and portable televisions could be used to keep the family and domesticity together even while people were on the go.⁶³ Spigel's conceptualization is a crucial model for understanding the history of the televisual; whereas Williams studied the emergence and lived experience of the televisual, Spigel instead highlights the possibilities and fantasies of the televisual as unplugged from the home. Established by the brief selection of portable television advertisements, discursive positionings of the portable television do emphasize freedom, mobility, dynamism, etc., yet simultaneously, retain a focus on the family unit and a mobile form of domesticity.

Although this portable and mobile relation to the televisual was not fully achieved through early attempts at the portable television, contemporary mobile streaming forms reengage the desire for mobile personalization and recreation of spaces beyond the domestic context of traditional television. The roots of a personalized and dynamic television experience trace back to the early years of the televisual, made apparent by the portable television. While promoted as empowering for viewers, the portable television served an economic purpose by capturing new times and spaces that might otherwise have been television free. Though these devices were limited in popularity, these examples yet again demonstrate the lineage of mobile personalization.

⁶³ Lynn Spigel, *Welcome to the Dreamhouse: Popular Media and Postwar Suburbs* (Duke University Press, 2001); Spigel, *Make Room for TV*, 75.

The Present and Future of Mobile Personalization

The goal of this chapter was not to argue that the drive-in cinema and portable television are direct antecedents falling along a neat evolution from early optical technologies all the way to mobile streaming. What we do wish to highlight is an alternative and oft-forgotten mobile and spatial history of the audio-visual. The drive-in and portable television are, in fact, different both materially and discursively from contemporary web-based mobile streaming content. Yet, despite these differences, we believe that these moderately successful historical examples do in fact inform our current moment of mobile streaming.

Although cinema began structuring urban spaces at the turn of the twentieth century, there are alternate origins to mobile streaming that did not receive as much attention. The example of drive-in cinema explains an alternative imagining that engages directly with mobility. In turn, this example disrupts the myopic linear narrative of cinema as fixed by providing historical ruptures and disjunctures that offer the possibility of a different relation to audio-visual practices.

Perhaps most importantly, this mobile-fixed dialectic highlights how the mobile and portable has often been subordinated over time. Conceivably in large part due to the rhetoric of the electrical sublime, as well as modernization, industrialization, and urbanization, the fixed space of the cinema gained legitimacy and primacy that was not always afforded to mobile forms. Cinema is also part of the rise of the mass audience, while mobile alternatives often catered to those not part of an urban modernity and the accompanying homogenization of twentieth-century mass media. The space of the cinema cannot be divorced from its historical period, and it should come as no surprise that the electrical novelty of cinema became enshrined in urban spaces through brick-and-mortar structures. In this way, the cinema created a spatial relation that structured movement to, from, and around urban space.

Cinema as fixed structure offered legitimacy via the immobility of the media form itself. The cinema space has long served as an arbiter of cultural forms; the movie theater is the first place of presentation for serious film productions, whereas lesser movie productions are relegated to “straight-to-home” formats. This same hierarchy often continues today, through the standard release order of theaters followed by airlines and hotels, home video, video-on-demand and pay per view, premium TV, and finally broadcast TV. Of

course, this model has been disrupted heavily by streaming providers, which in turn challenges not only strategies of monetization, but also legitimation.

The drive-in reveals that mobile capture of a desire for entertainment has been part of a cinema lineage that predates the movie theater itself. Material limitations notwithstanding, the fixed space of the cinema has been the gatekeeper of film since the turn of the twentieth century—a part of modernist regimes of spatial construction and control. It is the articulation of film to the internet that resurrects some of the same mobile desires of cinema lineage. Understanding mobile engagement with film via streaming media requires attention to this spatial, discursive, and material dialectic between fixed and mobile.

This dialectic is also apparent via the example of the portable television. For much of televisual history, domestic spaces defined the practice of watching television. Scholars such as Anna McCarthy have provided thorough examinations of the proliferation of screens as part of the construction of space and power relations, yet this fixed and site-specific nature of audio-visual media is just one vision that often coexisted with mobile alternatives.⁶⁴ From the televisual perspective, the portable television offered a vision of broadcasting free from the electrical infrastructure and fixed space of domesticity. Analyzed through this historical perspective, it becomes possible to view mobile streaming not as an entirely new media form, but instead as part of both continuities and discontinuities of discursive and material mobile technological forms that have existed for decades.

The examples of the drive-in and portable television reveal imaginings of cinema and the televisual that are quite different from the contemporary complexity of mobile streaming. Simultaneously, what unites these media forms is the function of recreating space as a capture of desire—what we call mobile personalization. As established in this chapter, such a capture of desire has precedents in terms of the cinematic and televisual. Perhaps the greatest departure of mobile streaming comes via the form of digitization. It is no secret that streaming providers rely on data collection, storage, and processing—elements that are not inherent in quite the same fashion to cinematic and televisual technologies. As such, mobile streaming is part of a lineage of film and television, but it also reanimates the trajectory of early media like census, clocks, and calendars, wherein the purpose of storage, data, and processing has once again become emphasized through the digital.⁶⁵ It

⁶⁴ McCarthy, *Ambient Television*.

⁶⁵ Jeremy Packer, “Epistemology Not Ideology OR Why We Need New Germans,” *Communication and Critical/Cultural Studies* 10, no. 2–3 (2013); John Durham Peters, *The Marvel-*

is precisely this digitalization and data collection that allows mobile streaming to become a hyper-efficient form of the capture of desire—and space and geographic data are part of this collection and subsequent algorithmic recommendation.

This chapter traced the origins of mobile personalization, with roots in the early optical technologies, experiments with the electrical and cinematic, the intertwining of emergent automobility and the cinema, as well as the promised dynamism and youth of the portable television. Our mobile personalization approach considers these sociotechnical changes not merely as linear evolution, but as moments that help trace the lineage of a mobile and portable desire to transform time and space through technologies.

In the next chapter, we examine the architectures of mobility that facilitate contemporary mobile streaming. Here, we go beyond the smartphone to also consider the infrastructures and platforms that enable mobile streaming media. This includes the role of platform models, wireless infrastructures, and how platforms and infrastructures of mobile streaming have come to merge. Moreover, we consider some of the key technical underpinnings of mobile streaming, like big data, algorithmic models, and APIs. Smartphones are certainly the point of access when it comes to streaming content, but infrastructures of data flows also play a pivotal role in creating, sending, and receiving streaming media.

CHAPTER TWO

Architectures of Mobility

Introduction

You're standing in line at the airport, halfway through an episode of your favorite Netflix series on your phone. You didn't plan ahead, just opened the app and hit play. Around you, someone's watching TikTok clips on repeat, another is livestreaming a concert on Instagram, and a kid nearby is deep into a YouTube rabbit hole.

No wires, no downloads, no friction.

It feels effortless.

But behind that casual scroll or stream is an enormous and largely invisible machine. Your video didn't come from Netflix directly; it may have been served from a local cache server tucked into a telecom provider's infrastructure. Your phone switched between 5G and Wi-Fi, managed by complex traffic prioritization protocols. The app collected data about what you watched, how long you watched, and whether you paused or skipped. It wasn't just entertainment—it was a data transaction. This everyday scene illustrates our core argument: Mobile streaming's ease and personalization rest on complex, hidden infrastructures and platform systems working in concert. And this infrastructure didn't emerge overnight. It is instead the result of a long trajectory of sociotechnical changes.

This moment in the airport line isn't just a modern convenience. It's the front end of a vast technical and economic structure designed to deliver

media in motion—what we call *architectures of mobility*, the technical systems and economic platforms that make such personalized, mobile media flows possible. In this chapter, we move from historical antecedents of mobile streaming to the complex mix of infrastructural apparatuses that developed to make on-the-go interactions like the above situation work. This chapter unpacks what makes seamless mobile streaming possible, and what it costs.

Mobile streaming media has become a pervasive mode of cultural interaction enabled by the entanglement of platforms and infrastructures. We often think of streaming in terms of our personal devices, such as the ubiquitous smartphone or tablet, but the true architecture of mobile media extends far beyond handheld screens; it encompasses global wireless networks, cloud computing systems, data centers, content delivery networks, and platform ecosystems that deliver video, music, games, and other media to users while they are on the move. Manuel Castells, a key theorist of networked mobility, posits that communication networks create a space of flows wherein information travels instantly across vast distances, reconfiguring how we experience space and time.¹ The “architectures of mobility” are the technical systems and economic platforms that make such flows possible. This chapter examines the infrastructures and platforms enabling mobile streaming media in primarily Western contexts, analyzing how technical underpinnings (wireless networks, cloud infrastructure, big data pipelines, algorithms, and APIs) merge with platform business models to structure the contemporary media landscape. Rather than a single “streaming industry,” these services represent an infrastructural logic spanning multiple media and tech sectors.

As streaming has moved off the desktop and into pocket, the once-separate domains of infrastructure and platform have increasingly converged. Digital platforms like YouTube, Netflix, Spotify, or TikTok present themselves as content delivery services or social hubs for users, but they also operate as digital infrastructures in their own right.² Nick Srnicek defines platforms at the highest level as “digital infrastructures that enable two or more groups to interact.”³ In other words, a platform such as a streaming service not only hosts media content and connects users with that content, but also provides the foundational technical base—storage, bandwidth, interface, and data processing upon which those interactions run. At the same time,

¹ Manuel Castells, “Communication, Power and Counter-Power in the Network Society,” *International Journal of Communication* 1 (2007).

² Jean-Christophe Plantin et al., “Infrastructure Studies Meet Platform Studies in the Age of Google and Facebook,” *New Media & Society* 20, no. 1 (2018).

³ Nick Srnicek, *Platform Capitalism* (Polity Press, 2016).

traditional communications infrastructures have been reshaped by the logic of platforms. Jean-Christophe Plantin, Carl Lagoze, Paul Edwards, and Christian Sandvig suggest that “platform-based services acquire characteristics of infrastructure, while both new and existing infrastructures are built or reorganized on the logic of platforms.”⁴ In effect, companies like Google, Amazon, and Facebook increasingly build out physical infrastructure (such as fiber optic cables, server farms, and distributed content caches) to support their platforms, while infrastructure providers (like telecom carriers) adopt platform-like models of operation. The result is a layered system in which platform and infrastructure are deeply interwoven.

Crucially, mobile streaming architectures are not just technical systems; they are also economic and organizational systems. In effect, the platform has emerged as a dominant business model for the digital economy, one oriented around the extraction and use of data.⁵ Streaming media platforms exemplify this model; they function as intermediaries between content producers and audiences, positioning themselves in the middle of every interaction so as to capture data and monetize user’s activity.⁶ Whether supported by advertising (YouTube, TikTok), subscription fees (Netflix, Spotify), or hybrid models (Twitch’s subscriptions and ads), these services all leverage algorithmic tools to maximize engagement and profit. The very freedom and mobile personalization that mobile media promises to users, such as the ability to watch or listen to anything, anywhere, anytime, is underpinned by an elaborate apparatus of pervasive data collection operating behind the scenes. Of course, such participation in networked digital media often comes with a tradeoff: Users gain convenient access to information and communication at the expense of the ownership of their data, which is claimed by the operator of the enclosure.⁷ In other words, platforms enable the widespread collection of user data that becomes property of the owner of the platform in use. This captured information becomes a key resource for platforms, fueling the algorithms that recommend content and the targeted advertising or personalized experiences that drive revenue.

To uncover the material and digital infrastructure that enable mobile streaming, this chapter will unfold over four sections. First, we examine the platform business models that characterize streaming services, drawing on

⁴ Plantin et al., “Infrastructure Studies Meet Platform Studies,” 294.

⁵ Srnicek, *Platform Capitalism*.

⁶ Srnicek, *Platform Capitalism*.

⁷ Mark Andrejevic, “Surveillance and Alienation in the Online Economy,” *Surveillance & Society* 8, no. 3 (2010).

theorists of platforms as infrastructures to understand how mobile media platforms create value by aggregating content and users. We then delve into the wireless infrastructures that enable mobility—the evolution of cellular networks (3G, 4G, 5G) and Wi-Fi—and how these networks interface with streaming platforms to deliver high-bandwidth content on the go. Following that, we explore the technical underpinnings of contemporary streaming, such as the big data pipelines, algorithmic recommendation systems, and application programming interfaces (APIs) that allow platforms to mediate and customize the flow of media. Finally, we consider the convergence of platforms and infrastructures, how companies merge roles to control both content and its distribution, and discuss what these architectures of mobility mean for mobile media practices and users.

Platforms, Business Models, and the Data Economy of Streaming

Underneath the user-friendly interfaces of Netflix, YouTube, Spotify, TikTok, and other streaming services lies a distinctive business logic best described as “complex governance arrangements.” These arrangements include not only the platform’s API, which enables the API to parse through incredible amounts of data, but also the material conditions that imbue the platform with such power.⁸ Relatedly, the platform has emerged as a logic that became prominent in the 2000s, characterized by its ability to extract, control, and organize immense amounts of data. Platform-based companies seek to gather diverse users, such as individual consumers, advertisers, content producers, developers, etc., onto their digital infrastructures and facilitate interactions among them, thereby capturing value from each transaction or engagement. Data is the fuel that drives platform business logics, and it has proven to be wildly successful, as it acts almost as a renewable resource that a platform’s users top off with every click.

In the realm of mobile streaming media, this platform model manifests in several ways. Streaming services operate multi-sided markets: For example, YouTube connects content creators, viewers, and advertisers, whereas Spotify connects listeners, music rights holders, and advertisers/sponsors. By serving as the enclosure for these interactions, platforms can collect tolls (subscription fees or ad revenue) and, just as importantly, collect data.

⁸ Anne Helmond, “The Platformization of the Web: Making Web Data Platform Ready,” *Social Media + Society* 1, no. 2 (2015).

A defining trait of platforms is that they transform activities into data via what's known as datafication, rendering of interactions into quantifiable data points.⁹ Every song streamed, every video searched for, every second watched, every "like" or comment is logged. Data is then leveraged in service of the business model, whether through targeted advertising, personalized content recommendations to drive more usage, or even decisions about content production.

The major streaming platforms illustrate the nuances of the platform business model while all following its core logic. Netflix, for instance, runs on a subscription model where users pay a monthly fee for unlimited access to a catalog of films and TV shows. On the surface, this might seem like a straightforward digital update of the cable TV or video rental model. But Netflix's operation as a platform is far more complex. It heavily employs big data analytics to inform both its recommendations and its strategic decisions.¹⁰ By analyzing the viewing habits of its more than two hundred million subscribers, including granular data like when viewers pause, rewind, or abandon content, Netflix gains the ability and knowledge to tailor the user experience to individual viewers, driving engagement in a way that only a platform can enable. This data-driven personalization is a hallmark of platform logic. In effect, the service improves as users generate more and more data, which is then compared against, in the most basic form, likes and dislikes to generate similarities between users, leading to network effects where a larger user base yields better recommendations, attracting even more users, and the process repeats. In essence, Netflix's competitive edge comes not just from having content, but from having an architecture that mediates between content and viewers in a data-centric way.

YouTube, owned by Google, operates on a predominantly advertising-supported model and epitomizes the multi-sided platform dynamic. It attracts content creators who upload videos in hopes of reaching an audience and possibly earning a share of ad revenue; it attracts viewers who consume free content; and it attracts advertisers who pay to show ads to those viewers. YouTube's role is to intermediate all these parties on its technical platform by hosting videos, streaming them efficiently to users, and integrating targeted advertising.

⁹ José van Dijck et al., *The Platform Society: Public Values in a Connective World* (Oxford University Press, 2018).

¹⁰ Danial Amin, "How Netflix Used Data to Create House of Cards: A Revolutionary Approach to Content Creation," *Medium*, March 6, 2023.

Spotify provides another variant of the streaming platform model in the music domain. Spotify offers both a free, ad-supported tier and a premium subscription tier, effectively balancing two revenue models. As a platform, it interfaces between music listeners, music rights holders (record labels, independent artists), advertisers, and an ecosystem of third-party developers who can integrate with Spotify's API for things like playlist sharing or music analytics apps. Spotify's value proposition relies heavily on personalization and convenient, on-demand access to a virtually unlimited library of songs. It pioneered popular algorithmic playlists such as "Discover Weekly" and "Daily Mix," which analyze a user's listening history and broader trends to serve up custom-curated music playlists. The success of "Discover Weekly," which reportedly significantly increased user listening time,¹¹ underscores how data-driven selection is central to the platform's appeal. This mechanism is classified as "selection," whereby the platform drives the user by providing carefully curated content.¹² In other words, rather than the user actively choosing every song, the platform pre-selects and guides the user's choices, based on what the data suggests the user will enjoy. Ostensibly, this enhances user experience but also serves Spotify's business by keeping users engaged longer, thus exposing them to more ads or extending the life of their subscription. Like other platforms, Spotify also treats its aggregated user data as an asset—providing their insights to record labels and artists about listening patterns and even using data to negotiate power within the music industry.¹³

The platform model also extends to short-form video streaming such as TikTok. TikTok is quintessentially mobile-first and uses an algorithmically-curated feed known as the For You Page (FYP) as its core feature. TikTok's business model mixes advertising, sponsorships, and e-commerce integrations, but the user experience is entirely driven by continuous scrolling through an endless stream of personalized videos and advertisements. What sets TikTok apart is the remarkable degree of real-time data-driven adaptation. The platform's algorithm rapidly learns from each swipe, pause, or re-watch, tailoring the next pieces of content to maximize user retention. This creates a feedback loop where the platform's AI effectively curates a unique channel for each user. From a business perspective, this sticky engagement trans-

¹¹ Nils Aoun et al., "Discover Weekly: How the Music Platform Spotify Collects and Uses Your Data," *Montreal AI Ethics Institute*, May 26, 2022.

¹² van Dijck et al., *The Platform Society*, chap. 1.

¹³ Katie Shonk, "Streaming Toward Win-Win Negotiation: Spotify Upgrades Its Dispute Resolution with Taylor Swift," *Harvard Law School*, April 22, 2025.

lates into massive ad inventory and the potential for viral trends that can be monetized: for example, hashtag trends and challenges that advertisers co-opt into their marketing. TikTok embodies an automation of watching, in which users are “leaning back” and letting the algorithm feed them, while the platform meticulously monitors every interaction in order to refine the feed further.¹⁴ The most dystopian critiques of this kind of data-driven personalization highlight how these platforms create a situation where “we’re little lab rats in th[e] online forum,” continually experimented on by algorithms that test which content keeps us hooked.¹⁵ The platform’s goal is deeply aligned with the infrastructure of data collection: More data yields better personalization, which yields more engagement, which yields more data, in a continuous cycle.

One cannot overlook Twitch, the Amazon-owned, ostensibly video gaming-centric live streaming platform. Twitch primarily monetizes through a combination of advertising and viewer subscriptions/donations to individual streamers, of which Twitch takes a cut. As a live streaming platform, its emphasis is on real-time community and interaction, such as live chat, emotes, donations, etc. This fosters a different dynamic from Netflix or TikTok that attempts to optimize content delivery to individuals. Twitch, on the other hand, has built an architecture that supports vibrant communal experiences at scale. In many cases this has paid off, as popular streamers might have tens of thousands of viewers chatting simultaneously, yet it is important to note that the vast majority of Twitch streamers (93.3%) have five viewers or less.¹⁶ This illuminates the allure of both the community-driven experience and of streaming’s potential (albeit very unlikely) to become a full-time job.

The technical challenges here become part of the platform’s competitive advantage: Ultra-low latency video delivery and synchronized chat are infrastructure feats that Twitch continuously works to optimize. From a business standpoint, Twitch’s data collection is not about recommending on-demand content, since live content is ephemeral, but it still gathers extensive metrics on viewership patterns, chat sentiment, etc., which can inform which streamers get promoted or which events are spotlighted. Indeed, there is a recommendation system similar to that of other stream-

¹⁴ Lisa Parks et al., eds., *Media Backends: Digital Infrastructures and Sociotechnical Relations* (University of Illinois Press, 2023).

¹⁵ Sarah Childress, “Mark Andrejevic: We Are All ‘Lab Rats’ Online,” *Frontline*, PBS, February 18, 2014.

¹⁶ Bill Cooney, “New Twitch Stats Reveal How Few Viewers Are Needed to Be a ‘Top’ Streamer,” *Dexerto*, March 6, 2021.

ing platforms; however, it is dependent upon channels that are, at that very moment, live. Moreover, Twitch demonstrates the diversity of content domains that streaming platforms can capitalize on. Streaming is not just pre-recorded media like songs, shows, and movies, but human activities, such as video gaming and performances in real-time.

While each platform has its own flavor, they share common strategies typical of platform capitalism: user acquisition through network effects, monetization through data-driven decision making, data-driven advertising or subscription at scale, and an ecosystem of control via APIs and platform governance. Three relevant types of platforms typify mobile streaming: advertising platforms (e.g., Google/YouTube, Facebook), cloud platforms (e.g., Amazon Web Services, which underlies many streaming services), and product platforms (e.g., eBay, Amazon).¹⁷ Many streaming services combine and employ various aspects of these or converge them into one ecosystem.

YouTube is an advertising platform; Netflix could be seen as a kind of content platform with a subscription base; Spotify is a hybrid; and Amazon is a special case that spans multiple platforms in that it runs both Twitch and AWS infrastructures, and Prime Video as a streaming service integrated into its e-commerce ecosystem. Across all these platforms, user data collected via mobile—location, app usage, listening/viewing habits, and interaction patterns are central. The mobile interface becomes a key site of data extraction, encouraging platforms to expand into more corners of user life. YouTube Shorts, Spotify’s podcast autoplay, and Netflix’s mobile games are all strategic moves to keep users within the app and to collect richer behavioral data. These examples show that “streaming” isn’t a single industry silo, but a cross-industry model that integrates entertainment, tech, and telecommunications into one ecosystem of on-demand content.

This logic often undermines user privacy and public interest. Mobile design features like infinite scroll, autoplay, and notification nudges are optimized to maximize engagement, but raise ethical concerns. A few dominant mobile platforms now control the bulk of mobile cultural consumption and communication. Their algorithms not only shape what is recommended, but also which creators succeed, which genres thrive, and which voices are heard. The result is a winner-take-all mobile ecosystem, where YouTube dominates mobile video, Spotify leads in music streaming, and Netflix holds firm in subscription-based entertainment. This consolidation hands enormous power to a few companies to shape cultural consumption and user data

¹⁷ Srnicek, *Platform Capitalism*, 49.

flows. Even well-resourced newcomers like Disney+ struggle to disrupt these entrenched mobile behaviors. In effect, the mobile streaming environment is consolidated, data-driven, and platform-governed—an infrastructure as central to contemporary media as broadcast towers and cable grids. In the next section, we turn to those underlying infrastructures—the wireless networks and mobile systems—that materially enable these platform-driven, personalized media experiences.

Wireless Networks and Mobile Infrastructure

We often don't think about mobile streaming via videocalls, but this relatively recent development has major consequences for platforms and wireless service providers alike. For example, Apple's FaceTime feature was initially restricted to Wi-Fi connections, as AT&T's 3G network couldn't support the bandwidth demands of video calls. This limitation underscored the inadequacies of existing wireless infrastructure in handling emerging streaming technologies. The challenges were further highlighted during Apple's 2010 Worldwide Developers Conference (WWDC), where Steve Jobs attempted to demonstrate FaceTime. The presentation faced significant Wi-Fi interference, causing delays and emphasizing the need for more robust wireless networks to support seamless streaming experiences. These events exemplify how the evolution of wireless infrastructure has been crucial in enabling the mobile streaming capabilities we often take for granted today.

If digital platforms provide the portals and economic engines for mobile streaming, wireless communications infrastructure provides the roads and highways those portals depend on. The notion of architectures of mobility directs our attention to the physical and code-as-protocol layers that enable users to access streaming media while on the move—principally, cellular and Wi-Fi networks that often undergird mobile connectivity. The evolution of cellular technology from 3G to 4G and now 5G has been a fundamental enabler of the mobile streaming boom. The value of mobile networks lies in providing ubiquitous connectivity, an “always-on” data stream wherever a user goes. This is the essence of mobile personalization—creating an environment in which mobile streaming media practices are not tied to a fixed place. Mark Andrejevic evocatively described the result as “a ubiquitous electromagnetic atmosphere—not merely extension across space, but a ‘filling-in’ of space, so that wherever one goes, the network can be accessed.”¹⁸ In practical terms,

¹⁸ Mark Andrejevic, “Media and Mobility,” in *The International Encyclopedia of Media Stud-*

this ubiquitous connectivity is achieved by the dense infrastructure of cell towers, underwater cables, data centers, exchange points, routers, and our own devices that blanket populated areas with wireless internet access.

The progression from 3G to 4G to 5G illustrates how each generation of wireless infrastructure opened new possibilities for streaming media. 3G networks (starting early 2000s) were the first to support reasonable data throughput on phones (~hundreds of kilobits per second up to a few megabits per second later in the life cycle). With 3G, one could do basic web browsing and perhaps stream highly compressed audio or low-resolution video, such as YouTube's mobile site in its early days. Yet 3G was often barely sufficient for smooth video and was limited by high latency. The rollout of 4G LTE (Long Term Evolution) around 2010 was transformative. Mobile bandwidth became comparable to home broadband, reaching tens of megabytes per second, with lower latency. This allowed users to stream 720p or, depending on their area of coverage, 1080p HD video on a phone or tablet via cellular, which had been impractical before. Telecommunications companies in Western markets experienced surging data traffic on their 4G networks largely due to video streaming, and by the second half of the 2010s, video dominated mobile data. One industry report noted that in North America, Netflix and YouTube alone accounted for about half of downstream mobile internet traffic by 2019.¹⁹

The network infrastructure had to keep pace. Carriers invested in upgrading backhaul (the connections from cell towers to the internet backbone), deploying more cell sites to increase capacity, and implementing technologies like LTE-Advanced with carrier aggregation to boost speeds. They also had to adjust data pricing models; many moved from unlimited plans to tiered data plans in the early 2010s to manage the explosion in use. Competitive pressures later brought back more "unlimited" offerings, though this often came with throttling caveats for heavy use or lower quality video streaming by default.

5G, the fifth-generation standard, began deployment in the late 2010s and is still being rolled out across the US. While 5G's real-world impact is still unfolding, it promises significantly higher speeds, in the gigabit per second range under ideal conditions, and, importantly for new media applications, much lower latency of 10 milliseconds or less in optimized scenarios. The

ies, ed. Angharad N. Valdivia, vol. 1, *Media History and the Foundations of Media Studies*, ed. John Nerone (Blackwell, 2013), 521.

¹⁹ Troy Dreier, "Video Will Be 78% of Mobile Data Traffic by 2021, Says Cisco," *Streaming Media*, February 7, 2017.

arrival of 5G has been heralded as enabling next-generation streaming experiences: ultra-high-definition 4K or 8K video on mobile, flawless VR/AR streaming, and cloud gaming with latency nearly as low as a local machine.²⁰ Early reports indicate that 5G networks “ensure flawless synchronization,” making it ideal for live sports, remote interviews, and virtual production.²¹ In essence, 5G is designed to make the mobile network feel “invisible”—so fast and responsive that users cannot distinguish between wired and wireless connectivity.

While 5G coverage and device uptake will take several years to mature, many streaming platforms have prepared by supporting higher quality streams and new features for 5G users. For example, Netflix and YouTube support HDR and 4K on mobile where the network allows, and cloud gaming services offered by Xbox, Steam, and PlayStation are poised to benefit from 5G’s low latency. Telecommunications providers often market 5G by highlighting streaming capabilities. Verizon and AT&T adverts show users seamlessly live-streaming or watching HD content on the go.

Control over these networks confers power, which is evident in debates over net neutrality and incidents like the 2014 Netflix–ISP peering dispute, where infrastructure owners could leverage quality of service to negotiate with content providers. Particularly in the United States and Europe, the principle of “net neutrality”—that internet service providers (ISPs) should treat all data equally without undue discrimination—has been hotly debated,²² ISPs have sometimes chafed at services like Netflix or YouTube consuming massive bandwidth on their networks without direct compensation. Conversely, platform companies argue that users have already paid the ISP for internet access and should get the content they request unimpeded.²³ This tension led to incidents like the 2014 peering dispute where Netflix traffic to Verizon and Comcast customers slowed down until Netflix agreed to pay for direct connections.²⁴

The debate and influence of those who regulate streaming environments can shape mobile streaming architectures. For example, zero-rating

²⁰ “Everything You Need to Know About 5G,” *What Is 5G?*, Qualcomm, accessed May 12, 2025.

²¹ “How 5G Is Transforming Live Broadcasting and Remote Production,” *Medialooks*, September 2024.

²² Rahul Awati and Kate Gerwig, “Net Neutrality,” *TechTarget*, May 2, 2024.

²³ Eriq Gardner, “Netflix Supports Net Neutrality Rules in Fight Against ISPs,” *Hollywood Reporter*, April 25, 2023.

²⁴ Nicholas Cartier, “Netflix’s System Architecture Represents a Rare Competitive Edge,” *Streaming Media*, October 24, 2023.

practices, where certain streaming services don't count against mobile data caps, have been used by carriers as a competitive tool. T-Mobile's "Binge On" program allowed customers in the US to stream from certain video services without using data, albeit at DVD quality, which arguably favored established partners and raised net neutrality questions. In the EU, some regulators cracked down on or scrutinized such practices to ensure a level playing field for content providers. These policies effectively determine whether telecom infrastructure remains a neutral conduit or whether it can act like a platform gatekeeper itself, prioritizing some content. The general trend in Western markets has been toward neutrality (especially in the EU), but with ongoing battles, particularly in the US, neutrality rules have waned with different administrations. For our purposes, the key point is that the design of mobile infrastructure governance directly affects streaming platforms: If an ISP throttles video or exempts one service from caps, it shapes user behavior and the competitive landscape of platforms.

Another aspect of wireless infrastructure is the integration of Wi-Fi and fixed networks into the mobile streaming experience. A large portion of "mobile" streaming occurs on Wi-Fi rather than via cellular data, since Wi-Fi is often less capped by connection speeds and faster. This has led to mobile apps typically allowing users to choose video quality based on network (lower on cellular, higher on Wi-Fi by default) to manage data use. Also, ISPs increasingly offer seamless bundling between cellular and Wi-Fi networks. For example, a smartphone may automatically switch to known Wi-Fi networks to offload traffic from cellular networks. Some cable companies in the US (like Comcast) leveraged their extensive Wi-Fi hotspot networks to create quasi-mobile networks, allowing their customers to stream outside the home without being on a traditional cellular plan. This convergence of Wi-Fi and cellular networks illustrates how "mobile infrastructure" is a patchwork of different network types. Nonetheless, from the user perspective, what matters is that a connection is available—whether via 4G, 5G, or Wi-Fi—to enable streaming. Castells's notion that being connected anywhere is the new normal of communication holds true, as people expect uninterrupted access to media as they move through different spaces.²⁵

During peak events, such as a major sports championship being streamed or a new Netflix series release, cellular networks may experience localized spikes in demand. Carriers deploy traffic management techniques such as quality-of-service (QoS) prioritization, compression, and sometimes throt-

²⁵ Castells, "Communication, Power and Counter-Power," 238.

ting heavy users to maintain overall network performance.²⁶ The back-end systems of telecommunication networks, like content-aware routers or caching within ISPs, also play a role. Many ISPs and mobile carriers deploy caches for popular streaming platforms—often as part of the platform’s own CDN program (e.g., Netflix Open Connect appliances or Google Global Cache for YouTube). These caches are infrastructure nodes that store copies of popular content within an ISP’s network, so that if multiple users in that region request the same video, it can be delivered from the local ISP cache rather than each request traversing the wider internet.²⁷ This significantly reduces bandwidth costs and improves latency. Such arrangements again blur the lines of platform and infrastructure. The platform (Netflix/Google) provides the caching servers and content, while the ISP provides power, housing, and connectivity for them, each benefiting from more efficient delivery. The architectural design here is collaborative—a layered system optimized for streaming quality.

Spectrum allocation and cell tower siting, while highly technical, are also part of the narrative. Spectrum allocation is the process of assigning specific frequency ranges to different types of wireless communication services (like mobile phones, radio, or TV) to avoid interference and ensure efficient use of the airwaves. Cell tower siting, on the other hand, is the process of selecting and approving locations for building cell towers, considering factors like signal coverage, zoning laws, and community impact. Mobile streaming’s success made video a must-consider traffic type when planning networks. 5G in the form of high-frequency millimeter wave bands offers huge capacity but short range, meaning it can support many 4K streams in a stadium or downtown (spaces of dense usage) but requires many small cells. Lower frequency bands cover wider areas but are more easily congested with video load. Carriers are spending billions at spectrum auctions (the US auctioned mid-band C-Band for 5G in 2021 for over \$80 billion,²⁸ for instance) to secure the frequencies needed to carry future streaming-intensive traffic. In a sense, streaming media has become a driver of infrastructure investment cycles.

Wireless infrastructure provides the pervasive connectivity that allows streaming platforms to reach users while they are on the go, and the continuous upgrades and policies of these networks enable mobile streaming’s

²⁶ Samir Marwaha, “Sandvine’s 2024 Global Internet Phenomena Report: Global Internet Usage Continues to Grow,” AppLogic Networks (blog), April 10, 2024.

²⁷ “What Is Caching?” *Fortinet*, accessed May 12, 2025.

²⁸ “Auction 107: 3.7 GHz Service,” Federal Communications Commission, accessed May 12, 2025.

growth. The relationship is symbiotic. User demand for mobile streaming drives network investment, and improved networks drive further innovative uses of streaming. Effectively, a network has been built where the movement of messages (media streams) are tracked alongside the movement of people. Smartphones constantly communicate with cell towers, negotiating bandwidth for the next segment of a video or song, creating a seamless illusion that media resides in the “cloud.” This always-on connectivity has effectively merged with our daily routines. Media consumption is no longer an activity bound to a living room or a wired desktop, but something that can occur in the car, on a walk, in a café, or anywhere we carry a device.

Taken together, the wireless infrastructure component of architectures of mobility underpins everything the platforms do. It is the less visible part of the system (users notice Netflix or YouTube, not TCP/IP packets, cell handovers, data centers, etc.), yet it is decisive in shaping what is possible. A limitation in infrastructure (e.g., data caps, congested networks, down towers or cables) directly limits the platform’s usage and revenue; conversely, advancements such as 5G open new strategic opportunities for platforms to offer varied types of media, including higher resolution content and interactive experiences. The next section examines how platforms capitalize on that connectivity—through data and algorithms—to personalize content and manage the constant stream of information.

Big Data, Algorithms, and APIs: The Backend of Mobile Streaming

We now turn our attention to the less-visible backend of mobile streaming: the big data pipelines, algorithmic systems, and APIs that power personalized content delivery. As of 2024, there are over five hundred hours of video uploaded to YouTube every minute, and the platform has over two billion logged-in monthly users worldwide.²⁹ To manage this deluge of content and connect viewers with videos that will keep them engaged, YouTube relies on algorithmic curation. Its recommendation algorithm and “Up Next” autoplay feature drive a large portion of views, learning from each user’s history and viewing/interaction patterns across similar users. Platforms like YouTube and Facebook do not merely mediate public discourse, “they constitute it”

²⁹ Robin Geuens, “How Many Hours of Video Are Uploaded to YouTube Every Minute?” SOAX, October 30, 2024.

through the design of their algorithms and interfaces.³⁰ By deciding what content is prominently recommended or filtered out, the platform architecture shapes audience tastes and trends and responds accordingly. Work on the “politics of platforms” emphasizes that companies like YouTube strategically frame themselves as neutral conduits or open marketplaces, yet in practice they continuously intervene, through moderation policies, recommendation systems, and interface nudges to organize content and user behavior in profitable ways.³¹ As such, YouTube’s business model is inseparable from its technical system of data collection and algorithmic sorting, which maximizes advertising opportunities (more watch time equals more ads viewed) but also raises concerns about filter bubbles and the promotion of sensational or extremist content as a side effect of engagement-driven algorithms. These hidden algorithms exert power by governing what audiences see (and don’t see), effectively impacting major aspects of public discourse.

Mobile streaming platforms are often praised for their, ostensibly, slick interfaces and vast content libraries, but their true power resides in the less-visible backend systems: the big data repositories, algorithmic recommendation engines, and APIs that together orchestrate what content is delivered to whom. These elements form the intelligent layer that sits atop the physical infrastructure. They enable personalization at scale, real-time interactivity, and the integration of third-party services, all of which are crucial to the modern streaming experience. This section examines how big data and algorithms operate within mobile streaming architectures, and how application programming interfaces function as both technical and governance mechanisms in the platform ecosystem.

Data collection is a core mechanism of platform ecosystems; it is the ongoing effort to convert every aspect of user activity into quantifiable data.³² Mobile streaming intensifies this process, as smartphones function not only as media devices but also as personal sensors. They capture contextual information such as location, device type, and social interactions in addition to content choices. Each time a user taps on a video or song, the system logs what was selected, when it was accessed, possibly where the user was, and whether the content was completed. Aggregated across millions of users, this generates big data characterized by high volume, speed, and variety. These data streams are typically funneled into cloud databases

³⁰ Tarleton Gillespie, *Custodians of the Internet: Platforms, Content Moderation, and the Hidden Decisions That Shape Social Media* (Yale University Press, 2018).

³¹ Tarleton Gillespie, “The Politics of ‘Platforms,’” *New Media & Society* 12, no. 3 (2010).

³² van Dijck et al., *The Platform Society*, 101.

and data lakes maintained by the platforms. YouTube tracks detailed viewing and search histories, Spotify logs second-by-second listening behavior, and Netflix monitors viewing patterns across time and demographics. This practice of datafication forms the basis of the “digital enclosure,” where user activity is continuously monitored and mined for informational value.³³ In the context of streaming, the enclosure is both literal within the confines of the app where every action is tracked, and metaphorical in reflecting the broader condition of having to accept dataveillance in exchange for access to services.

The implications of pervasive data collection are multifaceted. On the one hand, it enables unprecedented mobile personalization and convenience. When Netflix recommends a show that resonates with a user, or when Spotify surfaces a new favorite song, those moments of discovery are enabled by the analysis of vast data sets. However, this personalization comes with a tradeoff. Platforms hold immense knowledge about users, while users themselves have little insight into what has been collected or how it is interpreted. This disparity is often called the big data divide, highlighting how platforms gain strategic advantage through opaque data practices.³⁴ Users only see the output, such as a recommendation or playlist, but not the layers of surveillance and inference behind it. This data is often used for purposes beyond personalization, including targeted advertising, dynamic pricing, and the sale of anonymized trend data. Andrejevic warns that we now exist in a system of continuous observation where individuals effectively serve as experimental subjects, contributing unknowingly to A/B tests and machine learning models that refine platform behavior in real time.³⁵ This situation illustrates a power asymmetry: Platforms wield significant informational power over users, who remain largely in the dark about how their data is used. Streaming services regularly experiment with interface changes, algorithm adjustments, and content placement to see what drives engagement. Netflix, for example, tests thumbnail images and descriptions, while Facebook famously altered users’ news feeds in a study on emotional contagion.³⁶ These examples illustrate how platform governance increasingly relies on iterative, data-driven experimentation.

³³ Mark Andrejevic, “Surveillance in the Digital Enclosure,” *The Communication Review* 10, no. 4 (2007).

³⁴ Mark Andrejevic, *Infoglut: How Too Much Information Is Changing the Way We Think and Know* (Routledge, 2013).

³⁵ Parks et al., *Media Backends*, 13.

³⁶ Adam D. I. Kramer et al., “Experimental Evidence of Massive-Scale Emotional Conta-

At the heart of this experimentation by major streaming platforms is an algorithmic curation system that determines what content is shown or suggested to users. These algorithms are the computational manifestation of the platform's editorial logic. In earlier eras, media consumption was largely scheduled and curated by human editors or programmers (TV channel line-ups, radio DJs, etc.), but in the streaming era this role has increasingly been assumed by algorithms processing big data. Netflix's homepage is not a static catalog, but a personalized montage of rows and thumbnails uniquely generated for each viewer. YouTube and TikTok feeds are virtually endless, generated on the fly based on one's past behavior and real-time feedback. Spotify's playlists like "Discover Weekly" are assembled by algorithms that analyze millions of tracks and listening profiles. In short, algorithmic personalization is taking place at a mobile scale.

These recommendation engines typically combine several techniques. Collaborative filtering identifies patterns among users with similar tastes. Content-based filtering matches characteristics of previously liked content to suggest similar options. More recently, deep learning models have been introduced to detect subtle and complex correlations in user behavior.³⁷ YouTube's algorithm considers hundreds of signals, including video topic, watch time, click-through rate, and recency, to rank videos in the "Up Next" queue. TikTok also analyzes micro-behaviors such as how long a user watches a video, whether they replay it, or how quickly they scroll past. While these systems are designed to improve user satisfaction by surfacing relevant content, their deeper goal is to maximize engagement to support the business model. But remember, platforms do not simply mediate public discourse; they help constitute it.³⁸ Recommendation algorithms and interface features influence what people see and, in turn, what kinds of media rise to cultural prominence. For example, if Spotify's algorithm favors songs that are played to the end, artists may adjust their music to minimize listener drop-off. If YouTube rewards longer watch times, creators may extend video length or design more compelling openings. These algorithms quietly influence both user choices and the direction of creative production.

gion Through Social Networks," *Proceedings of the National Academy of Sciences of the United States of America* 111, no. 24 (2014).

³⁷ Paul Covington et al., "Deep Neural Networks for YouTube Recommendations," in *Proceedings of the 10th ACM Conference on Recommender Systems* (Association for Computing Machinery, 2016).

³⁸ Tarleton Gillespie, "Platforms Are Not Intermediaries," *Georgetown Law Technology Review* 2, no. 2 (2018).

While algorithmic curation can enhance usability by helping users navigate an overwhelming amount of content, it has also sparked considerable criticism. One prominent concern is the loop of content in which users become trapped, where algorithms become so effective at predicting preferences that they mostly deliver content aligned with a user's existing tastes or beliefs, thus creating filter bubbles and/or echo chambers. This limits exposure to differing viewpoints or unfamiliar material. The Proceedings of the National Academy of Sciences points out that YouTube's algorithm has frequently been accused of contributing to filter bubbles and guiding users toward increasingly radical content.³⁹ In the political sphere, critics argue that the recommendation engine can steer users toward more extreme or conspiratorial videos in its effort to maximize watch time. Even beyond political content, recommendation systems can narrow the range of media a user encounters. A listener who is repeatedly served the same style of music, or a viewer shown only one genre of television, may find their media habits becoming increasingly uniform. While platforms like Netflix and YouTube sometimes introduce unexpected or trending content into a user's feed to encourage exploration, personalized recommendations still dominate and largely shape the user's experience.

There are also concerns about fairness and transparency. Algorithms can unintentionally amplify certain types of content or creators while marginalizing others. For instance, if user behavior data suggests that one demographic generates higher engagement, the system may end up recommending content featuring that group more frequently, reducing visibility for others. Yet these decisions are largely invisible to users and creators because the algorithms are proprietary and in constant flux. Platforms treat them as trade secrets, offering only vague explanations through creator guidelines or occasional blog updates. This lack of transparency has led to growing calls for algorithmic accountability, especially as these systems shape the distribution and visibility of culture online.⁴⁰

In addition to shaping recommendations, algorithms also govern moderation and platform health. YouTube's Content ID system scans uploaded content for copyrighted material and can automatically block or demonetize

³⁹ Muhammad Haroon et al., "Auditing YouTube's Recommendation System for Ideologically Congenial, Extreme, and Problematic Recommendations," *Proceedings of the National Academy of Sciences of the United States of America* 120, no. 50 (2023).

⁴⁰ Frank Pasquale, *The Black Box Society: The Secret Algorithms That Control Money and Information* (Harvard University Press, 2015); Safiya Umoja Noble, *Algorithms of Oppression: How Search Engines Reinforce Racism* (NYU Press, 2018).

content on behalf of rights holders. Facebook and Twitch rely on automated systems to detect and flag potentially harmful material, such as hate speech or nudity, which is then reviewed by human moderators. These tools have a direct impact on what content remains available, affecting the cultural landscape users interact with. Indeed, moderation is a foundational function of platforms that is deeply tied to the values and business models of the platform owners. On TikTok, this becomes especially apparent as the algorithm promotes engaging videos while also suppressing content flagged as violating community guidelines. In some cases, TikTok has been accused of limiting the reach of political content or content from marginalized creators, raising further concerns about bias and suppression.⁴¹ Ultimately, algorithmic curation is not just about improving user experience, but it also reflects legal, commercial, and ethical decisions embedded in platform design.

Algorithmic curation is a defining feature of mobile streaming platforms. It enables personalization at scale, something no team of human editors could achieve for billions of users. However, it also introduces an automated, data-driven layer of governance over media consumption that is closely tied to corporate goals. These algorithmic systems are part of the broader architectures of mobility, not only through physical infrastructures like networks and devices, but also through the software and logic that govern information flow. Just as road design affects traffic patterns and accessibility, algorithm design shapes the visibility of content and influences what users see. Recognizing this influence is central to media literacy. Users must understand that what appears on their screens is not neutral, but the outcome of deliberate algorithmic processes. In response to growing concerns, scholars and policymakers have proposed interventions such as transparency reports and algorithmic audits. In the European Union, the Digital Services Act mandates that major platforms provide external researchers with access to their algorithms for the purpose of identifying systemic risks.⁴² These regulatory moves suggest a shift in how algorithms are viewed—not just as technical tools, but as components of public infrastructure that warrant oversight and accountability.

⁴¹ Robyn Caplan and Tarleton Gillespie, “Tiered Governance and Demonetization: The Shifting Terms of Labor and Compensation in the Platform Economy,” *Social Media + Society* 6, no. 2 (April 2020).

⁴² European Union Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on Contestable and Fair Markets in the Digital Sector and Amending Directives (EU) 2019/1937 and (EU) 2020/1828 (Digital Markets Act), *Official Journal of the European Union* L265/1 (2022).

In sum, controlling the backend has become as crucial as owning the network or the content. This blur between running a platform and running infrastructure leads us to the final issue: the full convergence of platforms and infrastructures, and its consequences.

APIs and the Merging of Platforms with Infrastructure

While algorithms manage the flow of content within platforms, application programming interfaces (APIs) govern how platforms interact with the outside world and with other software. APIs are the formal channels through which external applications or devices request data or services from a platform. They are a foundational element of modern digital ecosystems, including mobile streaming. For example, when a smart TV or Chromecast plays a Netflix show, it relies on Netflix's API to access the catalog and deliver the content. Similarly, a third-party website embedding a YouTube video uses YouTube's embed API. Spotify offers APIs that allow developers to create applications capable of retrieving song data or controlling playback, provided the user grants permission. In mobile streaming, APIs enable essential integrations such as content sharing to social media, single sign-on through services like Facebook or Google, and data reporting for analytics and advertising. These functions illustrate how APIs extend the reach of streaming platforms across devices, apps, and services.

Beyond their technical role, APIs also work as tools of platform governance and strategy. Anne Helmond argues that APIs are not just interfaces for data access but also mechanisms of control, allowing platforms to shape the broader digital ecosystem.⁴³ By controlling what data or capabilities are accessible—and to whom—platforms can guide third-party development in ways that align with their business goals. Twitter offers a clear example. In its early years, the company maintained a relatively open API that encouraged a flourishing ecosystem of third-party clients, which helped grow the platform. However, as Twitter shifted toward monetization through advertising within its own interface, it began restricting API usage. By 2023, it had fully revoked access for many third-party apps, effectively shutting them down.⁴⁴ This shift illustrates how platforms use API governance not

⁴³ Anne Helmond, "The Platformization of the Web: Making Web Data Platform Ready," *Social Media + Society* 1, no. 2 (2015).

⁴⁴ Mitchell Clark, "The Third-Party Apps Twitter Just Killed Made the Site What It Is Today," *The Verge*, January 22, 2023.

only to facilitate innovation but also to consolidate control in alignment with commercial interests.

In the realm of streaming media, similar patterns of platform control and governance play out. During its expansion phase, Netflix offered a public API between 2008 and 2014 that allowed external applications to display titles, manage queues, and build Netflix-based tools.⁴⁵ However, once Netflix had achieved wide adoption and could ensure consistent access through its official apps on all major devices, it shut down the public API in 2014. This move signaled a shift toward full control of the user experience, managed entirely within its own ecosystem. YouTube, in contrast, still maintains a relatively open API. Despite its openness, YouTube imposes clear limitations, including caps on requests and strict terms that prohibit alternative user-facing designs. These rules ensure that while YouTube's reach extends across the web, it retains core control. Videos must still play in YouTube's player, advertisements are served by YouTube, and any API client that violates policy risks losing access.⁴⁶ Spotify offers another example. Its API supports integration with smart home devices, enabling native Spotify functionality on speakers and other systems. It also connects to services like Last.FM and other social media platforms to display currently playing tracks. These integrations make Spotify more embedded in everyday user routines. However, Spotify carefully governs what data is made accessible. For many years, it withheld access to individual user's listening histories, citing privacy concerns and perhaps a desire to avoid enabling competitors or third-party data mining.⁴⁷ These examples illustrate how APIs function as selective gateways, tools for expansion and entrenchment, but also instruments of strategic control.

From a systems architecture perspective, APIs make mobile streaming scalable and modular. Whether users access YouTube via a smartphone, smart TV, game console, or browser, these different devices typically connect to the same backend API endpoints to fetch content and log interactions. This centralization allows platforms to update recommendation systems or content libraries in one place. APIs also enable cross-platform integration that contributes to infrastructural convergence. For example, authentication systems such as Google or Facebook login are made possible by those com-

⁴⁵ Ryan Lawler, "Netflix Will Shut Down Public API Support For Third-Party Developers On November 14," *TechCrunch*, June 13, 2014.

⁴⁶ "YouTube Data API Overview," Google Developers, accessed May 12, 2025.

⁴⁷ Bartosz Janota and Robert Stephenson, "Spotify's Event Delivery: Life in the Cloud," *Engineering* (blog), Spotify R&D, November 12, 2019.

panies' APIs. In streaming, ISP companies sometimes bundle subscriptions to services like Netflix or Disney+, with backend integration happening through APIs. In these cases, a carrier's set-top box may directly pull streams from the platform, creating a seamless experience that blurs the boundary between internet service provider infrastructure and the streaming service itself. For the end user, it often appears as one unified system.

Controlling the API also means controlling the terms under which others can interact with a platform's data or features. This control gives platforms significant leverage in shaping how their ecosystems function. This is accomplished when platforms increasingly assume infrastructural roles, as they can then impose their own technical and policy standards on third parties.⁴⁸ A clear example is seen in the mobile ecosystems governed by Google and Apple. Both companies provide APIs within Android and iOS that manage essential functions such as media playback, notifications, and in-app purchases. Streaming services must rely on these APIs to operate correctly on mobile devices, and in doing so, they often must comply with platform-specific rules.

For instance, Apple requires the use of its in-app payment system and enforces restrictions on alternative payment methods. It also mandates compatibility with specific video codecs supported by its operating system. In this way, the API becomes a key site where the platform power of app stores intersects with that of streaming services. The conflict between Epic Games and Apple offers a striking example. When Epic attempted to bypass Apple's in-app purchase API in the mobile version of Fortnite, Apple removed the game from its App Store. This response demonstrates the rigidity of these technical and economic architectures and highlights how tightly control over APIs is linked to broader questions of platform governance and market power.

The merging of platform and infrastructure is clearly visible in the cloud computing services that many streaming platforms depend on. Companies like Amazon Web Services (AWS), Microsoft Azure, and Google Cloud provide infrastructure-as-a-service through APIs. These services allow platforms to store data, rent servers, and use content delivery networks as needed. Netflix was one of AWS's most prominent customers for cloud computing, even as it developed its own content delivery network, Open Connect. Spotify similarly relies on Google Cloud for its backend infrastructure. In these arrangements, a platform such as Netflix or Spotify is built on top of another

⁴⁸ Plantin et al., "Infrastructure Studies Meet Platform Studies," 307.

platform's infrastructure, interacting entirely through API calls. This kind of layered dependence illustrates how some platforms effectively become the infrastructure for others.

The internet now functions through these kinds of layers. Authentication systems, mapping services, payment gateways, and analytics platforms all rely on APIs, meaning that much of an app's functionality may come from external services. The system is interoperable in theory, as any service can connect with another if APIs are available. For instance, a developer could create an app that queues a YouTube video using a voice command by combining YouTube's API with a voice recognition API. However, the system is also tightly controlled. Each API is owned and governed by a platform that sets specific usage terms, rate limits, and permissions.

The control of APIs is yet another consequence of platforms as infrastructures via the expansion of platforms into various sectors through APIs and modular integration. In mobile streaming, this is especially evident. News outlets distribute video through Twitter and Facebook APIs, sports apps incorporate YouTube highlights, and music platforms allow users to share songs directly to social media. The architecture of mobility now consists of a dense mesh of API connections layered on top of physical infrastructure. This architecture can be visualized as a stack: Physical infrastructure like fiber optics and cell towers forms the base; above that is cloud infrastructure and content delivery networks accessed through internet protocols; platform logic and data exist on the next layer and are accessed through APIs; and the top layer consists of the user-facing apps. Each layer communicates with the next through defined protocols, such as HTTP and JSON-based APIs.

By managing both algorithms and APIs, large streaming platforms control not only the internal user experience but also how their services connect with the broader digital ecosystem. Amazon is a prime example of this convergence. It owns Twitch as a content platform, operates AWS as a cloud infrastructure provider, sells hardware such as phones, tablets, TVs, etc., and runs the Prime Video service. This integration allows Amazon to coordinate across layers, giving priority to its own services on its hardware or using AWS data to enhance performance across its platforms. Google has a similar structure with the Android operating system (OS): as a content and algorithm-driven platform, with Google Cloud for infrastructure. This vertical integration has raised antitrust concerns,⁴⁹ but it also makes these platforms extremely resilient and difficult to challenge. A startup may create

⁴⁹ Lina M. Khan, "Amazon's Antitrust Paradox," *The Yale Law Journal* 126, no. 3 (2017).

an innovative streaming app, but without the economies of scale provided by major cloud providers or access to user data for recommendation systems, it faces significant structural barriers. Unless of course, the start-up is purchased by one of the conglomerates, which is sometimes the goal to begin with.

Platforms as Infrastructures: Convergence and Consequences

If there's one continual refrain this chapter, it's that the line between platforms and infrastructures has become increasingly difficult to define, and this convergence has profound consequences for mobile personalization and power. The technologies and business arrangements that allow a user to stream a movie on a smartphone while commuting reflect a convergence of roles that were once, ostensibly, separate. In the past, traditional infrastructures such as telephone networks or cable television systems functioned as carriers, typically seen to only transmit content from distinct service providers. While power relations in these delivery mechanisms certainly existed in earlier forms of mass media, these pale in comparison to the tangled web of platforms, wireless infrastructures, data centers, APIs, and proprietary algorithms of contemporary streaming. Today, companies like Google, Amazon, and Netflix operate not only the platforms that deliver content and services but also the infrastructure that supports global communication. These companies maintain data centers, undersea cables, and foundational services that are now integral to digital life.

Mobile streaming offers concrete examples of this convergence. Google functions simultaneously as a platform and an infrastructure provider. Through YouTube, it connects video creators and viewers, while Android supports app development and mobile services. At the same time, Google owns and operates undersea fiber-optic cables and one of the world's largest content delivery systems. It is important to note that the physical infrastructure, particularly undersea cables, was once maintained by a consortium of telecommunication companies; however, this infrastructure is increasingly privatized.⁵⁰ Android itself serves as a mobile infrastructure that other applications depend on to function. Similarly, Amazon operates Amazon Web Services (AWS), which powers not only its own streaming offerings but also those of direct competitors. Netflix, for example, has long relied on AWS for cloud services, effectively paying Amazon to host its content infrastructure

⁵⁰ Phil Gervasi, "Diving Deep into Submarine Cables: The Undersea Lifelines of Internet Connectivity," *Kentik* (blog), March 28, 2023.

while competing with Amazon Prime Video for market share. This layered interdependence reflects a defining feature of the current digital ecosystem. Companies that compete in the consumer marketplace may simultaneously rely on one another behind the scenes. The result is a complex and often contradictory system, where infrastructure and platform roles are deeply entangled, and where competitors can also be business partners due to the way modern digital services are built and delivered. In effect, streaming's infrastructural logic knits together what were once distinct industries (tech, telecom, and media) into a single, vertically integrated system.

Another facet of convergence is when infrastructure owners expand into providing content services directly. Telecommunication companies have tried this route. Verizon and AT&T in the US, for example, ventured into content. Verizon acquired AOL and Yahoo, and AT&T went as far as buying Time Warner in 2018, gaining HBO, Turner, and Warner Bros. with the intent to become a vertically integrated content-and-distribution powerhouse (HBO Max streaming service integrated with AT&T wireless deals). The underlying vertical integration theory was that owning both the infrastructure (wireless/broadband networks) and premium content would create synergies: AT&T could, say, zero-rate HBO Max for its mobile subscribers or use its data on network usage to inform content strategy. However, these moves have had mixed success. AT&T ended up spinning off WarnerMedia in 2022, suggesting that synergy was elusive, and that content and delivery infrastructure might be better separated. Nonetheless, the impulse behind the merger speaks to the recognition that infrastructures and platforms are interdependent.

From the user's perspective, the blending of platform and infrastructure can offer greater efficiency but also raises important concerns about the concentration of power. When a small number of companies control nearly every layer of the digital media ecosystem, the risks of anti-competitive behavior and systemic vulnerability grow. Regulators in Western countries have begun responding to this dynamic. The European Union's Digital Markets Act directly targets dominant firms such as Google, Apple, Amazon, Meta, and Microsoft. The legislation imposes obligations intended to reduce gatekeeping power, including requirements that Apple allow alternative app stores and enable greater interoperability across platforms. These measures represent an effort to loosen the grip of vertically integrated digital ecosystems.

Some observers have gone further, suggesting that large digital platforms should be regulated more like public utilities or common carriers, given their infrastructural importance. Castells noted that much of the public sphere

now operates through privately owned networks, meaning that any failure or bias within those systems can have far-reaching social consequences.⁵¹ A few high-profile outages have made this vulnerability clear. During the 2019 YouTube outage and the 2021 Fastly content delivery network disruption, major swaths of internet services became temporarily unavailable. These incidents served as a reminder that everyday media access often depends on a small number of centralized, privately operated systems.

On the other hand, the integration of platforms with robust infrastructure has enabled remarkable innovations and reliability. Streaming a live global event to millions of mobile users concurrently—something that would have been technically unfathomable not long ago—is now almost routine, owing to these architectures. The COVID-19 pandemic stress-tested the system in 2020, when home and mobile streaming surged. Networks held up generally, with only minor throttling (Netflix and YouTube temporarily reducing default stream quality in Europe to ease network load). The fact that the internet didn't buckle under unprecedented traffic is credited to continual upgrades by infrastructure providers and smart traffic management by platforms which can adjust bitrates adaptively. In a sense, platforms and infrastructure working in concert proved resilient. This resilience, however, often comes from centralized optimization like when Netflix decided unilaterally to lower bitrates globally for a period of time.

Culturally, the merging of platform and infrastructure means that the ways we create, discover, and distribute content are now deeply embedded in technical systems controlled by a few organizations. Whereas twentieth-century culture industries (film studios, record labels, broadcasters) certainly had concentrated power, today's tech platforms arguably wield an even more pervasive influence because they not only produce or license content but also own the channels and the data. A user watching a Netflix-produced show on a Netflix app delivered via Netflix's CDN on a tablet running Netflix's recommended calibration settings is experiencing a vertically integrated chain with very few external inputs. In essence, our deeply rooted user experience is made by possible by, and confined within, a vertically integrated chain controlled by one company. The platform's knowledge about that consumption (what you watched, when, if you liked it) stays with Netflix. In earlier eras, a TV manufacturer or a cable operator might not know exactly which show you were engrossed in, but now the feedback is precise and instant. This feedback loop enables personalization, but it also leaves the traces of

⁵¹ Castells, "Communication, Power and Counter-Power," 239.

our lives in proprietary silos. The consistent monitoring of behavior has been coined the age of surveillance and is central to economic extraction.⁵² Streaming platforms are a prime example, monetizing not just content but the matrix of its consumption.

Finally, the convergence raises the question of global versus local. Western streaming architectures, led by US-based corporations, have become global infrastructures in their own right. Netflix and YouTube reach into hundreds of countries, often dominating local media distribution. This has prompted some nations to push back, either through regulation (like European quotas for local content on Netflix, or Australia and Canada seeking to have YouTube/Facebook pay news outlets for content) or by fostering local platforms. In China, for example, the government kept out many Western platforms, leading to domestic equivalents (Youku for video, Tencent Video, iQIYI, etc., and a highly developed mobile streaming ecosystem behind the “Great Firewall”). Those Chinese platforms have their own integrated infrastructures; Tencent, for one, has massive cloud infrastructure and a constellation of services. So globally, there is a splintering, where Western platform-infrastructure complexes dominate in many regions, but alternative complexes exist in some markets, raising interesting issues of interoperability and digital sovereignty. The phrase “architectures of mobility” consequently also takes on a geopolitical meaning: Whose architecture, whose infrastructure, and whose platforms will structure mobile media access on a global scale?

Conclusion

Mobile streaming media is the product of a profound synthesis of technological infrastructure and platform organization. In this chapter, we traced how streaming services—from Netflix and Spotify to YouTube, TikTok, Twitch, and beyond—depend on and actively shape the underlying networks and systems that deliver content. This tangled web shows that mobile streaming is not just about smartphones as devices, but about entire platform ecosystems and wireless infrastructures working in tandem to enable new habits of media consumption. Throughout this chapter we’ve shown a progression toward greater integration. As networks became faster and more omnipresent, platforms evolved to leverage that connectivity with on-demand,

⁵² Shoshana Zuboff, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power* (New York: PublicAffairs, 2019), 3.

personalized services. This is mobile personalization at scale. In turn, the success of those services drove further network innovation and investment.

Streaming platforms may feel like tools of convenience and choice, but they operate within a much larger system driven by data extraction, algorithmic control, and economic incentives. These platforms are not just distributing content; they are actively shaping how we experience media. By tracking behavior, prioritizing engagement, and curating what users see, platforms play a central role in influencing attention, tastes, and even public discourse. What appears to be mobile personalization and user empowerment is often engineered for maximum profitability and platform growth.

The shift from traditional broadcasting to on-demand streaming represents more than just a technological upgrade. It reflects a broader transformation in how media systems function—favoring real-time feedback, user profiling, and content optimization over editorial judgment or public service. This has consequences not only for what we watch, but also for how culture circulates, how information is valued, and who gets to be heard.

Understanding streaming through this lens reveals that it's not just about entertainment. It's about infrastructure, power, and control. The platforms that dominate today are shaping the future of media in ways that often go unnoticed. Recognizing this helps us move beyond surface-level appreciation and toward a more critical awareness of how digital media environments are built, maintained, and monetized. In short, mobile streaming's architecture has reshaped media power dynamics: Personalization for the user is built on vast concentrations of power for the platform.

On a technical level, this chapter revealed that the material architecture of mobile streaming is layered and complex. It comprises physical elements (cell towers, servers, etc.), logical protocols (internet protocols, streaming protocols), software systems (CDNs, databases, AI algorithms), and user-facing applications—all coordinated to deliver a smooth experience. Each layer has its guardians: telecom companies, device manufacturers, platform operators, developers, sometimes all within the same corporate entity. The modularity provided by APIs is what allows these layers to interact and the entire edifice to function globally, but it is also what allows certain players to dominate multiple layers by setting the rules of interconnection.

Crucially, mobile streaming's architecture has reconfigured media power dynamics. Users have gained the ability to access a dizzying array of content on their own terms, fragmenting the old mass audiences of the broadcast era into personalized micro-audiences. Yet paradoxically, this individual

empowerment is delivered by highly centralized services—a few recommendation algorithms determining global viewing patterns, a few app stores controlling app distribution, a few network providers carrying the data. The promise of the early internet as a decentralized, democratized media space seems to have been a pipedream. The reality of mobile streaming is more akin to a network of centralized convenience. This does not negate the real benefits—we truly do have more choice and creative opportunity; anyone can start a YouTube or TikTok channel and potentially reach millions—but it contextualizes them to the platform and centralized expectations of the network they are operating within.

Looking ahead, the merging of platforms and infrastructure is likely to deepen with trends like metaverse visions and with technologies like the internet of things and 5G enabling new mobile media forms (immersive and interactive experiences in real physical space). The chapter's insights suggest that unless there are significant changes (regulatory or market-driven), the same few actors will extend their architectures to these new domains, embedding their data-driven, platform-centric logics into the next generation of media systems. However, there is also increasing awareness and pushback: antitrust cases, public pressure on privacy, and open-source or decentralized alternatives. For example, the ActivityPub protocol used by PeerTube for video or Funkwhale for audio aim to be “Fediverse” versions of YouTube/Spotify and seek to reimagine streaming without centralized monopolies. Whether these can offer viable alternatives remains to be seen.

The architectures of mobility that support streaming media are a marvel of modern engineering and enterprise, fusing brick-and-mortar data centers with satellites and subterranean cables with algorithms and attention economies. They represent a new kind of media infrastructure—one that is global, constantly evolving, and intimately entwined with our personal lives. Understanding these architectures is essential. As citizens and consumers, knowing how our media is delivered and shaped allows us to better navigate the contemporary media landscape. This chapter has provided an analysis of those underpinning structures and systems, shedding light on how platform business models, wireless infrastructures, big data, algorithms, and APIs all intersect to produce the mobile streaming experiences that many of us now take for granted. Armed with this understanding, we can better engage in discussions about the future of media, ensuring that the future balances innovation and convenience with fairness, openness, and respect for the human element at its core.

CHAPTER THREE

Monetizing Mobility

Walk into any coffee shop, airport lounge, or public park, and chances are your phone already knows where you are. And so do the companies behind your apps. Whether it's a push notification for a nearby fast food deal, a Spotify ad tailored to your current zip code, or a TikTok video that just happens to feature a local landmark, mobile streaming has become entangled with location-aware technologies like geofencing. These tools don't just deliver content; they deliver it strategically, turning your physical presence into a point of monetization. This chapter explores the convergence of mobile streaming and forms of monetization, situating the development within broader shifts in how media is produced, delivered, and capitalized on. From a teenager in Lagos scrolling TikTok, to a commuter in Mumbai listening to Spotify, to a Zoom meeting on a park bench in New York, billions of people engage with mobile media as a matter of routine. Platforms like TikTok, Twitch, Instagram, YouTube, and Apple Music function as global infrastructures, systems built not only to entertain and connect but also to shape behavior, collect data, and generate profit, in turn, consolidating power of media engagement. In short, this chapter is about how mobile streaming makes money. This occurs not just through ads, but also through architectures that increasingly know where you are, what you like, and when you're most likely to tap, swipe, or scroll. Such context-aware targeting exemplifies mobile personalization, the tailoring of content and ads to an individual user's data and situation, a concept central to this chapter's analysis.

Last chapter, we detailed how the model of the platform provides insight into how companies like ByteDance (TikTok), Google (YouTube), Meta (Instagram), Amazon (Twitch), and Apple (Apple Music) function as infrastructures of streaming built on data extraction, network effects, and an ecosystem of

control.¹ These services span social media, video, live streaming, and music and are unified by a common mobile personalization logic that turns user engagement into profit. Moreover, the network society situates mobile streaming in a broader transformation where digital networks constitute the social and economic infrastructure of society.² From this perspective, mobile streaming platforms are not just neutral technologies for content delivery; they are highly strategic entities that marshal technical infrastructure, user engagement, and data flows to accumulate capital and influence. This analysis highlights how features like cloud-based delivery, design, cross-platform integration, and the use of application programming interfaces (APIs) all serve to expand the reach and profitability of these platforms. Simultaneously, these same features reshape user experiences and behaviors, making continuous connectivity and consumption a normalized aspect of life.

If the last chapter was all about *how* mobile streaming operates, then this chapter focuses on the *why* as it relates to monetization and the role of mobile personalization in generating profit. Platform and infrastructure providers don't create all this labor and resource intensive architecture out of some form of social good. These companies, of course, operate toward a payoff. But how they do so varies widely. This chapter, consequently, follows the money of streaming and mobile streaming to unearth just how these platforms make a buck or two . . . or a couple billion.

By critically analyzing the monetization strategies of these mobile streaming platforms, we reveal a landscape of platformed media: a complex ecosystem where user attention, personal data, cultural content, and technological infrastructure are all interwoven into new commodity forms. First, we examine how mobile streaming became ubiquitous in everyday life. Through habitual engagement seemingly anytime and anywhere, users funnel their attention, time, and data into the silos of platforms. Next, we delve into five key monetization strategies: (1) mobile and locative aware advertising, (2) mobile subscription and premium services, (3) microtransactions and the creator economy, (4) monetized cross subsidies, and (5) APIs and platform capture. Through this structured analysis, we aim to provide a comprehensive understanding of how contemporary mobile streaming is not merely a set of popular apps, but rather a politically and economically significant development reconfiguring media industries and daily life on a global scale.

¹ Nick Srnicek, *Platform Capitalism* (Polity Press, 2016).

² Christian Katzenbach and Stefan Larsson, "Imagining the Digital Society: Metaphors from the Past and Present," *Digital Society Blog*, Alexander von Humboldt Institute for Internet and Society, May 15, 2017.

Mobile Streaming in Everyday Life: Ubiquity and Habitual Engagement

Before examining how platforms monetize, it's important to understand how deeply mobile streaming is woven into everyday life. The integration of mobile streaming into daily life has reached a point where, for many people, it is taken for granted that any moment of downtime or any routine activity can be augmented with streaming media. Global usage statistics underline just how deeply these platforms have penetrated our routines. For instance, as of 2024, TikTok users worldwide spent an average of ninety-five minutes per day on the app—far more than users spent on any other social network.

³ This is not limited to any single region; TikTok's over one billion monthly active users are spread across Asia, North America, Europe, and beyond. YouTube has over 2.7 billion users and reports that, collectively, people watch over one billion hours of video on YouTube every day.⁴ These figures convey how streaming platforms capture a large share of human attention on a daily basis. Furthermore, Instagram and Facebook together command billions of user visits daily, and services like WhatsApp or WeChat facilitate countless video and voice calls each day around the globe. Even live streaming-centric platforms like Twitch, while more niche, sustain around 35 million daily active users and at any given moment about 2.5 million people are watching a Twitch stream.⁵ Twitch viewers collectively watched over 20 billion hours of content in 2024,⁶ indicating that for a segment of the population, viewing live streams has become a regular part of leisure time.

These numbers translate into lived experiences that are remarkably common across different societies. It is not hard to imagine a morning train ride filled with commuters watching YouTube clips or scrolling Instagram Reels; teenagers coming home from school and broadcasting themselves live on Twitch or TikTok Live; families separated by distance having nightly video chats via WhatsApp or FaceTime; music streaming forming the soundtrack of daily exercise, study, or relaxation. Mobile streaming is embedded in daily rhythms. Many people start their day by checking social media videos or news clips on their phones and end their day watching streaming videos in

³ "TikTok Statistics You Need to Know in 2025," Backlinko, updated January 2026.

⁴ "YouTube Statistics 2025: Users by Country + Demographics," *Global Media Insight*, January 2025.

⁵ *TwitchTracker*, accessed May 12, 2025, <https://twitchtracker.com/>.

⁶ Dhanika Pineda, "Viewers Watched 20.35 Billion Hours on Twitch in 2024: 'Stream Big' Explores Why," *Central Florida Public Media*, December 18, 2024.

bed. In the interim, streaming fills all sorts of micro-moments—waiting in line, eating lunch alone, or taking a work break—as well as more extended periods like evening entertainment. Indeed, streaming media often accompanies activities that used to be free of media or, at least, dictated by broadcast schedules. People now listen to podcasts or music streams while driving or doing house chores, watch recipe videos while cooking, and scroll short videos while riding an elevator. The smartphone has become a constant companion, and the proliferation of high-speed mobile internet means entertainment and communication streams are constantly within arm's reach.

This deep embedding of media into everyday life is both a result of platform strategies and a driver of further growth. Platforms deliberately design their services to foster habitual use. Features like the infinite scroll, such as TikTok's never-ending personalized feed or YouTube's autoplay that queues another video after one finishes, encourage users to lose track of time and consume content continuously. In effect, these are major instances of mobile personalization keeping users hooked. Meanwhile, push notifications draw users back in. YouTube alerts about a channel's new upload, Twitch about a streamer going live, Instagram about a friend's live story, and so on. These notifications are carefully calibrated by algorithms to maximize re-engagement. In other words, the platform learns when and how to prod a user to open the app again. The result is what some commentators call the attention economy at work. That is to say, an environment where various services compete intensely for the finite hours of users' attention because that attention can be monetized.⁷ Social media and streaming platforms have become dominant players in this attention economy by captivating audiences with endless feeds of user-generated content. The more habitual and reflexive the usage becomes ("I'm bored, let me check TikTok" or "It's 8 p.m., time to browse Twitch"), the more stable the platform's audience commodity is for advertisers and the more opportunities for monetization via subscriptions or purchases.

The scale of this phenomenon is a testament to how mobile streaming aligns with certain universal aspects of human behavior, while also being tailored to local contexts. Smartphone adoption has been rapid in both developed and developing countries, which means billions now have a personal screen. In many emerging markets, cheap Android phones and affordable data plans led to a mobile-first internet culture—people whose

⁷ Richard A. Lanham, *The Economics of Attention: Style and Substance in the Age of Information* (University of Chicago Press, 2006).

first and primary way of accessing the internet is through mobile apps. For these users, streaming video on YouTube or local platforms, like India's Hotstar or China's Douyin, effectively replaced legacy media like TV. It goes without saying that large media companies have recognized these opportunities. YouTube, for instance, provides offline download features in markets with patchy connectivity, so users can download videos while on Wi-Fi and watch them on the go, and TikTok launched "TikTok Lite" for lower-end phones. By adapting to local bandwidth and device conditions, platforms further ensured their integration into daily habits across different socio-economic settings.

Moreover, streaming platforms often become integrated with social practices and cultural norms. TikTok's short-video format has given rise to global dance trends, memes, and challenges that engage youth culture; participating in or at least watching these trends becomes a social activity. Twitch and YouTube have created new forms of communal experience. For example, tuning into a live stream of a video game tournament or an influencer's Q&A feels akin to communal TV watching or even attending an event, except it can be done from anywhere on a phone. During the COVID-19 pandemic, usage of video calls and streaming skyrocketed, embedding these platforms even more deeply into work and family life. Zoom, practically a niche business tool pre-2020, became a household name and by April 2020 was hosting 300 million daily participants around the world as people used it for work meetings, remote classes, and virtual social gatherings.⁸ This period underscored how essential streaming communication had become. And even after the pandemic's peak, habits like working remotely via video and maintaining long-distance relationships with video calls have persisted.

The blurring of work and leisure through the same devices and networks is another notable aspect. A smartphone that receives office emails is also the device for watching TikTok; a laptop used for a Zoom meeting at noon might be used to stream Netflix at 9 p.m. This convergence can lengthen the time one stays "on" digital networks, giving platforms even more opportunity to capture user engagement in the margins of daily life (for instance, checking a social app in between tasks while working from home). From the user's perspective, the omnipresence of streaming can be double-edged. While it offers convenience and entertainment on demand, it can also lead to a sense of information overload, or an addictive pull to keep looking at

⁸ "A New Era of Growth: How the Zoom Partner Ecosystem Is Evolving," *Zoom Blog*, Zoom, October 2024.

the screen. Some studies have begun framing compulsive TikTok or social media use in terms of addiction, noting that the algorithmic feed is designed to be hard to put down.⁹

Such design is no accident. It stems from commercial imperatives. Keeping users engaged translates directly into profit, either via more ad impressions or higher likelihood of paying for premium features. This raises questions about the consequences of building daily routines around platforms whose primary goal is profit maximization. Critics argue that when engagement-at-all-cost is the business goal, platforms may encourage unhealthy usage patterns or promote sensational content that grabs attention, such as extreme challenges on TikTok, or outrage-inducing articles, Tweets, or videos, because these drive metrics up, a phenomenon documented across social media algorithms.

Despite these concerns, mobile streaming's integration into everyday life has a self-reinforcing quality. The longer people use these platforms and incorporate them into mundane practices, the more indispensable they feel. For example, a student might rely on YouTube tutorials daily for help with coursework, making YouTube an educational crutch; a professional might habitually listen to industry podcasts on Apple Podcasts or Spotify each morning; a family might develop a routine of a weekly video call across continents. In each case, what starts as a convenience can quickly become seen as a necessity or an assumed background to life. This cultural entrenchment makes it challenging for individuals to opt out of platforms, which, mind you, contributes to their monopoly power, and also creates lock-in on a societal level. In other words, public discourse, trends, and even news increasingly flow through these channels.

The patterns of use encouraged by the platforms—frequent, habitual, and extended engagement—are a direct outcome of their design and business objectives. As we turn next to examine the monetization strategies of these platforms, it is important to keep in mind that the casualness with which we treat ubiquitous streaming is in fact cultivated by design. The omnipresence of streaming is the fertile ground from which platforms harvest revenue. They have made themselves indispensable to users in order to more effectively capitalize on user activity. In short, the habitual engagement cultivated via mobile personalization sets the stage for monetization. Next, we turn to how platforms capitalize on this engagement through five monetization strate-

⁹ Li Gu et al., "What Drives Me to Use TikTok: A Latent Profile Analysis of Users' Motives," *Frontiers in Psychology* 13 (2022).

gies. We break down how exactly these platforms monetize the extensive engagement they have secured, and what new forms of commodification and profit are emerging in the mobile streaming era. This position in daily life underscores the power these platforms wield as gatekeepers of information and entertainment, a power base they will leverage in the monetization strategies that follow.

Monetization Strategies of Mobile Streaming Platforms

Behind the seamless streams of content that characterize the user experience lies a sophisticated array of monetization strategies. Mobile streaming platforms employ multiple business models to turn user engagement into revenue. These strategies include advertising, subscriptions, data monetization and sale, microtransactions and virtual gifting, content partnerships, and/or ecosystem plays. We view each of these strategies as a way to commodify different elements of the streaming experience, from user attention and user data to content and social interactions. This section analyzes the major monetization approaches, highlighting how they work on platforms like TikTok, Twitch, Instagram, YouTube, Spotify, and others.

Mobile and Locative Aware Advertising

Advertising remains one of the most dominant revenue models for mobile streaming platforms, especially those offering free content such as TikTok, Instagram, YouTube, and Twitch (for non-subscribed viewers, though the ad-free experience has slowly been deteriorating even for subscribed viewers). The logic of advertising-based platforms is a direct extension of mass media models from television and radio: Build a large audience and sell access to that audience's attention to advertisers. However, digital streaming platforms have turbocharged this model by leveraging personal data to target ads with precision and by reaching audiences on a one-to-one basis through their personal devices, providing a clear example of mobile personalization being used to maximize effectiveness. The result is a highly profitable system that, as noted earlier, treats user attention as a commodity to be sold. Notably, this personalized ad model spans across platforms, illustrating how a common monetization logic underpins diverse services rather than a single streaming industry. Dallas Smythe's critical insight on the subject is particularly apt: "The audience is the commodity that is bought

and sold and exploited to produce surplus value for advertisers and media companies.”¹⁰ In the mobile streaming context, the audience commodity is fragmented and individualized. Each user is packaged with a profile of data about their likes, habits, and demographics, and advertisers purchase impressions (or influencer partnerships, etc.) aimed at those specific users. In packaging and selling audience data, platforms exercise considerable power over cultural consumption, effectively deciding which voices or trends get visibility based on what’s profitable.

YouTube stands as a prime example. The platform generates tens of billions of dollars in advertising revenue annually by running ads before, during, or alongside videos that billions of people watch daily. Google’s advertising machinery ensures that the ads are tailored. By this we mean that someone who has been watching a lot of cooking videos might see ads for kitchen gadgets, while a heavy consumer of gaming videos might see ads for new video games. Google consistently derives the majority of its income from advertising.¹¹ This heavy dependence makes YouTube very much an advertising platform, meaning its primary business is extracting data and selling ad space. To keep this engine running, YouTube’s algorithms are tuned to maximize watch time and clicks, which in turn maximizes ad exposure. This has raised well-documented concerns about sensational or extremist content being algorithmically promoted—not out of ideological intent, but as a side effect of chasing engagement, which yields more ad impressions.¹² As such, the advertising model can skew what content gets produced and recommended, since creators often chase algorithmic trends to earn advertising dollars or to maintain visibility if the algorithm favors certain content that retains viewers.

Mobile tracking and location awareness are increasingly crucial components to the targeted ad machine. In 2017 US marketers spent more than \$16 billion on targeted mobile advertisements, with a projected growth to \$38.7 billion by 2022.¹³ Location-targeted advertisements have existed in various forms for years, as Facebook offered location-based advertisements for business beginning in 2014. This location-based targeting works through

¹⁰ Dallas W. Smythe, “Communications: Blindspot of Western Marxism,” *Canadian Journal of Political and Social Theory* 1, no. 3 (1977).

¹¹ “Earnings 2025,” Investor Relations, Alphabet Inc., accessed May 12, 2025.

¹² Alex Russell, “YouTube Video Recommendations Lead to More Extremist Content for Right-Leaning Users, Researchers Suggest,” *In Focus* (blog), UC Davis, December 13, 2023.

¹³ Janelle Nanos, “How Companies Use Geolocation Data to Target You,” *The Boston Globe*, 2017.

location sharing services that are built into certain phone features. Businesses provide advertising companies like Meta and Google with their address and preferred geographic radius, and these platforms, in turn, provide businesses with the ability to refine the search to target certain demographics.¹⁴ Though not exclusive to mobile streaming, forms of located-based advertisement are certainly more relevant to mobile devices than fixed, in that advertisements are able to “follow” the user through space. To this end, targeted streaming advertisement offers yet another critical evolution for mobile streaming practices. Put another way, platforms have utilized experiments in mobile streaming to capture user attention through emphasis on sociability, but the goal of these developments resides in strategies of monetization.

Critical to mobile models of advertisement-based video-on-demand is the collection of location, device, and geographic data. Global positioning data can pinpoint user locations, but location-based advertisement accuracy is often limited. However, not all location-aware applications function the same way. Some applications (such as weather apps) only take a snapshot of user location data, while others (including most streaming apps) require more frequently updating user data. For many streaming providers, this geolocative data is necessary to determine whether content can be viewed in that area; streaming platforms like Netflix have different content licenses in various countries, while streaming sports providers often must comply with local regulations and blackouts.

For social media streaming, location data is mainly used for advertising purposes. Potentially troubling is that users are often unable to opt out of disclosing their location, as is the case with Facebook.¹⁵ Early studies of GPS-enabled location-based advertisement noted that most users found the advertisements useful and were unconcerned or unaware of potential privacy issues accompanying location-data collection.¹⁶ Studies show that users believe targeted ads based on fine-grained data collection is incredibly invasive.¹⁷ A 2019 Pew Research Center study similarly found that 81% of US adults believed they had little control over their data, and 79%

¹⁴ Zach Miners, “Facebook’s Location-Based Ads Pitch You on the Store You Just Walked By,” *PC World*, October 7, 2014.

¹⁵ Alex Hern, “Facebook Users Cannot Avoid Location-Based Ads, Investigation Finds,” *Guardian*, December 19, 2018.

¹⁶ Eija Kaasinen, “User Needs for Location-Aware Mobile Services,” *Personal and Ubiquitous Computing* 7, no. 1 (2003).

¹⁷ Nicolette Conti et al., “When Did My Mobile Turn into a ‘Sellphone’?: A Study of Consumer Responses to Tailored Smartphone Ads,” in *Proceedings of the 26th Annual BCS Interaction Specialist Group Conference on People and Computers* (BCS, 2012).

were concerned about how companies were using data collected about their activities, with 72% reporting that they feel that almost all their mobile internet activity was being tracked.¹⁸ Taken together, these findings echo the claim that, for users, when context-aware advertising is made more tangible via smartphones and mobile devices, it is often perceived as an breach of personal space.¹⁹ Consequently, location-aware advertising—including new mobile streaming forms—are often perceived as an invasive threat to privacy for users.

Mobile-oriented platforms like TikTok and Instagram Reels similarly rely on advertising, particularly as they have rapidly grown user bases. TikTok's monetization initially lagged its user growth, but by the mid-2020s it had developed a robust advertising platform. Brands can pay for in-feed video ads that appear as one scrolls, which are often made to look like regular TikToks, or for hashtag challenges and other sponsored trends. The appeal for advertisers is the extremely high engagement and the algorithm's ability to pinpoint niche communities. TikTok collects detailed data on what each user watches, likes, or skips, enabling fine-grained targeting. While much of TikTok's algorithm is proprietary and has not been released, users have deduced that captions, lighting, and other details within the video itself also play a role in the algorithm. In effect, TikTok sells access to moments of user attention when they are most immersed. For instance, an ad might be inserted after a particularly engaging video when the user is likely to continue swiping. Instagram, integrated with Facebook's ad system, offers targeted advertising in Reels and Stories, benefiting from Meta's vast trove of user data across Facebook/Instagram/WhatsApp. Instagram's ad business is a cornerstone of Meta's revenues.²⁰

Advertising in live streaming (Twitch) and music streaming also deserves mention. Twitch inserts video or banner ads on streams, though it has a somewhat more complex relationship with ads because excessive ads can drive viewers away from a live experience, in that missing moments in a live broadcast due to an ad is frustrating for many viewers. Twitch has tried to balance this with features like picture-in-picture ads that run a small ad while keeping the stream visible. Nonetheless, advertising is a significant

¹⁸ Brooke Auxier et al., "Americans and Privacy: Concerned, Confused and Feeling Lack of Control over Their Personal Information," Pew Research Center, November 15, 2019.

¹⁹ Eric Gordon and Adriana de Souza e Silva, *Net Locality: Why Location Matters in a Networked World* (Wiley-Blackwell, 2011).

²⁰ Akash Sriram, "Instagram to Make Up More Than Half of Meta's US Ad Revenue in 2025, Report Shows," *Reuters*, December 18, 2024.

revenue component for Twitch, and it has recently pushed streamers to run more ad breaks by offering higher revenue share incentives.²¹ For a 2019 event called Twitch Sells Out: A Prime Day Special Event, the Amazon-owned streaming platform recruited streaming celebrities for a two-day live shopping showcase for products on sale during Amazon's Prime Day. Twitch derives a substantial percentage of views from mobile users. It is unsurprising, then, that Amazon would utilize Twitch for advertising and cross-promotion, as surveys have found that mobile gamers are particularly receptive to advertising about products they might want.²² Embedded mobile ads have become so prevalent that they have altered the length of content itself. A few years ago, YouTube videos began to increase average run time to facilitate more ads per video. This occurred despite user preferences for shorter content. Put another way, the inbuilt goal of advertising and monetization to YouTube's algorithm was, at least for a while, powerful enough to override human preferences for shorter content and persuade streamers to produce longer-form videos, even in an increasingly mobile environment.

On music platforms, advertising is typically used in a freemium model: For example, Spotify's free tier plays ads periodically, nudging users to pay for premium to remove ads. Apple Music notably eschews ads entirely in favor of a pure subscription model. Apple uses the absence of ads as a selling point and, likely, also as a way to differentiate itself as a more privacy-conscious service, aligned with Apple's device sales strategy. The absence or presence of advertising can shape user experience significantly. Ad-based services bombard users with commercial messages, whereas subscription ones promise uninterrupted content for a fee. As we'll see, many platforms now mix both models.

Mobile Subscription and Premium Services

The second major monetization strategy is subscriptions. Subscription-based monetization offers an alternative or complementary approach to advertising. Under a subscription model, users pay a recurring fee for access to content, often with added perks like no ads or exclusive material. This model has deep roots in older media, such as cable TV packages, premium channels, subscription newspapers, etc., but has been revitalized by streaming, partly

²¹ Taylor Hatmaker, "Twitch Plans to Cut Subscription Revenue for Some Top Streamers in Push Toward Ads," *TechCrunch*, September 21, 2022.

²² Robert Williams, "Twitch Readies 2-day Live Shopping Show for Amazon Prime Day," *Mobile Marketer*, July 1, 2019.

as a response to consumer fatigue with ads and the desire for a predictable revenue stream for companies.

Apple Music is a prime example of a pure subscription model in mobile streaming. Launched in 2015, it quickly grew to tens of millions of subscribers, reaching around 88 to 93 million subscribers by 2023,²³ making it the second largest music streaming service after Spotify. Apple Music does not offer a free tier aside from time-limited trials. In return, subscribers get an ad-free experience and a comprehensive catalog of music. Apple Music deliberately ties into Apple's broader strategy: The service reinforces the value of Apple's hardware ecosystem (phones, earbuds, operating system, etc.), and it generates recurring revenue that complements hardware sales. Even in subscription services, mobile personalization plays a key role. This is similar to Netflix's tailored recommendations or Spotify's custom playlists that add value for paying users by catering to individual tastes. Apple can afford not to chase advertising dollars here because its primary business is selling devices and services as a package. Indeed, Apple has become known for using one arm of the business to bolster another. By providing appealing services like music (or shows via Apple TV+), Apple adds value to its iPhones and retains customers within its closed ecosystem, even if these services largely break-even or even operate at a loss. It then profits when those customers buy the next device or additional paid services. This is a strategic use of subscription content to support hardware sales, a different approach from companies that rely solely on content for profit. In an age of platform mobility, Apple very well may be the last company to be able to offer a model such as this. The legacy and strong branding/marketing campaigns of Apple have cultivated a very niche and devoted audience.

YouTube has blended subscription with its primarily ad model by introducing YouTube Premium, formally known as YouTube Red. For a monthly fee, YouTube Premium offers an ad-free viewing experience, background play for videos on mobile, allowing continued audio while using other apps, and access to YouTube Music Premium. While the majority of YouTube's billions of users remain on the free, ad-supported tier, the premium option caters to a segment of users who are willing to pay for convenience and no ads. This dual approach lets YouTube monetize both groups: Ad-tolerant users generate advertising revenue, and ad-averse power-users generate subscription revenue. Similarly, Twitch employs Twitch Turbo for an ad-free experience and encourages viewers to subscribe to individual channels (more on that

²³ Joseph D'Souza, "Apple Music Statistics and Facts (2025)," *Electro IQ*, March 17, 2025.

in the next section), thereby mixing ad and subscription income. The move to incorporate subscriptions reflects an understanding that not all revenue should be left on the table for the sake of scale; some users can be “upsold” to higher-margin direct payments. It’s a diversification that can make the platform less dependent on fluctuations in ad markets.

From the user viewpoint, subscriptions can be both enabling and limiting. On one hand, they free the user from advertising and often provide higher-quality service, such as higher audio quality on music services and exclusive content on some video platforms. On the other hand, too many subscriptions can lead to subscription fatigue. As more platforms partition content behind paywalls, users must decide which ones are worth paying for. In the realm of streaming video, we see an analogous fragmentation with Netflix, Disney+, Amazon Prime Video, etc., though in mobile-centric platforms the fragmentation is more along functional lines. In other words, music versus video versus gaming content. This can potentially drive some users towards unofficial or pirated content if the paywalls become too high, but in practice many have shown willingness to pay for convenience and high quality, as evidenced by the growth of services like Netflix or Spotify.

For the platforms, a steady subscription base is appealing because it provides predictable revenue and reduces reliance on the volatile advertising market, which can fluctuate with economic conditions. However, to attract and retain subscribers, platforms must consistently offer value through original content, better features, or an ecosystem of benefits. Amazon is a prime example of this, as they bundle Prime Video with free shipping and other perks under a single Amazon Prime subscription. This creates a pressure to continually invest in content and features, which is why we see, for example, YouTube investing in original shows to entice Premium subscribers, or Apple bundling Apple Music in discounted bundles with other Apple services like Apple One.

Subscription models may seem more straightforward than advertising, but they still involve power dynamics and capitalizing on consumer behavior. Companies will price discriminate by offering student or family plans to maximize sign-ups and they will engage in lock-in strategies, such as playlists or content libraries that are hard to transfer if one leaves a service. And even if they are not selling ads, they will still collect data on subscribers to inform future offerings. Moreover, subscription services can concentrate cultural power. For instance, if certain music is only on Apple Music due to exclusive deals, that gives Apple gatekeeping power over that content.

Generally, though, subscription in mobile streaming coexists with free options, forming a tiered ecosystem of monetization. This model is widespread across streaming sectors: Video platforms like Netflix or Disney+ thrive on subscriptions, while music services like Spotify blend free and premium tiers, and even social media (e.g., YouTube Premium) uses subscriptions to upsell an enhanced experience.

Microtransactions and the Creator Economy

A third monetization strategy revolves around microtransactions and virtual gifting, central to the creator economy. Mobile streaming platforms, especially those centered around live content like Twitch, TikTok, YouTube Live, and Instagram Live, have embraced microtransactions and virtual gifting as core monetization strategies. These systems are central to what's now called the creator economy, where individuals earn income from their content, though often on terms heavily dictated by the platform.

For example, in addition to allowing users to add links to advertised products and services in their bios and posts, TikTok has also added the ability for audiences to purchase products featured in videos directly via the video.²⁴ Virtual gifts and in-app currencies let viewers spend real money to purchase coins, badges, or animated icons to send during live streams. TikTok coins, YouTube's Super Chat, Instagram's badges, and the aforementioned Bits on Twitch are all examples. These interactions serve dual purposes. For the creator, they reward and boost visibility or recognition for the viewer for donating to the creator. From the platform's perspective, this is quite lucrative as it monetizes user-to-user engagement while taking a sizable cut of each transaction. Importantly, this setup relies on parasocial relationships and fans paying for attention, validation, or affiliation. Crucially, the highly personalized nature of users' content feeds, particularly the constant surfacing of favorite creators, helps cultivate the kind of loyal engagement that fuels these microtransactions. This is another facet of mobile personalization as each user's feed is tuned to strengthen the parasocial bond that platforms then monetize.

Subscriptions to creators further expand the model. Twitch pioneered this approach, letting viewers subscribe monthly for perks like custom emojis and subscriber-only chat. YouTube and TikTok have since followed with their own versions. These subs are marketed as direct support for creators,

²⁴ Ivan Mehta, "TikTok Shop Officially Launches in the US," *TechCrunch*, September 12, 2023.

but revenue splits (often 50/50) and policy changes make it precarious. Even top streamers face sudden changes, like Twitch's decision to roll back 70/30 splits for high earners, citing infrastructure costs and increased ad revenue as compensation.

Social commerce and affiliate integration are also on the rise. Platforms like TikTok and Instagram now let creators tag products in content, turning livestreams or short videos into shoppable ads. The platform typically takes a cut, reinforcing a model where content, commerce, and monetization blur together. This strategy, sometimes called *shoppertainment*, is especially prevalent in live shopping streams.

These forms of monetization signal a shift. Instead of only selling audience attention to advertisers, platforms now monetize individual fan engagement and creator labor directly. Virtual currencies play a key role here. They introduce a layer of abstraction—users convert money into coins, which can't usually be withdrawn—encouraging spending and locking value into the ecosystem. In some ways, this resembles scrip systems which harken back to coal mine towns in the early eighteenth and nineteenth centuries, reinforcing platform dependency. Though framed as empowering for creators, these systems carry deep inequalities. Small numbers of creators earn a living; most do not. The labor of streaming, posting, and cultivating an audience generates significant platform value, but creators remain subject to algorithm shifts, policy changes, and commission structures which they do not control. This situation demonstrates a clear power imbalance. While individual creators are dependent on the platform's algorithms and policies, the platform wields structural power over who can monetize and how. The result is a precarious labor economy where platforms extract value from both sides, content creation and audience engagement, while retaining power over the entire system.

Monetized Cross-Subsidies

A fourth layer to monetization is how mobile streaming platforms often are part of larger corporate ecosystems, and how cross-subsidy strategies are used. We've touched on this with Apple (using services to bolster device sales) and can see it with others:

Amazon acquired Twitch in 2014 and integrated it into its Prime membership. Prime subscribers get one free Twitch channel subscription per month, which Amazon pays the streamer as if it were a normal subscription. This effectively means Amazon is subsidizing Twitch streamers to add value to

Prime, which is mainly justified by other revenue. Similarly, Amazon might accept lower margins on Twitch because Twitch runs on AWS, potentially driving business to AWS or benefiting from internal cost offsets. It's an example of using one platform's strength to support another. In other words, Amazon's retail profits and cloud profits can subsidize Twitch's growth.

Google distributes Android freely (open-source) and many of its services for free to ensure its presence on mobile—a strategy explicitly designed to undercut Apple and make Google the gatekeeper on mobile. In return, Google reaps massive advertising revenue from the mobile usage of YouTube, search results, and Android usage, for example. It's a cross-subsidy where a free mobile OS leads to more mobile internet usage, which leads to more searches and video views that Google can monetize. In a similar vein, YouTube itself is sometimes seen as subsidized by Google's deep pockets. For years YouTube operated at a financial loss or slim margin due to infrastructure costs and efforts to grow, but Google's other revenue streams allowed it to keep investing until YouTube became profitable. The free or inexpensive parts lure users in; the monetized parts extract the value.

These cases show that “streaming” isn't a standalone industry at all, but an infrastructural logic entwined with other industries (e.g., e-commerce, hardware, cloud services) as tech giants leverage one arm of their business to boost another.

In some arenas, particularly music streaming, content is seen as a loss leader. In the music streaming space, companies like Amazon and Apple use music as part of a larger strategy—i.e., to boost sales of Echo speakers and iPhones or encourage Prime subscriptions. They might have lower margins on music rights compared to a standalone company like Spotify, but they don't mind as much because the goal is an ecosystem lock-in. For Apple, someone on Apple Music might be less likely to switch to Android because their playlists and familiarity are in Apple's walled garden. For Amazon, offering Prime Music is a value-add to keep Prime subscriptions high. These indirect monetization effects are harder to quantify but are crucial in understanding why tech giants all want a piece of streaming media. It's not only about direct revenue; it's also about strategic control of user engagement, which can be turned into various forms of value.

By employing cross-subsidies, big platforms can outcompete smaller ones that do not have alternate revenue streams. This raises barriers to entry. For instance, a startup that makes a streaming app might struggle to monetize if users now expect free access because incumbents offer content

free, perhaps even despite knowing they monetize elsewhere. Once drawn in, the platform can monetize them in the domain where it has strength. This is why we often see tech giants entering each other's territories. Google in music and video, Amazon in video, Facebook in e-commerce, etc., each may offer something at low cost because they have another way to profit after capturing the user. This can lead to predatory pricing or unfair competition debates. For example, Spotify has argued that Apple's dual role as platform owner and competitor service is unfair, since Apple can afford to take a loss on music streaming which Spotify cannot.²⁵

In the contemporary mobile streaming ecosystem, monetization is multifaceted and constantly evolving. By the time you read this book, there is a high likelihood that these models have transformed in some way or new models have been introduced. Platforms rarely rely on a single source of revenue; rather, they build hybrid models to maximize profit: Ad-supported free usage casts the widest net for users and monetizes through selling attention. Premium subscriptions convert a fraction of users into direct payers for enhanced experience. Microtransactions and virtual economies tap into social and community dynamics to generate revenue from user interactions. Data-driven strategies improve all of the above and open potential for indirect monetization. Integration with commerce turns platforms into marketplaces (e.g., Instagram Shopping, TikTok integrating retail), further diversifying income. Ecosystem leverage and subsidies ensure that even if one service isn't highly profitable, it serves a strategic role in a larger profitability picture.

For users, this means their experience is often segmented. One can choose to "pay with attention" (watch ads) or pay with money for convenience; one can use the service freely but have one's data heavily mined or pay a premium that perhaps involves slightly less data exploitation (though the protections provided to such subscribers are incomplete at best). For creators, multiple avenues exist to earn (ad revenue shares, fan subscriptions, sponsorship deals facilitated by their fame on the platform), but all paths keep creators dependent on the platform's rules and revenue sharing frameworks.

From a critical perspective, one might note that these monetization methods rely on commodifying not just content but also audiences, data, and social interactions. The user base is segmented and sold to advertisers; user data is packaged into marketable insights; even the act of connecting with or influencing others becomes a revenue-generating service. In effect, mobile streaming platforms continually create new commodity forms out

²⁵ "The Case," *Time to Play Fair* (website), Spotify, accessed May 12, 2025.

of digital life. Whether through selling access to attention, charging for enhanced experiences, or skimming off user-to-user payments, platforms relentlessly seek to extract value from every moment of engagement. All of these strategies underscore a fundamental imperative of platform capitalism: *to leave no user activity unmonetized.*

APIs and Platform Capture

Lastly, beyond user-facing revenue models, platforms monetize through controlling technical integrations. This is what we call APIs and platform capture. Mobile streaming platforms do not exist in isolation. They function within broader platform ecosystems and often interconnect (or deliberately don't interconnect) with other services. The degree of integration versus isolation is a strategic choice that significantly shapes user experience and monetization opportunities. APIs, or application programming interfaces, are a key technical mechanism in this context, as they determine how easily different services can communicate and build upon each other. A deeper look at APIs and their integrations reveal how platforms balance openness, to encourage growth and use of their platform, with closure, to maintain control and exclusive benefits. It also highlights how big platforms leverage their ecosystems to fortify their market position, creating ecosystems that lock users in. While the previous sections focused on direct revenue streams, this section examines how controlling the very connections between services (via APIs) translates into monetization and market power.

Many mobile streaming platforms make use of integration to broaden their reach. YouTube, for example, provides embedding APIs that allow its videos to be played on countless websites and apps outside YouTube itself. This openness was instrumental in YouTube's early dominance. Users could encounter YouTube content on blogs, news sites, social media feeds, etc., with YouTube reaping the benefits of increased views and data collection and ads, if served. Similarly, Spotify and Apple Music have APIs that let other apps fetch song information or integrate music playback. For instance, Instagram allows linking a song snippet to a Story via Spotify. These integrations enhance user convenience and propagate the platform's presence across digital spaces. They also generate real or imagined network effects: If your favorite fitness app integrates with Spotify for workout playlists, you are more likely to subscribe to Spotify, for instance.

However, with greater maturity, platforms often turn toward closure and exclusivity to cement their power. Many contemporary strategies involve

making the platform a one-stop-shop so that users have less reason to use anything else. Facebook's integration of multiple functions (messaging, video, shopping) into one app family is a case in point. In the mobile streaming realm, we see analogous moves. Instagram and Facebook heavily promote their own Reels feature to compete with TikTok, even allowing cross-posting of Reels between the two Meta-owned networks, but they do not make it easy to share that content to outside platforms. TikTok, while it allows sharing of video links, prefers to keep users inside its app where the "For You" feed can continue to serve videos. It is important to note that if you are to click a TikTok link without the app, it often prompts you to download the app. Furthermore, platforms can impose technical or policy barriers. Twitter historically provided APIs for third-party clients but later restricted them to force users back to the official app where ads could be served; similarly, Facebook cut off some API access (like friend graphs) to outsiders, citing privacy but also arguably to prevent competition.

APIs shape monetization in that they define who can innovate on top of the platform and where value flows. An open API can stimulate an ecosystem of complementary apps. For example, Twitch's API enables a whole market of chatbot tools, stream overlays, and extensions that enrich the Twitch experience, making Twitch more attractive to streamers and viewers, which indirectly boosts monetization for Twitch. But open APIs can also create vulnerabilities. If third-party apps can show content without ads or capture user attention elsewhere, the platform might lose revenue. This could be pointed to as one of the reasons why Twitch, for example, has stopped counting third-party integrations toward viewers in any given stream. As such, this could be why many platforms gradually tighten API accessibility as they scale. We can interpret this as platforms wanting to internalize all interactions so that any profit potential remains in-house. The rise of closed-API apps in mobile is a broad trend. In practice, this means that each major company tries to channel user activity into its own app, curtailing interoperability that might let a rival benefit.

Cross-platform integration also involves hardware and software coordination. A salient example is how certain services are tightly integrated with specific devices or operating systems. Apple leverages this with features like Handoff/Continuity that make it seamless to move from watching or listening on an iPhone to doing so on a Mac or iPad—but only for Apple's own services or approved partners. By making the experience smoother within an ecosystem than outside it, companies incentivize users to stay within

their suite of products. Another example is exclusive compatibility. Apple's AirPlay casting feature works more readily with Apple's own streaming services, while Google's Chromecast was originally optimized for YouTube and Netflix, and Google limited Amazon Prime Video's casting until business disputes were resolved.²⁶ These technical choices have economic motives, as they subtly push users towards certain content sources over others.

We also see strategic non-integration as a form of competition. A case in point is how Amazon and Google feuded for years, resulting in YouTube being unavailable on Amazon's Fire TV devices for a time, or Amazon refusing to sell Google's Chromecast on its store. Effectively, each company was trying to deny the other access to its user base in order to bolster its own platform's edge. Likewise, Tencent's and Alibaba's ecosystems in China historically blocked each other's links to keep users from migrating out (e.g., WeChat blocking Douyin [TikTok] links). These battles demonstrate that integration or its refusal can be a tool of power: Controlling points of connection is as important as controlling content.

For users, a high degree of integration can be a double-edged sword. Integration brings convenience: the ability to log into many apps with one account (your Google or Facebook login), to share content easily between services, or to have a unified experience across devices. But it can also lead to diminished choice and privacy. If one company's ecosystem handles everything, that company wields tremendous influence over the user's media intake and captures a comprehensive data profile. When all your social contacts, entertainment, and communication occupy one platform, the cost of leaving that platform (the switching cost) becomes very high. This lock-in is exactly what companies aim for, as it secures their user base against competitors. Indeed, a further downside of this model is that it can be anti-competitive and stifle innovation; new entrants might be shut out because the dominant platforms either don't allow interoperability or can copy and integrate any new feature as Instagram did in copying Snapchat's Stories, or YouTube in cloning TikTok's short-form vertical video format with "Shorts." In essence, large platforms use their integration capabilities to create a moat around their user base.

The role of APIs in monetization is also evident in how third-party developers are allowed to participate in the platform's economy. Platforms like Facebook and YouTube share some ad revenue or allow third parties to develop for the platform (as with Facebook's Instant Games or YouTube's

²⁶ "Amazon and Google Deliver on TV Peace Deal," *BBC News*, July 10, 2019.

partner program), but they keep a significant portion of revenue and full access to data. If a platform instead kept its API entirely closed, it might slow growth but ensure any monetizable extension of the platform is built internally (thus capturing all value). We can see mixed models: Apple's App Store is a kind of API that lets developers create apps for the iPhone—Apple takes a 30 percent cut of any digital sales and has final say on what apps appear. This has been immensely profitable for Apple and extends the iPhone's functionality, but Apple carefully polices this ecosystem to prevent apps that would substitute its own services—for example, restrictions on game streaming apps, as mentioned, or enforcing in-app purchase rules. More insidiously, Blizzard's World of Warcraft is known for taking popular third-party add-ons and integrating popular features into the game itself with no recognition or compensation to the original developer.

In summary, cross-platform integration and APIs are not merely technical matters; they are planned and intentional business decisions. They determine who gets to build on the platform and who reaps the benefits. Open integrations can drive growth and enrich user experience, but too much openness can undercut a platform's monetization or allow competitors in; thus, most platforms adopt a strategy of selective integration. They open up enough to become pervasive but not so much that they lose control of user data or revenue flows. As platforms mature, there is a clear trend toward asserting more control through strategies such as tightening API use, cloning external innovations in-house, and promoting closed-loop usage. The endgame for each major player is to have users living their digital lives within that platform's ecosystem as completely as possible—which is the ultimate form of monetization and power, as it means the platform mediates a large share of social and media experience and can channel those experiences into profit.

Conclusion

Mobile streaming has swiftly transformed the media landscape and woven itself into the daily lives of billions, enabling an always-on, personalized stream of content that flows across devices and borders. This mobile personalization of media, where content is tailored to users in real time, is not just a user-friendly feature but the very engine of monetization that we explored in this chapter. By examining this phenomenon, we discern that its story is not just about technology or entertainment; it is fundamentally about power, profit, and control in the digital age. Platforms like TikTok,

Twitch, Instagram Reels, YouTube, Apple Music, and others exemplify the new breed of media companies whose operations hinge on data-driven, networked capitalism. They have embraced platform mobility—making content and data accessible everywhere—not simply to please users but to maximize engagement and revenue. They have turned user activities and social interactions into commodities, capturing attention and data and converting them into advertising placements, subscription fees, or transaction cuts. In doing so, they blur the line between leisure and labor, consumption and production, as users become unwitting workers in the generation of the very data that the platforms monetize.

Throughout this chapter, we have seen that monetization strategies are at the heart of why mobile streaming looks and feels the way it does. The endless feeds, the algorithmic recommendations, the push notifications are all optimized to serve the business model; typically advertising or engagement-based sales. Even as new models emerge, the pull of commercial imperatives remains. This raises the question: *To what extent should our daily communication and entertainment spaces be governed by the pursuit of profit?* When virtually every form of expression, from a TikTok dance video to a heartfelt personal vlog, takes place on corporate platforms, those corporations set the rules of visibility, acceptable speech, and revenue sharing. This chapter has not focused on free speech or content moderation issues, but it is implicit that the power to monetize is also the power to moderate. Platforms, undoubtedly, will allow or disallow content based in part on what is good for business. Avoiding scandal, pleasing advertisers, and complying with laws to avoid shutdown are always part of the conversation, if not the conversation itself. Thus, mobile streaming is intertwined with questions of democracy and autonomy in the digital public sphere.

The story of contemporary mobile streaming is a story of pervasive connectivity tangled with commercial agendas. It is a story of how our most casual daily habits—checking an app during a commute, sharing a video clip with friends, listening to a favorite song on the phone—are embedded in a global network of economic transactions and power relations. Throughout this chapter, we hope to have peeled back the screen to reveal the currents of capital and control that flow beneath, shaping the infrastructure, content, and uses of mobile streaming. This understanding is crucial for scholars, policymakers, and citizens alike, as we grapple with ensuring that the networked digital media, which now mediates so much of our social life, serves the public good and not only corporate interests. Mobile streaming has

unleashed incredible creativity and convenience, but it has also entrenched new forms of corporate power. This isn't one singular industry's story but a transformation cutting across entertainment, communication, and tech industries via a unified platform logic. Negotiating that duality, harnessing the benefits of ubiquitous connectivity while reining in its exploitative excesses, will be a defining challenge for years to come. Ultimately, recognizing mobile personalization and monetization as instruments of infrastructural power is crucial if we are to challenge and balance the influence these platforms wield moving forward.

CHAPTER FOUR

The Characteristics of Mobile Streaming

Streaming audio-visual media is popularly tied to Web 2.0, the era known as the participatory web.¹ The mid-2000s were a time where what we'd now consider relatively rudimentary versions of some still-stalwart websites originated; platforms like Facebook, Twitter, Wikipedia, and YouTube made their debuts alongside less popular, shorter-lived platforms like Foursquare, Tumblr, and Skype. Netflix as a company predates Web 2.0, originating in 1997 as a mail-delivery DVD rental service. It was in 2007 that Netflix launched its now ubiquitous streaming platform, which, along with YouTube and a growing number of competitors, added streaming media to this changing digital landscape. Contemporarily, streaming audio-visual content has become intertwined with countless platforms and websites. If one could go back in time and use the past versions of these platforms, the social networks of the early participatory web would now seem immensely dull in a landscape of the endless scroll, swipe, and stream of content. Audio-visual content is embedded in just about any website one visits—even if it is in the form of irritating auto-play popup ads. In short, if the early 2000s internet is defined by participation and user-generated content, then the contemporary web is all about streaming. One major challenge of the proliferation of streaming is that the term itself begins to lose some conceptual specificity. That is, it was easier to define streaming media when the practice was limited to a few major providers like Netflix,

¹ Gerard Goggin, "Mobile Web 2.0: New Imaginaries of Mobile Internet," in *Theories of the Mobile Internet: Materialities and Imaginaries*, ed. Andrew Herman, Jan Hadlaw, and Thom Swiss (Routledge, 2015); Tim O'Reilly, "What Is Web 2.0: Design Patterns and Business Models for the Next Generation of Software," *Communications and Strategies*, no. 65 (2007).

Hulu, YouTube, etc. Now that streaming has become entangled and embedded with all types of sites, platforms, and mobile devices, the concept itself becomes opaquer.

We've made the case for a more expansive history of mobile streaming media, what we call mobile personalization, in chapter 1 in tracing some audio-visual antecedent technologies that provide a mobile and portable orientation to entertainment, information, and monetization. In a narrower and more linear sense, streaming is often connected to the first livestreamed radio broadcast by ESPN SportsZone of a baseball game on September 5, 1995.² This broadcast was audio-only, yet this event marked the beginning of a series of legal battles for ownership of streaming media technologies. Streaming functions much like loading a website, in that the process does not involve a permanent download. We engage online through what's known as packet switching, in which a message is broken into smaller parts for transfer then reassembled at the destination.³ Packet switching works on the distributed network of the internet because packets can take the quickest of multiple routes to their destination, which allows the network increased resilience. TCP/IP protocols establish the rules of connection on the web, creating a form of standardization in the way packets are transmitted, routed, and received.⁴ While packet switching functions because packets can arrive in any order, streaming media requires a temporal continuity. Streaming video, then, utilizes additional protocols to ensure packets arrive and are assembled in the correct order, including real-time transfer protocol (RTP), real-time streaming protocol (RTSP), and real-time transport control protocol (RTCP), and HTTP-based adaptive streaming.⁵ Collectively, one way to define streaming is strictly technological, in that audio-visual streaming is based off layered transfer technologies and protocols.

Of course, we've worked to establish throughout this book that streaming is not just technical but sociotechnical; it is more than merely platforms, infrastructures, code, and devices. Our mobile personalization approach highlights how streaming is a major shift in how we watch, what we watch, and how we relate and connect to one another. It's also tied to major indus-

² Alex Zambelli, "A History of Media Streaming and the Future of Connected TV," *Guardian*, March 1, 2013.

³ Katie Hafner and Matthew Lyon, *Where Wizards Stay up Late: The Origins of the Internet* (Simon & Schuster, 1996); Janet Abbate, *Inventing the Internet* (MIT Press, 1999), 34.

⁴ Alexander R. Galloway, *Protocol: How Control Exists After Decentralization* (MIT Press, 2004), 8.

⁵ Chris Pollette, "How Streaming Video and Audio Work," *How Stuff Works*, May 1, 2024.

try shifts, wherein a mix of long-tail niche markets, digital piracy, changes in ownership by artists and creators, increased power of distributors, and data-driven infrastructures came together to disrupt traditional content distribution models for music, film, and television alike.⁶ In essence, defining streaming is a challenge. As explained by Ghislain Thibault, “For the television industry and television audiences alike, the term ‘streaming’ seems to mark the transition from analogue to digital television and the new types of small-screen consumption that accompany it.”⁷ Other scholarship has attempted to address what exactly is novel (or remediated) about this distribution model. Some focus on changing notions of live content, or the altered shared experience formerly felt in mass media. Others emphasize the dimension of time, in how streaming permits what’s known as time-shifting to alter rhythms of traditional viewing to fit the user. When it comes to new forms of mobile streaming, scholarship on hybridity of the physical, digital, and malleable spatial and temporal nature of screens is more relevant than ever. And it’s impossible to leave out the elements of data collection and algorithmic recommendation—for fixed or mobile streaming.

Existing scholarly accounts that attempt to make sense of the socio-technical shifts of streaming media fall back on how the medium alters or functions as a continuation of traditional audio-visual media habits. And that makes sense. Streaming in the era of Web 2.0 often took place in the fixed site of the living room on a web-connected television. Accordingly, streaming initially functioned as a direct parallel with the televisual. But this these practices have rapidly changed. Scholarship about streaming as spatially fixed has provided excellent conceptual anchors, yet it is critical to reassess and reconceptualize these attributes in the era of mobile streaming. In this final chapter, we set out to define the key characteristics of mobile streaming. In doing so, we engage with the question of how mobile streaming remediates prior understandings of flow, liveness and sharedness, screens and space, mobile and hybrid, and algorithms and big data. These concepts are all critical to understanding how mobile streaming is part of a longstanding trajectory of social, technical, and infrastructural shifts.

⁶ Michael D. Smith and Rahul Telang, *Streaming, Sharing, Stealing: Big Data and the Future of Entertainment* (MIT Press, 2016), 56.

⁷ Ghislain Thibault, “Streaming: A Media Hydrography of Televisual Flows,” *View* 4, no. 7 (2015): 110.

Mobile Streaming Flow

In 2016, the social media platform Instagram introduced “Instagram Stories,” a feature that allows users to share photos and videos in slideshow format. In contrast to the other content posted to Instagram, stories create a sequence that users can add to over the course of a day by using text and drawing tools to create overlap. As the user selects the story icon in the Instagram app, stories created by individuals, celebrities, organizations, and brands the user follows on Instagram are shown in unbroken succession, occasionally interspersed with advertisements based on Instagram’s location-aware model of data collection. As these stories are updated throughout the day, users are shown only new content that is added (and of course, new advertisements).

There is an inherent ephemerality to Instagram stories, as this content “disappears”⁸ after twenty-four hours, no longer accessible to other platform users. Instagram’s story feature is not necessarily a wholly original concept: Social media platforms such as Vine (a short-form video platform popular from 2012–2016) and Snapchat (a social media platform created in 2011 that allows users to share story-type content that similarly vanishes after a single viewing) provided the model for Instagram to appropriate in its own platform. At the same, what is relevant about Instagram’s co-option of the story feature is the platform’s reach and popularity. Instagram is among the top five most popular social media platforms in the world in terms of monthly users.⁹ The success of Instagram stories seems to coincide with users leaving Snapchat.¹⁰ Facebook’s billion-dollar acquisition of Instagram in 2012 is part of this platform recentralization as well, in that users can cross-share their stories to both social media sites.

Instagram stories are not “streaming media” in the colloquial sense, in that they are not the same as the streaming audio-visual media often positioned as part of a film and televisual lineage. Yet, simultaneously, these stories create a streaming “flow” that is influenced by antecedent communication technologies. Now, a few years later, Instagram has launched “Reels,” a direct short-form streaming competitor to TikTok. In a matter of fourteen years, Instagram morphed from a photo image-based platform to a streaming

⁸ It is well established that social media platforms collect and store even temporary and deleted content. Many, like Facebook and Instagram, store this content permanently.

⁹ “Most Popular Social Networks Worldwide as of February 2025, by Number of Monthly Active Users,” *Statista*, February 2025.

¹⁰ Sara Salinas, “Peak Social? The Major Social Platforms are Showing a Significant Slow-down in Users,” *CNBC*, August 8, 2018.

one. In essence, the platform, like many others, has attempted to recapture and reimagine flow.

Here, the work of Raymond Williams yet again functions as a conceptual mooring in considering the lineage of our contemporary streaming landscape. Williams explains that screen size, interruptions, participation, and a sequence of rhythms were all aspects of prior media forms remediated by television.¹¹ Pre-televisual habits and vocabulary, nevertheless, were shaped by the experience of discrete, isolated, and temporary events.¹² It is the concept of flow—how the sequence of program content and commercials create the real and unbroken experience of watching television as unbroken broadcast—that marked a critical departure for this cultural technology. The experience of television, then, captures viewers through this flow and readies them for the planned interruptions of commercials. Flow is a question of power via altered temporal rhythms and a capture of attention. This concept is something that is very much alive in mobile streaming as well.

That being said, flow is an inherently slippery concept—something that's perhaps more sensed and felt than measured. Flow explains the difference between watching a play and the novel experience of the televisual; for example, while a play has a planned intermission that functions to break up content, the televisual instead operates through an industrialized organizational schema in which planned interruptions create an unbroken experience of watching content. We could perhaps quantify flow by studying the temporal patterns of content and advertising over time. What's harder to capture, however, is the massive experiential change that accompanied the arrival of televisual broadcasting and the decades that followed. And the novel forms of mobile streaming add in aspects that were completely foreign to the early televisual era; location-awareness, algorithms, direct user participation and curation, temporal duration, and mobility remediate the familiar experience of flow. Perhaps one of the reasons that flow as a concept endures despite these changes is that Williams himself conceived of flow as flexible and mutable, in that he believed television itself would be subject to technological convergence.

Scholars have taken heed and updated Williams's enduring concept. Williams conceived of flow as a more passive process for the viewer, whereas streaming often requires a push-pull between suggested content and a more

¹¹ Raymond Williams, *Television: Technology and Cultural Form* (Fontana, 1974), 87–88.

¹² Williams, *Television, Technology and Cultural Form*, 87–88.

active selection.¹³ Stalwart televisual-style streaming platforms like Netflix function by promoting seemingly limitless choice for audiences, while at the same time narrowing these options via algorithmic and data-driven personalization. Netflix has experimented with automatically playing content (either when users hover over a choice or when they finish episodes). Binge-watching, the act of watching multiple episodes in a row of unbroken content, was a defining quality of the early turn toward audio-visual streaming media. This push-pull between active and passive viewing also applies to data collection, including in the era of the “second screen.”¹⁴ Data collection by streaming platforms entails both active and passive data—what we watch, for how long, where we watch, what device we use, when we pause or give up on content, what we like, what we reshare—the list goes on. In essence, the passivity of Williams’s televisual flow is mediated and remediated by streaming and then more mobile orientations to the process.

While not specific to mobile streaming, Sheenagh Pietrobruno reexamines Williams’s flow in the context of YouTube.¹⁵ In an era of algorithmically suggested content, flow takes on a different meaning. That is, instead of a programmed flow driven by one-way dissemination of content that is directly controlled by broadcasters, YouTube and other streaming platforms do not determine exactly what content is viewed by users. This process instead functions as more of a “staccato flow,” wherein the user self-selects short video content that is shaped by underlying algorithms. YouTube’s algorithms—designed for monetization—can both reinforce dominant cultural narratives and enable alternative perspectives to circulate. Elements like audience engagement, video rankings, and metadata contribute to the visibility of varied cultural representations, indicating YouTube’s potential to both homogenize and expand the kind of content that circulates. Stated differently, flow is altered by streaming in multiple ways; the flow of content looks different, it feels different, and the patterns of “unbroken” content have changed as well. Perhaps most importantly, flow is no longer merely the purview of television producers, but instead algorithmically defined.

¹³ William Uricchio, “The Future of a Medium Once Known as Television,” in *The YouTube Reader*, ed. Pelle Snickars and Patrick Vonderau (National Library of Sweden, 2009), 32.

¹⁴ Hye Jin Lee and Mark Andrejevic, “Second-Screen Theory: From Democratic Surround to the Digital Enclosure,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 40.

¹⁵ Sheenagh Pietrobruno, “YouTube Flow and the Transmission of Heritage: The Interplay of Users, Content, and Algorithms,” *Convergence* 24, no. 6 (2018).

We've highlighted the importance of algorithms and big data to streaming in previous chapters of this book, and we'll come back to the significance of these concepts to mobile streaming in a few pages. Suffice it to say, other scholars have connected streaming media, algorithms, and flow. One such theorization comes from Christopher Cox, who defines the contemporary logics of streaming flow—the televisable and invisual.¹⁶ Televisable emphasizes the importance of search bars, menus, autoplay screens, and other platform elements that foster a sense of user control of streaming while downplaying the latent industrial control by platforms. The invisual refers to unseen functions like code, software, algorithms, and data that pull users toward what appears on their screen. Together, these concepts help complicate aforementioned issues of agency within streaming flow, indicating the complex weave of users, platforms, content, and technology that creates the experience of watching.

Flow is clearly a durable concept of the televisual that is remediated by streaming; but where does the mobile element fit in? Kathleen Oswald and Jeremy Packer update the concept of flow for a mobile, screen-saturated landscape, emphasizing how streaming complicates both televisual and filmic traditions, as well as recent studies of digital media. These scholars instead focus on the emergence of “tele-practices” enabled by shifting formats, devices, and mobilities, where content moves fluidly across multiple screens. In a web-connected world, flow no longer binds viewers to a single broadcast but instead captures desire through advertising and spatial-temporal consumption patterns.¹⁷ Screens structure movement and behavior, making individuals responsible for managing flow. Engaging with Williams's theorization of televisual flow—how disparate content forms a seamless narrative—the authors argue that contemporary flow operates across platforms, keeping subjects “fixed in transmission” rather than fixed on one transmission.¹⁸ What matters is less what we watch than the fact that mobile devices transform reality into a “magical space of mobile desire,” promising empowerment but also demanding self-regulation. Mobile screens are part of a reconfiguration of time and space, but these devices and their attendant practices create a flow in the form of governance.

¹⁶ Christopher M. Cox, “Programming—Flow in the Convergence of Digital Media Platforms and Television,” *Critical Studies in Television* 13, no. 4 (2018): 439.

¹⁷ Kathleen F. Oswald and Jeremy Packer, “Flow and Mobile Media: Broadcast Fixity to Digital Fluidity,” in *Communication Matters: Materialist Approaches to Media, Mobility, and Networks*, ed. Jeremy Packer and Stephen B. Crofts (Wiley (Routledge), 2010). 278.

¹⁸ Oswald and Packer, “Flow and Mobile Media,” 277.

The wealth of scholarship updating flow largely focuses on fixed streaming, whereas Oswald and Packer instead emphasize the importance of the materiality of screens. Though an interesting and generative account, missing from remediations of flow is the space in between—the practices of mobile streaming. Here again, mobile streaming is not simply technical. These practices certainly are undergirded by devices, infrastructures, and platforms. Yet, mobile streaming is also the content, the connectivity, the interaction, and the sociability of it all. The flow of the early three-channel televisual era is certainly not an experience we are able to go back to and reexperience now in an age of constant connection and information overload. Yet the affective dimension of capturing and keeping audience attention through an unbroken stream is, perhaps, more important than ever.

Flow is not scaffolded by screens; it is accessed in a variety of spaces and fills all sorts of time in between activities. We reach for our phones instinctively. And younger people are most likely to open platforms and begin watching flows of short video content. Flow is also algorithmic and data-driven, another remediated aspect of mobile streaming. In short, contemporary examples—from Instagram stories to TikTok, from popup video ads to embedded video on news sites, and from YouTube Shorts to streaming music platforms—all function as a remediated flow.

To summarize, flow—how the rhythms of content and planned interruptions create an unbroken organizational schema of watching—remains relevant to streaming and mobile streaming. Considering the multifarious, embedded, and ever-changing ways that streaming is now brought to us via our mobile web-connected devices, nevertheless, necessitates a reconsideration of flow. As previously mentioned, flow is something that is experienced. It's hard for someone born in the early 2000s to comprehend flow in the halcyon era of cable television, just as 90s kids had to trust our parents explaining what the early era of the televisual felt like. What is clear is that something is *different* now. Instead of a neatly planned corporate scheme, flow is now structured by algorithms. Instead of existing solely in the domestic site of the home, flow now fits into the cracks of our lives when we mechanically reach for our phones in moments of boredom. And instead of tuning in at set times, we now experience flow in an individual sense. Mobile flow is, accordingly, an essential element present in the televisual through the contemporary mobile landscape, even if it does appear and feel a bit different.

Seemingly at odds with this last point, nonetheless, is the communal element that has been reignited by the personalized and individualized nature of

mobile streaming. Elements of liveness and sharedness—long critical to the televisual—are remediated by streaming and then again by mobile streaming.

Liveness, Sharedness, and Mobile Streaming

The first academic work one of this book's authors ever published is a chapter about the changing experience of communal television in the nascent years of Netflix's streaming platform. Originally conceptualized in 2015, the chapter centers on the relationship that was at that time developing between streaming platforms, audiences, and shared online conversations about content. At the heart of this change was Netflix upending the traditional content release model by providing viewers "all at once" access to entire new seasons of the platform's original programming like *Arrested Development*, *House of Cards*, and *Orange Is the New Black*. This release model turbocharged the idea of binge-watching, with viewers rushing to finish new seasons faster than ever. Audiences seemed to both love and hate this schedule. On the one hand, there was no waiting a week in between new episodes; on the other hand, something seemed "lost" in the lack of anticipation. Websites that covered popular culture often lamented this model too. If one watched half the season the day it was released, were they "ahead" or "behind"? After all, is watching content all that fun if you can't talk about it with others?

Now over a decade later, the traditional televisual release model of weekly new episodes has largely returned. There is likely no single reason for this reversal. For platforms like Netflix, the positive buzz of all-at-once seasons was outweighed over time by the need to keep audiences paying for the platform month after month. Younger readers might not remember the brief period when cord-cutting cable could be achieved by a meager \$10 subscription. The contemporary proliferation of televisual and cinematic subscription platforms now feels like cable costs all over again, and legacy cable providers like Comcast, Spectrum, and Verizon offer their own streaming subscription platforms, joined by cable-like streaming platforms that include Fubo, Sling, YouTube TV, and Hulu Live. Releasing an anticipated new show in a single day means that platform audiences might sign up for a month and then cancel their subscription. From the platform provider perspective, it's better to string them along over ten weeks.

Just as importantly, it's worth considering if binge-watching new content is what audiences truly want. Much has been written about the death of monoculture, or the end of the golden and platinum eras of television.

But it's clear that even now, there is something to the shared experience of televisual-like content. Live sporting events are now viewed via the technical process of streaming, but the togetherness they provide never went away, even as formats changed. The slowed pace of televisual releases similarly works to give audiences a chance to engage in conversations, fan theories, and speculation each week. These conversations, once the purview of comment sections on pop culture think-piece and recap articles, have now migrated to short-form social media streaming videos and podcasts. Streaming about streaming. And that's before considering just how much viewing time of televisual and cinematic content has been outright replaced by streaming on social media platforms, accessed on mobile devices.

In essence, this situation speaks to concepts that are mediated and re-mediated by streaming and then mobile streaming: liveness and sharedness. Put differently, liveness and sharedness are at the elusive heart of streaming. Personalization is desirable for viewers, but too much personalization can be an isolating experience. Clearly, there is something to watching live (or some approximation thereof) and engaging with shared content. And this desire for live and shared connection is remediated by mobile streaming.

Historically, one-to-many communication models facilitated both asynchronous and synchronous interaction. Asynchronous media such as books, newspapers, and early websites allow users to engage with content at their own pace. In contrast, synchronous media such as broadcast radio, television, and cinema demand simultaneous engagement, underscoring the significance of temporal coordination.¹⁹ Within these synchronous formats, concepts such as liveness and sharedness emerge as fundamental to understanding media viewing and engagement patterns. Liveness, as conceptualized by Nick Couldry, plays a pivotal role in reinforcing social coordination around media “centers” that disseminate images, narratives, and information in near real-time.²⁰ While there are rival forms of liveness, especially when accounting for streaming media, Couldry argues that liveness—in the general sense of continuous connectedness—is likely to remain an essential aspect of new media. This is because media's role in shaping temporal and spatial organization of the world, along with notions of shared reality and what is important, is still valued by users.²¹

¹⁹ Klaus Bruhn Jensen and Rasmus Helles, “Speaking into the System: Social Media and Many-to-One Communication,” *European Journal of Communication* 32, no. 1 (2017): 17.

²⁰ Nick Couldry, “Liveness, ‘Reality,’ and the Mediated Habitus from Television to the Mobile Phone,” *The Communication Review* 7, no. 4 (2004): 353.

²¹ Couldry, “Liveness,” 357.

Film is an early visual (and then audio-visual) medium that allowed for one-to-many dissemination and shared experience via storage. That is, film is captured and stored for presentation at certain times in limited spaces of the theater. Liveness, in a televisual sense, is connected to notions of simultaneity and storage. But viewed retrospectively, there exist debates on exactly how “live” the televisual ever really was. For instance, William Uricchio contends that the televisual is dominated by a center to periphery model that focuses on either simultaneity (and liveness) or storage.²² While Uricchio believes simultaneity and storage become conflated via television, Doron Galili instead highlights that the desire for simultaneity and storage oscillates within the development of the televisual, as both are present and separate in a goal of “mastering time.”²³ Regardless of these details, what is clear is that liveness and sharedness have long been a critical aspect of audio-visual media. And elements of liveness and sharedness are deeply entangled with storage, space, and time.

The example of Netflix’s release model highlights a brief period of disrupted liveness and sharedness. As mentioned, more traditional televisual release patterns have seemingly returned, but not before some interesting experiments that highlight how streaming media converged with social media. One experimental feature that brought together the elements of location-awareness, liveness, sharedness, and plasticity is Facebook’s 2018 feature called Watch Party. Watch Party allowed Facebook group administrators to select a public video and share it to other group members to create a simultaneously streaming “shared viewing experience.”²⁴ According to Facebook Head of Video Fidi Simo, “Watching, commenting and reacting to the same moments together . . . helps build the kind of community and engagement we’ve seen with Live.”²⁵ Within Watch Party, viewer comments and emoji reactions are overlaid on the video stream, like the setup of mobile Facebook livestreams. This shift aligns with remarks from Facebook’s VP of News Feed, Adam Mosseri, who noted that publisher content distribution would likely decline as the platform reorients itself toward direct social engagement.

Over time, Facebook’s parent company Meta significantly increased its emphasis on live streaming as a core feature, largely in response to com-

²² Uricchio, “Television’s First Seventy-Five Years,” 287.

²³ Doron Galili, *Seeing by Electricity: The Emergence of Television and the Modern Mediascape, 1878–1939* (Duke University Press, 2020), 224.

²⁴ Josh Constine, “Facebook ‘Watch Party’ Lets Groups View Videos Simultaneously,” *TechCrunch*, January 17, 2018.

²⁵ Constine, “Facebook ‘Watch Party.’”

petitors. Although Facebook is not an exclusively mobile platform, trends indicate that mobile access constitutes its primary mode of engagement. This shift is exemplified by Facebook's 2012 acquisition of Instagram and the subsequent integration of Instagram Stories across both platforms, reflecting the company's strategic investment in mobile streaming technologies. One early and related example of this emphasis comes from Facebook's continued support of Watch Party, wherein the company began testing a simultaneous live viewing feature embedded within the Facebook Messenger just ten months after the rollout of Watch Party.²⁶ Yet again, this form of streaming engagement is not inherently mobile, but the fact that Facebook overhauled its mobile-oriented Messenger application to include Watch Party shows the increasingly mobile dimension to streaming media. Furthermore, such features emphasize not a death of liveness and sharedness, but new forms of mobile capture of the desire for collective temporal engagement. The Watch Party feature is one strategy to address concerns that the live experience of viewing would be lost via the advent of streaming.²⁷ Here, sharedness, liveness, mobility, and temporal flexibility meet via new streaming practices.

Research into streaming practices finds that despite the scheduling freedom and time-shifting potential that streaming media provides, viewers remain aware of the viewing habits and patterns of others.²⁸ Others similarly establish that streaming viewers still desire a sense of togetherness and develop new strategies for social cohesion despite disrupted viewing practices.²⁹ This so-called plasticity refers to the way that individuals accessing streaming media are able to fill the cracks of their leisure time through the malleable nature of streaming, illustrating how streaming media integrates seamlessly into the fragmented moments of daily life. Examples from the past decade validate these findings in action: The initial disruptions of streaming as something that lacked structured temporality and community have rubber banded back toward something that is both the same and different, particularly when factoring in new forms of mobile streaming. That is to say that while contemporary liveness and sharedness might not look like the nuclear family sitting together in the living room watching three-network television

²⁶ Mallory Locklear, "Facebook is Testing a Watch Party-Like Feature for Messenger," *Engadget*, November 16, 2018.

²⁷ Couldry, "Liveness"; Lilly Irani et al., "Rhythms and Plasticity: Television Temporality at Home," *Personal and Ubiquitous Computing* 14, no. 7 (2010).

²⁸ Irani et al., "Rhythms and Plasticity," 624.

²⁹ Yu-Kei Tse, "Television's Changing Role in Social Togetherness in the Personalized Online Consumption of Foreign TV," *New Media & Society* 18, no. 8 (2016): 1550.

options alongside millions of other households, we still cling to elements of shared experiences via what we stream. Well-articulated by Grame Turner, liveness and sharedness still exist “outside the box” of traditional television: “If liveness includes a sense of the shrinking temporal gap between oneself and the rest of the world, as well as a palpable sense of immediacy, then this is something we can find as readily online as in television.”³⁰ Accordingly, the contemporary landscape of always available mobile streaming connectivity only furthers this notion of liveness.

Streaming and mobile streaming practices are possible through arrangements of technologies, platforms, infrastructures, and individuals that allow temporal and spatial coordination via new forms of liveness and sharedness. Despite initial prognostications that streaming would lead to the death of liveness, along with other consequences for a shared sense of content, mobile streaming reveals how these concepts are more relevant than ever. As streaming has become entangled with social media, it has simultaneously become a means to view and listen to content on the go. And the elements of live viewing and shared experiences have come to form the bedrock of what it means to stream. TikTok and Instagram Reels, among other platforms, are exemplars of a new orientation to streaming that reactivates the televisual and cinematic theories of liveness and sharedness. In the next section, we discuss how screen and spatial scholarship is remediated by mobile streaming.

Screens, Space, Streaming

In past comments about Netflix, both CEO Reed Hastings and Chief Content Officer Ted Sarandos have commented on mobile viewing and screen size, explaining that the company has no plans to hunt for “shorter content or asking producers to change their shows to suit specific viewing situations.”³¹ Explicitly stating “We’re not trying to adapt to the size of a screen,” both figures acknowledged that young people tend to watch via mobile devices.³² However, Hastings was quick to note that users watch content across multiple screens, as the convenience of being able to pause and pick back up with content in another location via a different device is important. Just a few years later, comments in Netflix’s 2019 fourth-quarter earnings report

³⁰ Grame Turner, “‘Liveness’ and ‘Sharedness’ Outside the Box,” *Flow* (2011).

³¹ Chris O’Brien, “Netflix’s Reed Hastings and Ted Sarandos Explain Why Screen Size Doesn’t Impact Their Shows,” *Venture Beat*, May 4, 2016.

³² O’Brien, “Netflix’s Reed Hastings.”

stated that “We earn consumer screen time, both mobile and television, away from a very broad set of competitors. We compete with (and lose to) ‘Fortnite’ more than HBO.”³³ Fortnite is a massively popular online multiplayer survival game, available on a variety of devices that includes smartphones. The comment from Netflix articulating Fortnite as a direct competitor shows a breakdown between fixed and mobile, wherein the screen now occupies a variety of spaces beyond the living room. Better understanding this contemporary moment of mobile streaming necessitates a focus on two interrelated aspects of scholarship: space and screens.

Prior to what’s known as the spatial turn of the 1970s and 80s, the concept of space was subordinated to a focus on time; space was often treated as dead, fixed, and immobile, whereas time was rich, lively, and dialectic.³⁴ Spaces were often conceived of geometrically as an empty area and as a container that was given meaning by the activities taking place there.³⁵ Scholars like Michel Foucault, Henri Lefebvre, and Edward Soja, among others, instead began rethinking space not as static, but instead lively. Space is a dynamic arrangement, something at once geographic, economic, demographic, sociological, ecological, political, commercial, national, continental, global.³⁶ Doreen Massey defines spaces as the product of interrelations, as spheres of possibility and heterogeneous multiplicity, and as always under construction.³⁷ Space, then, cannot be considered static, but instead as an ongoing assemblage of diverse local and global flows.

Critical to these theorists is the notion that space is relational, and that these relations contribute to the construction of values, meanings, knowledge, control, and production. This book emphasizes how the material and discursive work in tandem with one another. One aspect that’s important to note about the spatial turn is that, under these conceptualizations, space is not subject to essentialisms nor ontological dualisms. That is, the materiality of space may be given, but its organization and meaning is a product of dynamic and ongoing relations.³⁸ Here, the materiality of space is not

³³ Kevin Webb, “Netflix Says It’s More Worried About Competition From Video Games Like ‘Fortnite’ than Other Streaming Services,” *Business Insider*, January 18, 2019.

³⁴ Michel Foucault, “Questions on Geography,” in *Power/Knowledge: Selected Interviews and Other Writings, 1972–1977*, ed. Colin Gordon (Pantheon, 1980), 61.

³⁵ Henri Lefebvre, *Everyday Life in the Modern World* (Routledge, 1984).

³⁶ Lefebvre, *Everyday Life*, 4.

³⁷ Doreen Massey, “For Space,” *Geografiska Annaler: Series B, Human Geography* 89, no. 1 (2007): 9.

³⁸ Edward Soja, *Postmodern Geographies: The Reassertion of Space in Critical Social Theory* (Verso, 1989), 122.

denied, nor is it considered separate from social space; instead, social relations via discursive formations (and technologies) give meaning, definition, and organization to the matter of space.

Insights on how technologies are part of the ongoing organization of space, and by extension, regimes of power and control have been taken up in a wealth of scholarly literature on the cinematic³⁹ and the televisual.⁴⁰ One such specific example comes in James Hay's historical account of the televisual as part of spatial arrangements for the production, maintenance, and management of social relations.⁴¹ Hay considers how, over time, the television is brought into productive relationship with other environmental elements, in turn decentering the television itself while still considering how media matters at various sites. What is key for Hay is how television is both a site and a network, part of sociospatial arrangements of interconnectivity that help manage how individuals sense, experience, and move through the world. In an account more specific to the domestic site of the home, Lynn Spigel notes the way that televisual technology provided new connection and collapsed together what had formerly been separate public and private spaces. With television, the home is no longer divorced from public world; instead, the outside world is filtered in via broadcast technology, albeit in the form of recreation and consumerism.⁴² In turn, the relationship between television and space is a question of how regimes of power and control develop and are managed across sites throughout history.

Other more specific accounts on the relationship between the television and public space include Anna McCarthy's examination of television's pervasive role in public space, emphasizing its materiality and its influence on spatial experiences, public-private boundaries, and institutional power structures.⁴³

³⁹ Bill Marshall, "Montreal Between Strangeness, Home, and Flow"; Adrian Fielder, "Poaching on Public Space: Urban Autonomous Zones"; J. Paul Narkunas, "Streetwalking in the Cinema of the City: Capital Flows Through Saigon"; Rolando B. Tolentino, "Cityscape: The Capital Infrastructuring and Technologization"; and Geoffrey Nowell-Smith, "Cities: Real and Imagined," in *Cinema and the City: Film and Urban Societies in a Global Context*, ed. Mark Shiel and Tony Fitzmaurice (Blackwell, 2001).

⁴⁰ Jeremy and Kathleen F. Oswald, "From Windscreen to Widescreen: Screening Technologies and Mobile Communication," *The Communication Review* 13, no. 4 (2010); Paul Vonderau, "Beyond Piracy: Understanding Digital Markets," in *Connected Viewing: Selling, Streaming and Sharing Media in a Digital Era*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014).

⁴¹ James Hay, "Locating the Televisual," *Television & New Media* 2, no. 3 (2001): 212.

⁴² Lynn Spigel, *Make Room for TV: Television and the Family Ideal in Postwar America* (University of Chicago Press, 1992), 18.

⁴³ Anna McCarthy, *Ambient Television: Visual Culture and Public Space* (Duke University Press, 2001), 1.

She argues that television screens are embedded in local power dynamics and everyday environments, shaping public spaces in distinct ways. In later work, McCarthy refines this perspective, conceptualizing the screen not as a static object or network but as something that “takes place” within social and spatial contexts.⁴⁴ By contrast, Ginette Verstraete frames screens as active components within heterogeneous power relationships that shape public space through consumer integration and technological assemblages.⁴⁵ This distinction between screens as seamlessly embedded in daily life (McCarthy) versus screens as dynamic elements within broader sociotechnical systems (Verstraete) defines a key debate in spatial media studies. The rise of mobile streaming further complicates this discourse, as streaming platforms add another layer of spatial experiences through user practices, infrastructures, and augmented space.⁴⁶ These spaces, overlaid with dynamic data flows, intensify surveillance, immersion, and user movement. Mobile streaming, by extension, reinforces a bidirectional but asymmetrical relationship, where users gain new forms of engagement in exchange for their data.

One element that’s crucial to consider is that a mobile device does not automatically equate to streaming “on the go” nor in a variety of spaces. In fact, some of the earliest corporate imaginings of mobile streaming–adjacent practices were, paradoxically, largely confined to the home. Some initial attempts to incorporate the smartphone into audio-visual entertainment came via experiments such as the 2012 iPhone app “Avengers: A Second Screen Experience,” which allowed Marvel fans to sync their Blu-Ray views of *The Avengers* (2012) with additional content via behind-the-scenes videos, games, galleries, and information about the movie accessed simultaneously via their phones. Often dubbed “the second-screen experience,” novel attempts to capture audience attention by combining smartphones and more traditional televisual setups included companion viewing apps, social networking apps, franchise apps, and TV everywhere apps.⁴⁷ Second-screen adaptations were often a corporate vision of interactivity that forces audience’s attention and

⁴⁴ Anna McCarthy, “From Screen to Site: Television’s Material Culture, and Its Place, in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017), 93.

⁴⁵ Ginette Verstraete, “Media, Places, and Assemblies: From Videowalls to ‘Smart’ Environments,” *Space and Culture* 19, no. 4 (2016): 312.

⁴⁶ Lev Manovich, “The Poetics of Augmented Space,” *Visual Communication* 5, no. 2 (2006): 220.

⁴⁷ Ethan Tussey, “Connected Viewing on the Second Screen: The Limitations of the Living Room,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 204.

labor, while ignoring mobile possibilities in hopes of re-entrenching familiar ways of engaging with the televisual.⁴⁸ Stated differently, it's easy to forget these apps existed precisely because of how managed and top-down these trials in audience engagement truly were. Because many connected viewing apps required the use of television, there was still a focus on the audience and the living room instead of new ways that technology might work beyond the traditional context. Furthermore, while these apps all provide different ways of watching, they did mirror the traditional TV experience; even "extended content" for television programs was not always designed for mobile use.⁴⁹ Even for a platform like Netflix, in which corporate leadership appears keenly aware of growing mobile habits, advertisements for the service often depicted the living room as the site of streaming. What the rise (and fall) of second-screen applications for use alongside television programs establishes, then, is that the living room remained an important imagined site of televisual engagement—even as audiences increasingly began engaging with mobile devices to stream.

The iPhone, previously noted as a key milestone in the history and development of mobile streaming, is now well over fifteen years old. As mobile internet infrastructures improved from 3G to far more powerful 4G and 5G connectivity, individuals and platforms embraced streaming in a variety of spaces never before possible. By extension, a large part of the mobile streaming shift is interaction with new forms of the screen that may be untethered from the domestic spaces of the televisual. Key here is the importance of televisual screen size, noting that the notion of the television "set" has disappeared. We now speak of screens, monitors, and platforms, with a new focus on functionality and technical interdependence.⁵⁰ As emphasized in the discussion of a changing definition of flow, Jeremy Packer and Kathleen Oswald call for more attention to the history of screen technologies in relation to mobile media, noting how screen technologies function as a method of networked control exerted by and through each node, as mobile users move through space.⁵¹ Mobile screens, then, are part of ongoing types of biopolitical management, as the role of the screen in public life becomes more ubiquitous and mobile.⁵² As this brief subset of

⁴⁸ Tussey, "Connected Viewing," 214.

⁴⁹ Tussey, "Connected Viewing," 210.

⁵⁰ Paul Frosh, "The Face of Television," in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017), 307.

⁵¹ Packer and Oswald, "From Windscreen to Widescreen," 336.

⁵² Sean Cubitt, "Current Screens," in *The Screen Media Reader: Culture, Theory, Practice*, ed.

literature on screens shows, investigations of the screen often examine the materiality of the screen in relation to social and cultural forms, as well as questions of power and control.

Inspiring to our own work is recent scholarship from Barbara Maly-Bowe, who examines how Netflix influences the relationship between media consumption and the concept of the home. Similarly coming from a place of reinterpreting Raymond Williams's notion of mobile privatization, Maly-Bowe introduces the concept of "private mobilization." Here, Netflix redefines the home in three key ways: (1) It disrupts and reclaims traditional associations between television, mobility, and domestic life; (2) It links personalization and diversity to a sense of belonging, using "home" as a narrative tool for inclusion; and (3) It commodifies the private sphere, aligning with neoliberal market forces by transforming home into a space of both consumption and surveillance. Netflix, then, fosters a sense of personal belonging through diverse storytelling and flexible viewing, while also reinforcing corporate control over personal data and cultural narratives. We've examined elements of this increased corporate control in previous chapters. In addition, our own mobile personalization emphasizes the longstanding desire to monetize and make productive new times and spaces via mobile and portable technology.

Relevant overall in this section is that there is an enduring relationship between screens, space, and audio-visual entertainment. As explained in chapter 1, the screen of the cinema and its precursors structured urban and rural spaces. The televisual, by contrast, was a far less public but incredibly critical technology, introduced into the domestic space of the home at a time when families were becoming more mobile and withdrawn. The screen as technology has continued for decades to structure public spaces of stores, offices, schools, and more. However, the mobile screen of the smartphone allows a new orientation to space—one that is commodified and personalized. If Maly-Bowie focuses on private mobilization in terms of how the home is redefined by Netflix and other platforms, we instead want to trace the trajectory of mobile personalization. Key here is how the mobile screen of the smartphone can be used to transform all kinds of times and spaces for the user, while also working toward monetization and commodification via data capture. Relatedly, practices of mobile streaming animate scholarship on mobility and hybridity of space.

Mobile and Hybrid Streaming

For decades, the dominant vision of the television operated within the confines of the home and the imagined (and real) site of the living room. The mobile aspect of the televisual enters in the form of imagined mobility. Audiences were able to “travel” around the world thanks to new audio-visual coverage of places, people, and events previously unseen and unheard. Of course, as we examined in chapter 1, there were simultaneously mobile and portable orientations to the televisual and the cinematic that coexisted with the prevailing vision of static and domestic television. Our overarching argument in that chapter is that there is a long lineage of a mobile and portable desire for entertainment and information in a variety of spaces and times via technology. Viewed from that lens, mobile streaming is a continuation of an existing trajectory and not something wholly new.

The “mobile” is a concept that’s at once separate from but also deeply intertwined with space. The prior section discussed the spatial turn, a moment of increased scholarly attention to spaces as multiplicitous, relational, and always under construction. In the 1990s, the mobilities turn represented another moment of scholarly emphasis, this time on the increasingly important dynamic of mobility. The 1990s were a period of greater global connection, driven, in part, by new international trade agreements, the fall of the Soviet Union, and the nascent commercial web. Scholars interested in the understudied element of mobility to local and global social formations emphasized questions of globalization, the production and reproduction of territories and nation-states, as well as the implications for identities and belonging.⁵³ Other key mobility theorists highlight how mobilities are intertwined with the social. Mobility simultaneously produces and is produced by social relations.⁵⁴ Furthermore, mobility studies include the act of physical movement, representations of movement that give mobility shared meaning, and the experienced and embodied practice of movement.⁵⁵ As such, “the mobile” is not only the physical act of movement, but also how movement is represented in cultural forms, as well as the ontological and phenomenological dimensions of mobility that require attention.

⁵³ Mimi Sheller and John Urry, “The New Mobilities Paradigm,” *Environment and Planning A* 38, no. 2 (2006): 211.

⁵⁴ Tim Cresswell, “Towards a Politics of Mobility,” *Environment and Planning D: Society & Space* 28, no. 1 (2010): 21.

⁵⁵ Cresswell, “Towards a Politics of Mobility,” 19.

What is important about the mobilities turn is that a conception of space and the social require attention to movement itself, which has implications for technologies and practices that include mobile streaming. Stephen Wiley and Jeremy Packer clarify that the mobilities turn allows scholarship to move away from a media-centric approach (and the problematic questions of media effects, technological determinism, and audience agency). Instead, emphasis on questions of spatial production and mobilities allows scholars to situate communication technologies and discursive frameworks within a broader social field that encompasses networks of circulation, mobilities and immobilities, infrastructures surveillance control, and social organization.⁵⁶

Taken together, the mobilities turn is critical to the study of streaming media for several reasons. First, the prevalence of mobile streaming might not “replace” fixed streaming forms, yet users around the world are increasingly engaging with the mobile, with younger individuals heavily gravitating toward portable and mobile streaming. Second, the new forms of streaming possible via mobile technologies and infrastructures—not only watching cinematic and televisual content but also captured and shared mobile streaming content—requires an expansion of what it means “to stream.” To this end, the spatial, mobile, and portable dimensions of streaming make mobile streaming a critical object of scholarly examination.

Examples show how location and mobility have become key aspects of new streaming practices. One study of early developing mobile engagement found that viewing habits of university students were strongly shaped by the location of viewing.⁵⁷ Other similarly early studies found that different platforms are amenable to different viewing styles, finding that mobile media is more participatory and “lean forward” in terms of viewing practices and content consumption.⁵⁸ This active or passive demarcation mirrors the conceptualizations of streaming as fixed versus mobile dialectic and is increasingly relevant to consider when taking into account streaming not only through audio-visual content platforms but also mobile streaming and livestreaming through social media. As mobile streaming takes on new forms, including streaming through social media, it is critical to emphasize

⁵⁶ Stephen B. Crofts Wiley and Jeremy Packer, “Rethinking Communication After the Mobilities Turn,” *The Communication Review* 13, no. 4 (2010): 264.

⁵⁷ Chuck Tryon and Max Dawson, “Streaming U,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 218.

⁵⁸ Sharon Strover and William Moner, “The Contours of On-Demand Viewing,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 248.

how space and mobile devices remediate a static history of streaming largely connected to the fixed and domestic spaces of the televisual. In other words, mobile streaming expands these activities to new times and spaces that were previously only imagined or perhaps just briefly experimented with.

Finally, the mobile nature of contemporary streaming practices further complicates divides between physical and digital spaces. Scholarship and popular culture depictions in the 1990s and early 2000s positioned the web as a “cyberspace” that individuals would enter or leave via fixed devices.⁵⁹ Accompanying this bifurcation between the digital and physical were prognostications that included the dissolution of cities, the reduced need for mobility, spatial homogenization, and the unimportance of the corporeal body.⁶⁰ Predictions about the future of technology often get it wrong, and these instances are no exception. Instead, mobile streaming is best considered as part of hybrid spaces, merging the physical and digital when it comes to social environments and created in large part by the mobility of individuals connected via mobile devices.⁶¹ Hybrid spaces examine the mutual imbrication of the digital and physical via location-aware mobile technologies. This concept of hybridity is a crucial departure of mobile streaming from the televisual, cinema, and even more fixed forms of streaming. The aspects of mobility, portability, pervasive web connectivity, and location-awareness make mobile streaming a novel media practice with substantial implications for how the user senses, experiences, and moves through space. This is also a major question of power—specifically, how mobile streaming and web-connected screens guide the user through space and make productive their time.

This brief overview of space, mobility, and hybrid space foregrounds the importance of space as a relational, ongoing, and multiplicitous construction with social and communication dimensions. Moreover, mobility and movement—both physical and digital—are important aspects of the construction and reconstruction of spaces as hybrid. The spatial and mobile turns, as well as understanding the elision between the physical and digital via mobile technologies as a form of hybrid space is critical to understanding mobile

⁵⁹ Helen Couclelis and Harvey J. Miller, “Misses, Near-Misses and Surprises in Forecasting the Informational City,” in *Societies and Cities in the Age of Instant Access*, ed. Harvey J. Miller (Springer Netherlands, 2007), 72.

⁶⁰ Francis Cairncross, *The Death of Distance: How the Communications Revolution Is Changing our Lives* (Harvard Business School Press, 1997), 10.

⁶¹ Adriana de Souza e Silva, “From Cyber to Hybrid: Mobile Technologies as Interfaces of Hybrid Spaces,” *Space and Culture* 9, no. 3 (2006): 262.

streaming. Mobile streaming practices take a variety of forms, yet it is the new connections to and constructions of space, the movement through space in concert with audio-visual streaming content, and the hybridity of space due to data collection and the new relevance of location, geographic, and device data to streaming that require new attention.

Mobile Streaming, Big Data, and Algorithms

It's a story told so many times that it's become streaming legend. Netflix's "gamble" on rebooting *House of Cards*, one of the platform's early hit series, was not a risk in the traditional sense. That is, the development cycle for Netflix's original programming heavily departed from the production cycle of television networks. When networks relied on some level of spotty ratings data and past trends, the decision to greenlight or pass on a show was largely driven by gut instinct. Netflix, conversely, had gobs of audience data points to work with. The company knew there was an audience who loved the original *House of Cards*, they knew there was an audience who would watch actor Kevin Spacey, and they knew there were dedicated fans of director David Fincher. What mattered was looking at points of overlap between these groups to create some level of empirical evidence that an audience existed for a new series.⁶² In other words, big data has been key to content production in the streaming era.

Perhaps the most significant departure of streaming media from preceding forms of audio-visual engagement comes in the form of big data and algorithms. As previously mentioned in our focus on streaming industry shifts, streaming media is part of a mass media alteration from a transmission model to one that also includes forms of data collection, storage, and processing. This utilization of data is critical in understanding streaming media. Furthermore, the omnivorous nature of big data collection, as well as increasing engagement with mobile forms of streaming magnifies the importance of data points for device type, location, and mobility.

The ideology of collecting any and all data points for use in decision making and predictive analysis now undergirds just about any contemporary field—from streaming video to social media, from medicine to industrial production, and from government to corporate decision making. For streaming specifically, the switch from analog to digital cable, then to DVR, and

⁶² Blake Hallinan and Ted Striphas, "Recommended for You: The Netflix Prize and the Production of Algorithmic Culture," *New Media & Society* 18, no. 1 (2016): 128.

finally to streaming permitted a “quantum leap” in tracking and monitoring user behavior.⁶³ Early on, this included what shows audiences stored on their DVRs, when they viewed content, when they paused, rewound, or fast-forwarded. This data was supercharged by streaming platforms like Netflix, which tracks what we watch, when, where, with what device, when we pause, rewind, or skip ahead, and when we abandon a show entirely.⁶⁴ Consequently, Netflix and other streaming platforms can parlay user meta-data into decision making regarding original programming. Netflix, like other providers, allows user data to be shared between partners like internet service providers, mobile phone carriers, and voice assistant platforms.⁶⁵ Because these corporations have access to viewer preferences regarding genre, actors, directors, bingeing penchants, and more, streaming providers can use algorithms to create programming with a built-in, near-guaranteed audience. Algorithms, moreover, deliver and promote content based on previous user activity, ensuring a more efficient, curated, and personalized relationship to audio-visual media, as well as the potential of a filter-bubble effect in terms of suggested content.

Big data and algorithms matter from a mobile streaming perspective as well. TikTok’s own recommendation algorithm is based on factors that include user interactions (videos that are liked or shared, followed accounts, comments posted), video information (captions, sounds, hashtags), and device and account settings (language preference, device type, country settings). Further, TikTok’s machine learning models primarily rely on metadata like description, tags, time, and location of video upload.⁶⁶ In essence, contemporary streaming practices build off the data structures of more fixed streaming antecedents; however, elements of location and device type become more magnified as streaming becomes mobile and social.

While “the algorithm” has become a bit of a catch-all buzzword by virtue of social and streaming media platforms, what the term refers to is merely a mathematical or logical set of instructions.⁶⁷ Layers of algorithms can be used to make sense of large datasets via what’s known as machine learn-

⁶³ Mark Andrejevic, “Surveillance in the Digital Enclosure,” *The Communication Review* 10, no. 4 (2007): 308.

⁶⁴ Kate O’Flaherty, “All the Ways Netflix Tracks You and What You Watch,” *Wired*, October 10, 2021.

⁶⁵ O’Flaherty, “All the Ways Netflix Tracks You.”

⁶⁶ Will Knight, “This Technique Can Make It Easier for AI to Understand Videos,” *Wired*, October 15, 2019.

⁶⁷ Lucas Introna and David Wood, “Picturing Algorithmic Surveillance: The Politics of Facial Recognition Systems,” *Surveillance & Society* 2, no. 2–3 (2004): 180.

ing. And the use of neural networks constitutes a more powerful form of machine learning known as deep learning. Machine and deep learning are subsets of what we'd call artificial intelligence, most commonly used now to identify relationships, detect and uncover patterns, classify and transcribe, and even to decide or recommend in an automated fashion.⁶⁸ In turn, recommendation algorithms are critical to what we see when we stream—no matter if that occurs while watching Amazon Prime on a smart TV in one's living room or streaming TikTok on a smartphone while riding the metro.

Equally important is the aforementioned aspect of big data. Big data has been defined numerous ways via some key characteristics. Certainly, the idea of collecting and storing data about the world is nothing new. Even ancient Sumerians, the first human civilization, utilized record keeping. What sets the big data era apart includes the digitization of records for computational readability, unprecedented storage capacity to house data digitally, and new modes of algorithmic analysis to help make sense of data.⁶⁹ Big data allows entities to harness information in novel ways, producing insights of significant value. This entails the ability to analyze enormous amounts of data instead of smaller (likely randomized) sets; a willingness to embrace data messiness instead of precision; and a growing respect for correlations.⁷⁰ Finally, big data is omnivorous. Platforms attempt to collect any type of information they can from user practices and interactions for analysis and insights.⁷¹ This omnivorous appetite of big data for ever-increasing types of data is increasingly relevant to mobile streaming, as device type, location, and geographic data add to this insatiable mode of data collection. Taken together, big data is as much an ontological shift as it is an epistemic one: This process entails an intensified digitization of the longstanding process of collecting, storing, and processing data, along with a reverence for a worldview that if we collect enough data, we can access and predict all aspects of reality itself. Big data is correlative and predictive, yet these correlations are often opaque to a point beyond human comprehension.

While big data and algorithms largely structure streaming media practices, it is worth noting the asymmetrical aspects of this platform-user relation-

⁶⁸ Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective* (MIT Press, 2012); Ian Goodfellow et al., *Deep Learning* (MIT Press, 2017).

⁶⁹ Mark Andrejevic and Kelly Gates, "Big Data Surveillance: Introduction," *Surveillance & Society* 12, no. 2 (2014): 186.

⁷⁰ Viktor Mayer-Schönberger and Kenneth Cukier, *Big Data: A Revolution That Will Transform How We Live, Work, and Think* (Houghton Mifflin Harcourt, 2013), 12.

⁷¹ Mark Andrejevic, "The Big Data Divide," *International Journal of Communication* 8 (2014): 1674.

ship. There exists a growing divide between those with access to big data and its attendant infrastructure and those without.⁷² We hear about big data living in “the cloud.” The cloud merely refers to tons of servers housed in a warehouse owned by large companies. The terminology of the cloud invokes a lightness and freedom that runs contra to the reality of data collection; this metaphor obfuscates the surveillant aspects present in all forms of streaming media.⁷³ Put another way, platforms like TikTok have significant power, and this has resulted in geopolitical controversy when it comes to attempts at regulation. This situation remains ongoing and is covered thoroughly in other publications. Nevertheless, global bans of TikTok by a number of countries demonstrate how platforms vie for power via reach, users, data, and influence.

As scholars have noted, the conjunction between big data and algorithms has had significant impacts on what we watch, how we watch, and what gets made in the first place. We’ve mentioned in this section how big data collection is a far more efficient model of measuring viewership than previous forms when compared to prior systems of assessing television ratings through a variety of technologies used for representing, and calculating audiences, including face-to-face interviews, phone surveys, and diaries.⁷⁴ The use of big data has led to a shift from intuition-based decision making by studio executives to data-driven models in streaming platforms, where audience engagement continuously strengthens algorithmic predictions. These algorithms, though largely proprietary, are central to how platforms mediate cultural authority.⁷⁵ Blake Hallihan and Ted Striphas examine this dynamic through the case of *The Netflix Prize*, a rare look into how the Netflix algorithm operated through an open competition held by the company.⁷⁶ The authors explain that streaming algorithms often work to mediate more divisive outlier content; rather than evaluating cultural worth, Netflix moderates atypical works to fit broader patterns, displacing cultural judgment into technical processes. This reinforces a closed loop of recommendation, where content increasingly reflects user preferences rather than challeng-

⁷² Andrejevic, “Big Data Divide,” 1674.

⁷³ Mark Andrejevic, “Surveillance in the Digital Enclosure,” *The Communication Review* 10, no. 4 (2007): 304.

⁷⁴ Jérôme Bourdon and Cécile Méadel, “Inside Television Audience Measurement: Deconstructing the Ratings Machine,” *Media, Culture & Society* 33, no. 5 (2011): 792.

⁷⁵ Michael D. Smith and Rahul Telang, *Streaming, Sharing, Stealing: Big Data and the Future of Entertainment* (MIT Press, 2016), 211.

⁷⁶ Blake Hallinan and Ted Striphas, “Recommended for You: The Netflix Prize and the Production of Algorithmic Culture,” *New Media & Society* 18, no. 1 (2016).

ing them. Consequently, cultural value is no longer anchored in human interpretation alone; systems now play a decisive role in what enters and shapes the cultural field.

Of course, this book is all about the mobile aspects of streaming media. Both big data and algorithms take on new forms via mobile practices. As previously noted, mobile devices initially became central to streaming media through the rise of the second-screen experience. The second screen facilitated increased real-time monitoring, customization, and targeting—key elements of the interactive commercial economy.⁷⁷ Framed as enhancing viewer agency, second-screen apps support a “push” model of data collection, capturing user input through designed interactions. This model reaffirms the value of live viewing and advertising while contributing to a broader system of data capture.⁷⁸ Notably, this viewing paradigm was grounded in the domestic space of the living room, where mobile devices are woven into fixed media habits. Other studies have found that college student viewing habits are shaped by place, social setting, cost, and access, signifying that *what* and *how* students watch are factors of *where* they watch.⁷⁹

Similarly, other scholars underscore the role of mobility, temporality, and platform in shaping viewing behavior. They note that mobile viewing tends to be more participatory and “lean forward,” necessitating a rethinking of viewership in terms of devices, platforms, and context.⁸⁰ Together, these accounts highlight the entanglement of mobile streaming with big data and algorithmic governance. Streaming platforms rely on pervasive data collection to optimize both content production and targeted distribution. As audiences increasingly stream via mobile devices, data tied to geography, behavior, and device becomes critical to predictive analytics, reinforcing the commercial imperative for constant growth. In essence, mobile streaming represents an emergent frontier of data capture, particularly because of how well it captures user attention across new times and spaces.

Taken together, big data and algorithms are pervasive aspects of both streaming and mobile steaming media. Big data and algorithms have been a significant factor in the more efficient production of content as well as content distribution via suggestion to targeted audiences by streaming

⁷⁷ Lee and Andrejevic, “Second-Screen Theory,” 42.

⁷⁸ Lee and Andrejevic, “Second-Screen Theory,” 45.

⁷⁹ Tryon and Dawson, “Streaming U,” 248.

⁸⁰ Sharon Stover and William Moner, “The Contours of On-Demand Viewing,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 248.

platforms. Moreover, the goal of continual profit expansion aligns with the omnivorous mode of big data collection: More data allows for more correlative predictive potential, and device, geographic, and location data are increasingly important as individuals engage in mobile streaming practices.

Conceptualizing Mobile Streaming

In this chapter, we've engaged with a major question: How does mobile streaming remediate prior understandings of flow, liveness and sharedness, screens and space, mobile and hybrid, and algorithms and big data? This is far from a one-size-fits-all situation, especially considering just how embedded streaming has become. Immensely popular mobile-focused platforms like TikTok and Instagram operate alongside audio-visual media providers such as Netflix, Amazon, and Hulu. With users more and more likely to engage in streaming practices via a mobile device, it's critical to consider how the increase in user captured and shared streaming content, designed for the short-form temporal duration and comparatively small screen size of mobile viewing, necessitates reexamination of what it means to stream.

As we've highlighted in previous chapters, there's a historical dimension at play. The desire for a mobile and portable orientation to entertainment and information anytime and anywhere is far from a new goal. While historical examples help situate this trajectory, we've also discussed changes in the material technologies and infrastructures needed to enable the kinds of mobile streaming engagement we've come to integrate into our lives. And finally, we've looked at the platform model as key to changing mobile streaming monetization.

This trajectory has led us to this final chapter in which we've reconsidered how scholarly accounts on televisual, cinematic, and fixed streaming are challenged, remade, and remediated by mobile and portable streaming. These concepts, deeply intertwined with pre-streaming mass media and fixed streaming alike, are found just as readily in mobile streaming practices. Nevertheless, the myriad of novel ways to stream anytime and anywhere does necessitate a critical reassessment of the unbroken experience of watching, of how mobile streaming is live and shared like never before, how mobile screens are part of the spaces where we watch, how streaming involves movement and hybrid orientations to space, and how new and old forms of data and algorithms are part of these experiences.

In addition, the temporal, flexible, and rhythmic aspects of streaming and mobile streaming make clear that the concepts of liveness and sharedness that largely structured televisual and other synchronous mass media engagement continue to be part of streaming media as well. While some of these features have been abandoned by the audio-visual streaming providers commonly associated with streaming (Netflix, Amazon, etc.), the adoption of and increased emphasis on livestreaming by social media platforms demonstrate the attention paid to a shared streaming experience. While the importance of big data and algorithmic engagement has long structured the production and economic efficiency of streaming, the capture of location, geography, and device information is critical to new evolutions of mobile streaming. What is critical here is that location-aware features are gradually made salient within new mobile streaming alternatives.

Together, this chapter has contextualized new forms of streaming media by taking into account novel mobile streaming practices. Although more fixed forms of streaming media have not been rendered irrelevant nor replaced by mobile streaming, the growing amount of mobile engagement—including the overtaking of fixed streaming by mobile—necessitates increased scholarly engagement with this streaming evolution. Consequently, this chapter examined the increasingly mobile dimensions of streaming regarding some affiliated sociotechnical practices such as industry shifts, liveness, sharedness, plasticity, and algorithmic engagement. These are all elements of mobile personalization. Mobile and portable technologies are critical components in the ongoing transformation of times and spaces, as they provide entertainment and information for users while simultaneously operating toward the goals of monetization and control. Critical here is that the local is not erased but made salient through mobile devices, infrastructures, streaming practices, location-aware technologies, and data collection. Taking seriously the shift to mobile streaming means reconsidering the increasingly dominant method of audio-visual media interaction as a dynamic rather than static process in which location and geography matter.

CHAPTER FIVE

The Future of Mobile Streaming

Everybody's on a barge floating down the endless stream of great TV.

—The Arctic Monkeys

Attempting to predict the future is an exercise that often ages poorly. Tech CEOs see stocks benefit from bombastic hysteria and editorials gain traffic from hot takes. But academic work should tread more carefully. We previously mentioned George du Maurier's 1897 comic depicting an imagined Edison invention called the telephonoscope, where two individuals can view, listen to, and interact with a public setting from the comfort of their own home. This playful fiction certainly seems prescient of a certain type of streaming now well over a hundred years later. But for every technological prognostication that comes true, there are countless others that miss the mark. Science fiction like *Blade Runner* had audiences thinking that synthetic human replicants would be born in 2016. *2001: A Space Odyssey* showed humans traveling to Jupiter, when manned spacecraft have not ventured past the moon now some twenty years later than the movie's setting. Even *Neuromancer*, a sci-fi novel lauded for its prescience on the internet and AI, still imagined characters going through their lives in ways that echoed the 1980s context in which it was written; no one is glued to a screen in their pocket.

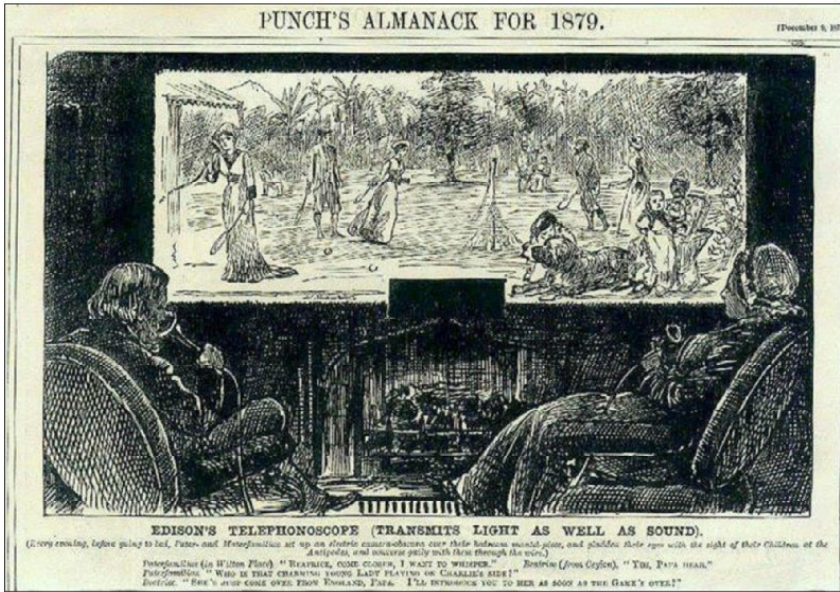


Figure 5.1. George du Maurier's telephonoscope comic, 1897. Uploaded to ResearchGate by Benjamin Burroughs.

Science fiction might seem like an easy target to pick on (lovingly, of course), but even more impactful and bombastic prognostications are seen time and time again in contemporary pitches by big tech. Tesla has promised the imminent advent of self-driving cars for almost a decade.¹ Companies pitching the future of AI and the accompanying global disruptions seem to be more vaporware than actuality.² Hype has become critical to courting tech investors, even if the claims are often dubious.³ In short, it's probably safer to assume big tech has it wrong than the other way around.

Streaming media is no exception when it comes to slightly wrong and even outright incorrect predictions. In 2014, for example, Netflix CEO Reed Hastings correctly asserted that Nielsen ratings could not capture accurate audience numbers for Netflix programming because of the amount of phone and iPad views.⁴ While he was correct about the growing number of mobile views, he also predicted the fairly imminent death of traditional television,

¹ Trisha Thadani and Rachel Lerman, "7 Times Elon Musk's Predictions About Self-Driving Cars Fell Short," *The Washington Post*, October 10, 2024.

² Emily Stewart, "We're Entering the Age of AI BS," *Business Insider*, March 27, 2024.

³ Gemma Milne, *Smoke & Mirrors: How Hype Obscures the Future and How to See Past It* (Robinson, 2020), 3.

⁴ John Hect, "Netflix Chief Downplays Nielsen Plans to Measure Streaming Service Viewership," *Hollywood Reporter*, November 24, 2014.

declaring, “It’s kind of like the horse, you know, the horse was good until we had the car . . . the age of broadcast TV will probably last until 2030.”⁵ It is true that a majority of US households now do not have cable TV, and that “cord cutting” is increasing yearly.⁶ In 2021, 69 percent of surveyed respondents stated that the high cost of cable and satellite was the driving factor to their switch to streaming-only.⁷ Like many other sociotechnical disruptions, however, what started out with utopic promise has become more of a mixed bag. The freedom of cord-cutting cable services for one cheap streaming alternative has come to look a lot more like cable than ever before. The promise of liberty from inconveniences such as advertisements and time-specific content has also come crashing back to earth. And even when it comes to user-generated content streaming platforms, the social optimism of Web 2.0 has given way to the dangers of mis/disinformation and mental health issues.

As noted throughout book, delineating a singular origin date for streaming media is challenging. In a popular sense, the streaming revolution for audio-visual content began to take off in 2005 and ensuing years, which means we have been living with streaming as a normal part of everyday life for roughly two decades. All of this is to say that in a chapter about the future of mobile streaming, we are wary of going too far out on a limb. Chances are that streaming in all its forms is here to stay. Individuals—and particularly younger audiences—seem to prefer mobile and short-form content. But that doesn’t mean that cinematic and televisual-style content is not long for this world. If there is any trend we can safely assume will continue, it’s that individuals will keep streaming.

Up until this not-so-bold claim, we’ve spent this book looking to the past and present of mobile streaming for what we believe to be good reason. Our project is inspired by media archaeological accounts, particularly those attentive to understudied portable and mobile technologies. We eschew the search for a “eureka moment”—a simple narrative of a lone inventor changing the world. Instead, we’ve attempted to show the complex lineage of mobile streaming, as well as the complex mix of technological changes and social imaginings that underpin our contemporary practices. We did so by looking at material changes in technologies—not just the screen of

⁵ Hect, “Netflix Chief Downplays Nielsen Plans.”

⁶ David Satin, “Report: 52% of US Households Do Not Have Cable; Nearly 70% of Streaming Users Plan to Cycle,” *The Streamable*, February 13, 2023.

⁷ Lee Rainie, “Cable and Satellite TV Use Has Dropped Dramatically in the US Since 2015,” Pew Research Center, March 17, 2021.

the smartphone, but also the infrastructures, protocols, wireless networks, APIs, and platforms that make streaming anytime and anywhere possible. We've also highlighted changes in the discursive—that is, how powerful forces shape the idea of mobile streaming. Rather than merely accepting that mobile streaming is a relatively new idea, we've traced the emergence of a desire for a sociotechnical orientation to time and space as something where individuals could access information and entertainment, while corporate interests could extract value. We call this lineage mobile personalization.

We began chapter 1 by tracing the historical roots of mobile personalization. Streaming media is inexorably linked with the history of the televisual and cinematic—something that's reinforced through statements by representatives of major streaming platforms. Our focus was not to outright reject the relevance of the film and television to contemporary mobile streaming, but to instead complicate these narratives. Cinema and later television were critical technologies to the temporal and spatial reorganization of industrial modernity and mass society. But alongside the fixed, brick-and-mortar structures of the cinema and the domestic site of emergent televisual consumption by the nuclear family were imaginings of these technologies intertwined with the mobile and portable. A media archaeological approach highlights the drive-in cinema and the portable television as oft-ignored technologies on the margins of commonly accepted historical narratives. Most critically, these mobile and portable takes on the televisual and cinematic demonstrate that the desire to expand spaces of entertainment and information while also monetizing spaces through novel technologies existed long before the smartphone.

Chapter 2 considered the hidden infrastructures and platform systems that make mobile streaming possible. What seems effortless—a video on your phone in an airport line—is actually the product of a vast, layered architecture of networks, cloud systems, platforms, and data pipelines. Mobile streaming depends not just on smartphones or apps, but also on global telecommunications networks, data centers, algorithmic curation, and platform business models built to extract data and monetize attention. We showed how platforms like Netflix, YouTube, Spotify, TikTok, and Twitch operate as both services and infrastructures, using APIs, wireless networks, and big data to shape media access, recommendation, and consumption. These architectures of mobility have shifted media from static, scheduled delivery to a personalized, always-on, real-time system deeply entangled with surveillance and commercial interests. Power is increasingly concentrated

in a few platform companies that control both the means of delivery and the flow of information. This convergence reshapes culture, narrows choices through algorithmic curation, and extends corporate influence into daily life. Recognizing how these systems operate, and who benefits, is essential for understanding the stakes of the mobile streaming era. Streaming media is not just about content; it is about control, infrastructure, and the future of digital culture.

Chapter 3 continued this trajectory through an examination of how mobile streaming has reshaped everyday life and redefined media industries. Platforms like TikTok, YouTube, Twitch, Instagram, and Apple Music have built seamless, always-on ecosystems where content and user data move fluidly across devices. This convenience, however, is engineered to serve deeper commercial goals such as maximizing user engagement, extracting data, and monetizing every interaction through advertising, subscriptions, microtransactions, and platform integration. Platform mobility and cloud infrastructure work together to tether users to corporate ecosystems, blurring the line between leisure and labor. Mobile streaming has become embedded not just through technology, but also through strategic design choices that normalize continuous connection and consumption. These platforms are not neutral; they shape behaviors, speech, and cultural norms based on business imperatives. The global reach of mobile streaming also highlights inequalities in infrastructure, regulation, and platform power. As mobile streaming becomes the dominant mode of media consumption, the challenge ahead is to ensure that these powerful systems serve broader societal needs, not just corporate profit. Understanding the commercial and political architecture behind mobile streaming is essential if we hope to navigate, challenge, and reshape the digital environments we increasingly live within.

In the final chapter, we considered how the move to mobile streaming complicates some of the key academic literature on audio-visual media. Specifically, mobile streaming necessitates reconsideration of elements like flow, liveness and sharedness, screens and space, mobile and hybrid, and algorithms and big data. These concepts originated in scholarship about audio-visual media, including more recent accounts of streaming. However, mobile streaming remediates some of these academic understandings. Flow, the sequence of programming that creates an unbroken experience for audiences, is remade in the era of the endless video scroll. Liveness and sharedness, aspects of mass media that seemed endangered by early streaming, are made immensely relevant to mobile streaming via social connectivity. Mobile

streaming practices also foster new relationships to screens and space, as well as mobile and hybrid engagement practices. And finally, the elements of algorithmic suggestion and big data collection, storage, and processing are shaped by mobile devices and location. The goal, of course, is to consider what it means to “stream” in a mobile world.

Taken together, we’ve spent a lot of this book looking backward to better understand the present moment of mobile streaming. Just a few short years ago, a platform like TikTok was an unrealized idea. Now, both conventional streaming platforms as well as rival social media platforms chase TikTok’s runaway success. Short-form streaming was not realized through the corporate releases of Quibi, but instead through social media content. At the same time, audiences watch a host of other content via mobile devices, be it shows, parts of movies, advertisements, game streams, adult content, news, and advertisements. It’s almost become redundant to say “mobile streaming,” considering just how much streaming engagement now comes via mobile devices.

The plurality of mobile streaming practices aligns with our material-discursive archaeological approach. That is, we aim to decenter one simple narrative of mobile streaming, as well as to avoid deterministic thinking. We aren’t sure where streaming goes next. And it’s difficult to decipher whether what’s next will be “good” or “bad.” Social media streaming allows people to come together, to share information and be entertained. It also allows misinformation and disinformation to circulate persuasively and effectively. Streaming platforms increasingly optimize for mobile, but as noted throughout the book, these same platforms are data hungry and engage in a type of surveillance and subtle manipulation. Not to mention the fact that the production of content has gone from taking risks to data-based in a way that can often feel soulless. As examined in chapter 3, the layers of technologies that underpin streaming are subject to corporate consolidation. Just what happens when major forms of communication are controlled by a small number of actors is a question with some serious consequences. And how consolidation plays out across geopolitical borders, such as the example of multiple countries banning TikTok, remains to be seen.

Mobile streaming is no longer novel but a seamless part of daily life, threading through our routines in ways we barely notice. We carry it with us from morning commutes to late-night wind-downs, expecting entertainment anywhere. That ubiquity is a mixed blessing. It can enrich moments with music or video, but it can also fragment our attention, shape our habits,

and quietly trade away privacy and time. Streaming services now offer both promise and puzzle, connection and distraction, all woven together in the background of our days.

Taken together, while we might not know exactly what streaming will look like five, ten, or twenty years from now, we can speak to a few certainties. Streaming is likely to continue to change, intermix with new contexts, and further integrate into media ecosystems. If previous trends hold true, the distinction between mobile and fixed is likely to become blurrier and perhaps less meaningful. Most of the world, particularly younger individuals and those in the Global South, are far more likely to access the web via mobile devices. For them, mobile is the norm. Short-form mobile streaming, at first something trendy, hip, and cutting edge via vanguard platforms like TikTok, is now an integral part of more legacy platforms like Facebook, which largely trends toward older users. In essence, what mobile streaming looks like will change and further integrate into everyday contexts. However, the perspective of mobile personalization allows us to understand that these changes go beyond the merely technological and are instead sociotechnical in nature. New constellations of mobile streaming are not isolated but rather a part of a long lineage. And that lineage is driven by the desire to capture new times and spaces for entertainment, information, production, and monetization.

These are just some of the many developing aspects of mobile streaming. Mobile streaming encompasses such a range of platforms and practices, and it looks to remain a key modality of audio-visual interaction for the foreseeable future. We hope that by looking to the past development of mobile streaming, we can better understand our present moment. Ultimately, the trajectory of mobile personalization is not novel, even if what we're watching and listening to, and on what platform, is continually changing.

Looking ahead, the future of mobile streaming remains open-ended. It will continue to evolve alongside new devices and services but will probably remain the constant hum of life. It's neither an unalloyed blessing nor a curse; it is simply one of the many ways technologies are intertwined with culture and everyday life. Rather than feeling optimistic or pessimistic, perhaps the wisest stance is to stay curious and mindful of how the next stage unfolds. In the end, mobile streaming is here to stay and will keep reshaping the fabric of daily life—a flowing thread of contemporary existence, carrying our hopes and contradictions into whatever comes next.

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People were watching and listening on the move long before the iPhone's launch. The world of always-on streaming, accessible anywhere, has a pre-history: magic lanterns, drive-in theaters, and portable radios.

In *Mobile Personalization*, Justin Grandinetti and Charlie Ecenbarger place Netflix, YouTube, Spotify, TikTok, and other corporate claimants in a lineage that includes the build-out of sprawling, invisible infrastructures—undersea cables, server farms, and adtech. These “architectures of mobility,” together with their Silicon Valley underwriters, make seamless, algorithmic streams possible and profitable. The upshot is an evolving regime of mobile personalization, with entertainment for consumers and data-driven monetization for companies.

Mobile Personalization recasts foundational ideas from Raymond Williams, including “flow” and “mobile privatization,” to account for the algorithmic delivery of targeted video and audio to lucrative audiences. As this important new book argues, the mobile streamers have delivered convenience, but also audience fragmentation and behavioral capture.

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